



Open Chiplet Atlas Harness (OCAH)

Integrator Guide

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This guide provides instructions and best practices for integrating OCAH subsystems into your chiplet design. Each section covers integration considerations, interfaces, and configuration requirements for the respective subsystem.

Revision History

This document tracks major revisions and updates to the OCAH Integrator Guide.

Table 1. Revision History

Version	Date	Author	Description
0.0.1	2026-06-17	Alessandro Ottaviano	Initial integrator guide draft
0.0.2	2026-08-13	Alessandro Ottaviano	Completed the documentation migration: added primitive-cell replacement and CDC constraint guidance, corrected DTP/JTAG integration, and aligned SMC/SEP paths, GPIO and DFX status descriptions, and cross-guide references with the current tree.
0.0.3	2026-08-21	Ben Melton	Added a Project Defines section covering compile-time defines that apply across OCAH, starting with SIMULATION : how to set it, the testbench hooks and models it enables, and the flows it must not reach.
0.0.4	2026-08-31	Ben Melton	Documented the SMU IC_RESET TDR slice layout (SMC, SEP, external) and marked the external slice as adopter-defined.
0.0.5	2026-09-02	Alessandro Ottaviano	Consolidated immutable ROM FW integration guidance, including the generic weak-symbol override method and SMC, SEP, and Key Manager ROM FW integration sections.
0.0.6	2026-09-03	Colin McKellar	Added the initial version of the trust-aware User IP Integration guide for SMC-managed peripherals, SEP security extensions, and application-owned accelerators, including a security-contained NPU hardware/software example and an evidence-based integration checklist.

Chapter 1. SMU Top-Level Integration

The System Management Unit (SMU) comprises three peer subsystems: the System Management Controller (SMC), the Debug & Test Ports (DTP), and the Security Processor (SEP). The subsections below describe each subsystem's interfaces independently, followed by the [SMU Composition](#) section that explains how they are wired together inside the SMU and how to select between the `smu.sv` logic-level core and the `smu_wrapper.sv` reference integration.

1.1. Address Map Integration

The OCAH address map has three levels, and the assignment contract splits across the integrator, SEP firmware, and SMC firmware. Assign and program them from this section; the deeper subsystem and crossbar sections referenced below give the per-block detail.

- **Chiplet envelope (integrator).** The SoC/chiplet address map assigns each subsystem a global base and reserved size, and the adopter's fabric decoder routes those ranges to the SMU. The SMU crossbar preserves offsets and does not constrain envelope size, so the integrator owns non-overlap and containment at the SoC level.
- **SEP and SMC chiplet apertures (firmware).** Each subsystem exposes a chiplet-visible aperture whose base and size are programmable at runtime. Inbound transactions to an aperture are remapped global-to-local before they reach the subsystem.
- **Local-alias views (fixed in RTL).** Internally each subsystem addresses its own resources through a fixed local map: the SMC at `0xC000_0000` (see [Address Map and Aperture](#)), the SEP through its own local map. These are RTL-fixed and not integrator-assigned.

Table 2. Address-map assignment contract

Level	Owner	Assigned / programmed by
Chiplet envelope	Integrator	SoC/chiplet address map + adopter fabric decode
SEP chiplet aperture	SEP firmware	<code>SEP_GLOBAL_BASE_ADDR / SEP_REGION_SIZE</code> (the <code>sep_cpu_ctrl</code> CSRs)
SMC chiplet aperture	SMC firmware	<code>GLOBAL_BASE / REGION_SIZE</code> (the SMC CPU Control CSRs at <code>0xC001_0000</code>)
SEP / SMC local view	Fixed (RTL)	Offsets within each subsystem's local map (SMC local base <code>0xC000_0000</code>)

Programming order and constraints: firmware programs the SEP and SMC aperture CSRs during early boot, **before any cross-subsystem AXI traffic is issued**, and must keep the two apertures non-overlapping and inside the chiplet envelope the integrator assigned. The full runtime crossbar behaviour and the exact CSR plumbing are in [Firmware programming order](#) under [SMU Composition and Internal Wiring](#).

1.1.1. Register collateral

The machine-readable address map ships in the repository tree; integrators consume it from the release checkout or tarball rather than a separate package:

- System tops: `hw/sys/<sys>/regs/<sys>.rdl` (e.g. `hw/sys/sep/regs/sep.rdl`, `hw/sys/smc/regs/smc.rdl`).
- IP blocks: `hw/ip/<ip>/regs/<ip>.rdl`.
- Vended blocks: `vendor/.../overlay/regs/<ip>/regs/<ip>.rdl` or `vendor/.../overlay/rdl/<ip>.rdl`.
- Generated outputs per block: `<block>/regs/gen/{c,sv,svh,py,ral,adoc,html,ipxact}/` — C headers, SystemVerilog packages, Python models, UVM RAL, documentation, and IEEE 1685-2014 IP-XACT component descriptions.

For the architecture-level responsibility split between the integrator, firmware, and the internal SEP/SMC maps, see

1.2. Immutable ROM FW Integration

OCAH contains three immutable ROM FW images: the SMC production ROM FW, the SEP production ROM FW, and the Key Manager production ROM FW. Each image has two integration contracts. The adopter connects and configures the corresponding ROM macro, and the final ROM FW image must be present before its processor leaves reset. Where ROM FW must initialize or access an adopter-owned peripheral outside OCAH, the image may also expose build-time integration hooks.

1.2.1. Weak-symbol override method

An immutable ROM FW image may provide selected vendor-neutral integration hooks as weak symbols. Supply an adopter implementation as a strong definition of the same symbol before the final image is frozen. The strong implementation must program the same adopter-owned IP instantiated by the RTL integration, and its signature, ABI, return values, and documented semantics must match the weak declaration.

This mechanism applies only when the ROM FW needs to interact with a peripheral outside OCAH. It is a build-time image-composition mechanism, not a runtime extension mechanism. Include the adopter implementation in the final ROM FW build so that it replaces the weak definition at link time.

Inspect the final ELF or linker map after linking. For example, `riscv64-unknown-elf-nm` must report each selected implementation as a strong symbol rather than `W`; successful linking alone does not prove that the weak default was replaced.

1.2.2. SMC ROM FW integration

Connect `rom_intf_req_o` and `rom_intf_rsp_i` to the adopter's ROM implementation. In the current four-core configuration, the request provides the ROM clock, enable, and 14-bit 64-bit-word address, and the response returns one 64-bit word. This is a direct memory interface, not AXI-Lite. The 128 KiB ROM FW image must be present before the first boot; its build is under `hw/sys/smc/bootrom/prod/`.

Connect the ROM macro configuration input reached as `rom_cfg_i` and `mem_cfg_i` according to the selected macro's contract. OCAH forwards these 11 bits without assigning technology-specific semantics; the reference `smc_ip_integration.sv` ties them to zero. Set the eFuse shadow value that drives the CPU's `rom_flip_endianness_i` path to match the byte order of the image and ROM macro. Deasserted passes each 64-bit read word unchanged; asserted reverses its eight bytes.

Table 3. SMC ROM FW override inventory

Hook	Weak default	Rationale	Override and integration action
<code>smc_pll_init()</code>	Returns <code>SMC_PLL_SUCCESS</code> without programming PLL hardware.	PLL implementation is technology- and vendor-specific, so OCAH does not include a vendor-specific PLL driver.	Override when <code>BL0_PLLCLK</code> is asserted and the adopter's PLL requires ROM-time configuration. Provide the same <code>smc_pll_result_t smc_pll_init(void)</code> signature in the final ROM FW build. Any result other than <code>SMC_PLL_SUCCESS</code> is fatal: the ROM FW records an interface error and remains in <code>wfi</code> .

Hook	Weak default	Rationale	Override and integration action
<code>i3c_release_reset()</code> , <code>cfg_ps()</code> , <code>init_i3c_ctrl()</code> , and <code>I3C_GetDriverInstance()</code>	Permit the ROM FW to boot without an active I3C implementation; I3C data operations report a hardware error.	OCAH includes an I3C implementation, but the adopter may select a different I3C target and driver for the final system.	When the ROM FW must use an active I3C target, select the open HCI implementation with <code>I3C_CORE=swap</code> , or connect the adopter's I3C target through the SMC external AXI-Lite master (<code>smc_external_req_o</code> and <code>smc_external_resp_i</code>) and link strong implementations that conform to <code>drivers/include/i3c_target_driver.h</code> .

1.2.3. SEP ROM FW integration

The SEP exposes the `sep_boot_rom` memory interface for its production ROM FW image. The detailed ROM-macro configuration, image-loading contract, and adopter override inventory are not yet documented here.

Table 4. SEP ROM FW override inventory

Hook	Status
<i>To be documented</i>	No adopter override requirement is defined here yet.

1.2.4. Key Manager ROM FW integration

The Key Manager exposes the `km_rom_mem` interface through SEP for its production ROM FW image. The detailed ROM-macro configuration, image-loading contract, and adopter override inventory are not yet documented here.

Table 5. Key Manager ROM FW override inventory

Hook	Status
<i>To be documented</i>	No adopter override requirement is defined here yet.

1.3. System Management Controller (SMC) Integration

The System Management Controller (SMC) is a subsystem responsible for power management, clock control, reset sequencing, and peripheral interface coordination. The `smc.sv` module exposes logic-level interfaces; all technology-specific components (PLLs, PVT sensors, eFuses, memory macros, GPIO padrings) are instantiated externally by the adopter. See the [SMU Module Variants](#) section for how the reference wrappers demonstrate this integration.

For detailed architectural documentation about the SMC internal design, refer to [hw/sys/smc/doc/](#). The public integration examples are [hw/top/smc_wrapper.sv](#) and [hw/top/smc_ip_integration.sv](#).

1.3.1. Clock and Reset

The SMC requires three external clock inputs: `clk_smc_i` (primary system clock driving the CPU cluster, fabric, and peripherals), `clk_periph_i` (peripheral clock driving the peripherals), and `clk_ref_i` (reference clock for timing-sensitive operations). All must be generated externally by the adopter's PLL infrastructure; the SMC provides AXI-Lite control outputs for PLL configuration (see [Peripheral Interface Connections](#)). The peripheral clock is used to drive the I3C interface, I2C interface, UART interface, and AVSBus interface.

Hold `rst_cold_ni` asserted (low) while power supplies and clocks stabilize. Assert `powergood_i` once supplies reach nominal, then deassert `rst_cold_ni` (drive high) to start the boot sequence.

The SMC generates AASD (async-assert / sync-deassert) reset outputs: `rst_primary_smc_clk_no` for the main clock domain, `rst_primary_periph_clk_no` for the peripheral clock domain, and `rst_primary_ref_clk_no` for the reference clock domain.

These ensure clean reset release without metastability across the system hierarchy.

Reset domain overrides are delivered through a single typed input, `jtag_reset_ctrl_i` (`smc_pkg::jtag_smc_reset_ctrl_t`), which carries all SMC reset override enables and active-low values. The DTP drives this input through the JTAG IC_RESET mechanism (see [SMC-DTP Internal Interconnection](#)); most integrations leave all override enables deasserted.

1.3.2. Address Map and Aperture

The SMC has a chiplet-visible aperture and an internal local address view. Internally, the SMC uses a fixed local base address of `0xC000_0000` and a fixed local window size of 16 MiB. SMC-local addresses (registers, memories, remapped regions) are defined as offsets from that base. The chiplet-visible aperture is adopter-defined: the chiplet or SoC address map assigns the global base and size that external masters can use to reach SMC through the SMU crossbar.

When integrating the SMC into a chiplet or SoC, the adopter assigns a system base address and reserved size for the SMC window. The system base (e.g. `0x0800_0000` in a chiplet address map) may be different from the SMC-local base `0xC000_0000`. Firmware programs the SMC `GLOBAL_BASE` / `REGION_SIZE` CSRs during early boot to match that assignment. The adopter's fabric or address decoder must route the reserved system range to the SMC block so that offsets are preserved: system base + offset maps to SMC-local `0xC000_0000` + offset.

Table 6. Address Base Mapping

View	Base Address	Description
SMC-local (internal)	<code>0xC000_0000</code>	Local alias address (fixed). Used by SMC CPU, firmware, and fabric.
System (adopter)	Adopter-defined	Base of the chiplet/SoC aperture assigned to SMC.

The SMC fabric uses three registers in the CPU Control block (`0xC001_0000`) to distinguish local from remote addresses:

- `LOCAL_BASE` (read-only, fixed at `0xC000_0000`): The local alias base. Together with `REGION_SIZE`, this is the address space the SMC CPU and internal fabric use to access SMC resources.
- `GLOBAL_BASE` (read-write, defaults to `0x4000_0000`): The system-level base address. Firmware must program this during early boot to match the adopter's system address assignment, enabling the SMC to issue outbound transactions using global system addresses.
- `REGION_SIZE` (read-write, resets to `0x0100_0000` / 16 MiB): The size of both the local alias window and the global aperture. It sizes the SMC input fabric's local/global split as well as SMU-level routing and SEP-to-SMC decode. Firmware must not program it to zero, and must keep `LOCAL_BASE + REGION_SIZE` inside the 32-bit SMC-local address space.

For full details on alias remapping and outbound routing, see the SMC Fabric documentation ([hw/sys/smc/doc/fabric.adoc](#)).

For a detailed breakdown of all components within the SMC local map, refer to the SMC Memory Map documentation ([hw/sys/smc/doc/memmap.adoc](#)).

1.3.3. AXI Interface Configuration

The SMC provides three distinct AXI interface groups serving different access paths. The `sys_axi_in` interface accepts requests from external system masters requiring access to SMC-managed resources. This interface uses a 56-bit address space with 64-bit data width and 6-bit transaction IDs. Connect this interface to your system fabric for normal operational access from application processors or DMA controllers.

The `sep_axi_in` interface provides dedicated access from the Security Processor (SEP) subsystem. This interface shares the same sizing parameters as the system input but operates independently to maintain security domain separation. Use this interface only when integrating alongside the SEP subsystem. The interface may remain unconnected if your integration does not include SEP.

The `output_axi` interface carries filtered and remapped requests from the SMC to external system resources. This interface uses an 8-bit transaction ID field to accommodate merged traffic from multiple input paths. Connect this interface to your system interconnect to allow SMC components access to external memory and peripherals. An AXI VIP or equivalent BFM is a suitable stand-in during integration verification before the real system fabric is available.

The `jtag_axi_in` interface provides debug access through JTAG mechanisms. This interface uses reduced 2-bit transaction IDs and connects to your JTAG debug infrastructure.

1.3.4. GPIO Pad Integration

GPIO pad integration requires careful attention to function assignment and pull requirements. The GPIO table below lists each signal, bump location, function, recommended pull, and integration notes. Default input/output direction is defined by `smc_padrng_pkg::DefaultDirectionMap`, not by the table.

The public `hw/top/smc_wrapper.sv` and `hw/top/smc_ip_integration.sv` examples show the pad connections. For bidirectional pins, implement proper tri-state buffers controlled by the `core2pad_en_o` signals. Connect multiplexed UART, I2C, I3C, and SPI pins to external devices according to the function assignments in the GPIO table.

Configuration straps are captured separately from the GPIO table. Establish them before cold reset deassertion and keep them stable during capture. The generated reset-unit `STRAPS_LO` and `STRAPS_HI` register descriptions define the decoded bit assignments, including `BYPASS_SRAM_REPAIR` at `STRAPS_LO[13]` and `TEST_EN` at `STRAPS_LO[14]`.

1.3.5. Peripheral Interface Connections

The SMC exposes logic-level interfaces to several technology-specific subsystem components that the adopter instantiates externally. The subsections below describe each connection in turn.

Adopter Peripheral Window

The `smc_external_req_o` and `smc_external_resp_i` signals form a 32-bit AXI-Lite interface to the adopter's own peripheral register maps. Any technology- or product-specific block that SMC firmware needs to reach sits behind this window rather than on a dedicated per-peripheral port.

The window is the recommended home for the control/status registers of logic that the SMC must configure or monitor but that the SMC itself does not implement—padrng configuration logic, clock generation, on-die sensors, memory-technology SHIMs, and similar adopter-owned blocks are typical examples. For example, runtime reconfiguration of pin directions, pull settings, and peripheral multiplexing is expected to be reachable here. The SMC decodes the aperture and forwards the transaction; what lives behind it is entirely adopter-defined.

The window is split into a mandatory region, whose contents and placement the SMC memory map fixes and every integration must implement, and a supplementary region for anything else the adopter chooses to expose. For placement and the full register layout, refer to the SMC Memory Map documentation ([hw/sys/smc/doc/memmap.adoc](#)).

eFuse Interface

The eFuse interface consists of two parts:

- `efuse_bank_ctrl` — a 32-bit AXI-Lite interface used to access the SHIM control CSRs (for example, foundry-specific bank timing such as the initialization delay before sensing can begin).
- `efuse_shim_command` — the custom fuse command interface used to issue read, program, and program/read-back operations to the eFuse bank.

Adopter responsibility. The `efuse_shim_command` interface is intentionally generic and technology-neutral. The adopter must implement an Interface SHIM that converts these generic commands into the hardware signals required by their OTP macro. The conversion is driven by the adopter’s own read and program state machines inside the SHIM; OCAH only defines the command/response contract, not the macro-side signaling.

The command and response payloads are exchanged as packed structs:

```
// Controller -> SHIM
typedef struct packed {
    addr_t          address;           // Target address (in bits)
    logic           program_data;      // Data to program (when programming)
    access_length_words_t access_length_words; // Number of WORD_WIDTH words to access
    command_t       command;           // READ / PROGRAM / PROGRAM_READ_BACK
    logic           valid;             // Request valid
} fuse_command_req_t;

// SHIM -> Controller
typedef struct packed {
    data_t data; // Read data (one WORD_WIDTH word)
    logic status; // Operation status (0 = OK, 1 = error)
    logic valid; // Response valid (qualifies 'data' for one cycle)
} fuse_command_resp_t;
```

Fuse sensing protocol. During automatic sensing the controller issues a single read with `access_length_words` set to the entire bank (all `NUM_WORDS` words). The SHIM is expected to stream the bank back one `WORD_WIDTH` word at a time: for each OTP access it drives the read `data` onto `fuse_command_resp_o.data` and asserts `valid` for **exactly one cycle**. The controller consumes one word per cycle that `valid` is high, advancing to the next eFuse word on each assertion.

WARNING

`valid` must be asserted for only a single cycle per returned word. If the SHIM holds `valid` high for two cycles, the controller will sample the same `data` value a second time and count it as the data for the **next** eFuse word, shifting all subsequent words and corrupting the sensed shadow contents. The adopter’s SHIM FSM is responsible for producing a clean one-cycle `valid` strobe per `WORD_WIDTH` access.

The `shadow_regs_o` output provides read-only access to the sensed fuse values. During bring-up the SMC automatically senses the fuse contents and propagates them through the shadow registers; the `fuse_sense_done_o` signal indicates completion of this sensing operation.

1.3.6. Memory Subsystem Integration

The SMC includes an embedded RISC-V CPU cluster with associated memory interfaces. See [Immutable ROM FW Integration](#) for the ROM FW integration contract.

SRAM and Cache Array Integration

Scratch RAM interfaces (`scratch_ram_intf_req_o` and `scratch_ram_intf_rsp_i`) connect to 32 independent RAM banks totaling the scratchpad memory space. Cache memory interfaces for L1 instruction and data caches (`l1_icode_*` and `l1_dcache_*`) connect to tag and data RAM banks as specified in the port table.

The `disable_sram_auto_init_i` input controls automatic memory initialization behavior. When deasserted, the SMC automatically zeroes all SRAM contents during bring-up. The `init_mem_done_o` output indicates completion of this initialization sequence. The `SRAM_AUTO_ZERO_DISABLE` strap (`STRAPS_LO[26]`) can disable this automatic zeroing behavior if your integration requires preserved SRAM contents across resets.

1.3.7. Memory Repair Integration

The SMC implements Built-In Redundancy Analysis (BIRA) memory repair to address manufacturing defects in SRAM arrays. Repair uses redundant memory elements configured with data stored in eFuse memory, replacing defective cells detected during manufacturing test.

Memory repair executes automatically during boot after eFuse sensing completes and before MBIST operations. The process must complete successfully before the CPU cluster can safely access SRAM arrays. Repair is required on every cold reset but can be bypassed during Function Level Reset (FLR) operations for faster recovery.

Control Signals:

- `fuse_sense_done_o`: eFuse controller output indicating fuse sensing completion. External DFT logic monitors this signal to trigger repair operations.
- `skip_mem_repair_o`: Reset unit output indicating repair should be bypassed. It is driven by the isolation request inputs (`isolate_req_pin_sync_smc` or the SMC isolation register), not directly by the strap. When asserted, external repair logic should skip operations and immediately assert completion status.
- `ext_boot_seq_done_i`: External signal indicating the external boot sequence (memory repair and shadow register override) is complete. Held low to pause reset release until the external agent gives the go-ahead; required before the system can continue its boot sequence

Repair Data:

Repair data is stored in the eFuse BIRA shadow register as a 16-kilobit `repair_data` field. The eFuse controller reads this data during fuse sensing and makes it available through shadow registers. The external memory repair controller must read repair data from `shadow_regs_o` and apply it to SRAM arrays according to your memory architecture.

Status Monitoring:

Repair completion and success are reported through `DFX_CTRL_STATUS.STATUS_SMU` at `0xC000_B800`. The `mem_repair_done` bit indicates completion regardless of success, while `mem_repair_success` indicates successful repair. Both bits are set by the external repair controller. The SMC ROM FW monitors these bits during initialization to verify repair completion.

Bypass Configuration:

The `BYPASS_SRAM_REPAIR` configuration strap (`STRAPS_LO[13]`) is captured into the reset-unit strap registers. Keep these three mechanisms distinct:

- No RTL path drives `skip_mem_repair_o` from this strap directly. `skip_mem_repair_o` is driven from the isolation request path.
- Production ROM reads the strap and, when asserted, skips repair-status validation and writes

`MBIST_STATUS_MEM_REPAIR_BYPASSED` to scratchpad register 15 (`SMC_SCRATCH_MBIST_STATUS`, `0xC003_90F8`). It does not cause the external repair controller to skip physical repair.

- External repair logic independently honors `skip_mem_repair_o` to decide whether to bypass physical repair operations.

Use bypass mode only when SRAM arrays are guaranteed defect-free, such as during development or with known-good components.

Integration Requirements:

Provide an external memory repair controller that monitors `fuse_sense_done_o` to initiate repair, respects `skip_mem_repair_o` for bypass control, reads repair data from eFuse shadow registers, applies repair configuration to SRAM arrays, asserts `ext_boot_seq_done_i` and updates DFT Control Status registers with completion status. The controller operates independently of the SMC core and must complete before the CPU cluster accesses SRAM.

For direct access to the eFuse besides from the shim, i.e. from a DFT fuse controller, adopters must ensure the integrated logic will not interfere with accesses from the OCAH's efuse interface.

During FLR operations, the isolation request path asserts `skip_mem_repair_o` to bypass repair for rapid reset sequences. Normal cold reset sequences always execute repair unless isolation is already asserted.

1.3.8. Interrupt Integration

The SMC aggregates interrupts from multiple sources into `cpu_interrupts_o`, which feeds the PLIC of the embedded RISC-V CPU cluster. The interrupt vector is assembled in `hw/sys/smc/rtl/smc_base.sv` and `hw/sys/smc/rtl/smc_peripherals/rtl/smc_peripherals.sv`. The authoritative bit-level map (1-core and 4-core columns, RTL trace, and PLIC source ID note) is maintained in a single place:

See the *SMC CPU interrupt vector map* in the Technical Reference Manual.

The SMC PLIC and CPU interrupt vector are internal. `sync_irq_o` is **not** the interrupt aggregate—it is the software-controlled `SYNC_REG.sync` global synchronisation bit in the reset unit, unrelated to PLIC interrupt delivery. Do not connect an external interrupt controller to `sync_irq_o`.

External interrupt sources visible to the adopter are `ext_interrupts_i` (SMC external interrupt input vector, width `NUM_INT_TO_SMC`), `ext_mailbox_interrupts_o` (outbound mailbox notifications, width `NUM_MAILBOXES`), `gpio_interrupt_o` (GPIO edge/level events, width `NUM_GPIO_WRAPS`), and `uart_interrupt_o` (UART TX/RX events, width `NUM_UART`). For the full PLIC source map see the SMC Technical Reference Manual.

1.3.9. Telemetry and Debug Integration

For system-level debug and telemetry, the SMC implements ATB (Advanced Trace Bus) receiver interfaces. The telemetry interface group (`telemetry_atdata_i`, `telemetry_atid_i`, `telemetry_atvalid_i`, `telemetry_atready_o`) operates in a dedicated `clk_telemetry_i` clock domain with separate `rst_telemetry_ni` reset. Connect these signals to ATB trace sources for debug data collection.

At the SMC boundary, `ext_debug_bus_i` is 512 bits wide; the SMU concatenates the adopter-visible 128-bit input with the SEP's internal 384-bit bus before driving SMC. The typed `xtrigger_ss_i` / `xtrigger_ss_o` ports are the only cross-trigger signals at the SMC boundary. The adopter-visible SMU cross-trigger interfaces are `xtrig_ctm_*`, `xtrig_clk_stop_req_i`, and the CTP groups.

Trace memory interfaces (`trace_mem_req_o` and `trace_mem_resp_i`) connect to trace RAM instances for storing captured

debug data. `smc_ip_integration.sv` shows an example set of RAM instances that can be adapted to your debug requirements.

1.3.10. Bring-Up Sequence

The recommended bring-up sequence begins with asserting `rst_cold_ni` while stabilizing power supplies and clocks. Once all supplies reach nominal voltages, deassert `rst_cold_ni` to initiate the SMC boot sequence. The SMC immediately begins fuse sensing to read configuration values and security credentials.

Monitor the `fuse_sense_done_o` signal to detect completion of fuse sensing. Configuration straps sampled during this phase determine boot modes and operational settings. After fuse sensing completes, the SMC releases internal subsystem resets based on the programmed reset sequencing configuration.

The CPU cluster begins executing the ROM FW after reset release. The ROM FW performs additional initialization including peripheral configuration, memory testing (unless disabled), and system health checks. The `init_mem_done_o` signal indicates completion of memory initialization if automatic SRAM zeroing is enabled.

To support Function Level Reset (FLR), assert `cfg_flr_pf_active_i` to trigger a warm reset sequence. The SMC asserts `isolate_req_o` bits for the affected subsystem indices. The adopter's power controller drives `ss_reset_complete_i` high once the subsystem reset is complete; the SMC synchronizes this into the readable `SS_RESET_COMPLETE` register. Isolation remains asserted until cleared by software: SMC firmware must poll `SS_RESET_COMPLETE` and then write `ISOLATE_REQ_SMC_REG` to clear the FLR isolation latch. Do not assume isolation releases automatically on `ss_reset_complete_i`.

See `hw/top/smu_wrapper.sv` and `hw/top/smc_ip_integration.sv` for reference bring-up wiring examples covering clock, reset, strap, and peripheral connections.

1.3.11. SMC Port Interface

The SMC exposes a comprehensive set of interfaces for system integration. The following table defines all SMC top-level ports including clocks, resets, AXI interfaces, peripheral connections, and control signals. Refer to the detailed SMC documentation for internal signal descriptions and timing requirements.

Table 7. SMC Port Declaration

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
<code>clk_smc_i</code>	logic	Input	1	Primary SMC clock domain. All SMC internal logic operates synchronously to this clock.	PLL output	N/A
<code>clk_ref_i</code>	logic	Input	1	Reference clock domain. Used for timing-sensitive operations and peripheral interfaces.	PLL reference clock	N/A
<code>clk_periph_i</code>	logic	Input	1	Peripheral clock domain.	PLL peripheral clock	N/A
<code>powergood_i</code>	logic	Input	1	Power-good indication. Assert after supply voltages stabilize.	Power management controller	N/A
<code>powergood_stable_o</code>	logic	Output	1	Stabilized power-good output. Debounced version of <code>powergood_i</code> .	Reset unit → External systems	N/A
<code>rst_cold_ni</code>	logic	Input	1	Cold reset input (active low). Asynchronous assertion, synchronous deassertion.	System cold reset	N/A
<code>rst_primary_ref_clk_no</code>	logic	Output	1	Primary reset synchronized to reference clock domain (active low).	To subsystems using ref clock	N/A
<code>rst_primary_smc_clk_no</code>	logic	Output	1	Primary reset synchronized to SMC clock domain (active low).	To subsystems using SMC clock	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
sys_axi_in_req_i	smc_sys_in_56_64_6_12_axi_req_t	Input	-	System AXI input request (56-bit addr, 64-bit data, 6-bit ID, 12-bit user).	External system masters → Input Fabric	N/A, tie to '0 if unused
sys_axi_in_resp_o	smc_sys_in_56_64_6_12_axi_resp_t	Output	-	System AXI input response.	Input Fabric → External system masters	N/A, leave floating if unused
jtag_axi_in_req_i	smc_jtag_56_64_2_12_axi_req_t	Input	-	JTAG debug AXI request (56-bit addr, 64-bit data, 2-bit ID, 12-bit user).	JTAG debug port → Input Fabric	N/A, tie to '0 if unused
jtag_axi_in_resp_o	smc_jtag_56_64_2_12_axi_resp_t	Output	-	JTAG debug AXI response.	Input Fabric → JTAG debug port	N/A, leave floating if unused
axil_smc_otp_jtag_req_i	smc_pkg::smc_axil_32_32_req_t	Input	-	eFuse OTP/JTAG AXI-Lite request.	JTAG → eFuse controller	N/A, tie to '0 if unused
axil_smc_otp_jtag_resp_o	smc_pkg::smc_axil_32_32_resp_t	Output	-	eFuse OTP/JTAG AXI-Lite response.	eFuse controller → JTAG	N/A, leave floating if unused
sep_axi_in_req_i	smc_sep_in_56_64_6_12_axi_req_t	Input	-	SEP AXI request (56-bit addr, 64-bit data, 6-bit ID, 12-bit user).	SEP → Input Fabric	N/A, tie to '0 if unused
sep_axi_in_resp_o	smc_sep_in_56_64_6_12_axi_resp_t	Output	-	SEP AXI response.	Input Fabric → SEP	N/A, leave floating if unused
output_axi_req_o	smc_sys_out_56_64_8_12_axi_req_t	Output	-	Filtered and remapped AXI output request (56-bit addr, 64-bit data, 8-bit ID, 12-bit user).	Output Fabric → External system resources	N/A, leave floating if unused
output_axi_resp_i	smc_sys_out_56_64_8_12_axi_resp_t	Input	-	Filtered and remapped AXI output response.	External system resources → Output Fabric	N/A, tie to '0 if unused
efuse_bank_ctrl_req_o	smc_pkg::smc_axil_32_32_req_t	Output	-	eFuse bank control AXI-Lite request.	eFuse controller → eFuse Shim	N/A
efuse_bank_ctrl_resp_i	smc_pkg::smc_axil_32_32_resp_t	Input	-	eFuse bank control AXI-Lite response.	eFuse Shim → eFuse controller	N/A
efuse_shim_command_req_o	smc_efuse_pkg::fuse_command_req_t	Output	-	eFuse command interface request.	eFuse controller → eFuse Shim	N/A
efuse_shim_command_resp_i	smc_efuse_pkg::fuse_command_resp_t	Input	-	eFuse command interface response.	eFuse Shim → eFuse controller	N/A
shadow_regs_o	smc_efuse_pkg::fuse_map_t	Output	-	eFuse shadow register output.	eFuse controller → External systems	N/A
pll_clk_obs_i	logic	Input	1	PLL clock observation input.	PLL → GPIO padding (muxed)	N/A
pll_clk_obs_en_i	logic	Input	1	PLL clock observation enable.	Control logic → GPIO padding	N/A
pvt_process_clk_obs_i	logic	Input	1	PVT process monitor clock observation input.	PVT → GPIO padding (muxed)	N/A
pvt_process_clk_obs_en_i	logic	Input	1	PVT process monitor clock observation enable.	Control logic → GPIO padding	N/A
cat_therm_i	logic	Input	1	CAT thermal sensor input.	CAT thermal sensor → PVT wrap	N/A, tie to '0 if unused
lsio_interface_select_o	logic	Output	[NUM_GPIO_WRAPS-1:0]	LSIO interface selection for each GPIO wrap.	GPIO padding → GPIO wraps	N/A
pad2core_i	logic	Input	[NUM_GPIO_WRAPS-1:0]	GPIO input data from pad ring to core.	GPIO padding → SMC	N/A
core2pad_o	logic	Output	[NUM_GPIO_WRAPS-1:0]	GPIO output data from core to pad ring.	SMC → GPIO padding	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
pad2core_en_o	logic	Output	[NUM_GPIO_WRAPPS-1:0]	GPIO pad-to-core path enable.	SMC → GPIO padding	N/A
core2pad_en_o	logic	Output	[NUM_GPIO_WRAPPS-1:0]	GPIO core-to-pad path enable.	SMC → GPIO padding	N/A
rst_cool_n_from_pin_i	logic	Input	1	Cool reset input from GPIO pin 61.	GPIO pin 61 → Reset Unit	N/A
clk_telemetry_i	logic	Input	1	Telemetry clock domain. Separate clock for ATB telemetry data collection.	Telemetry clock source	N/A
rst_telemetry_n_i	logic	Input	1	Telemetry reset (active low). Resets telemetry receiver logic.	Telemetry reset source	N/A
telemetry_atdata_i	telemetry_receiver_pkg::telemetry_data_t	Input	[NUM_TELMETRY_RECEIVERS-1:0]	ATB telemetry data input array.	ATB sources → Telemetry receivers	N/A
telemetry_atid_i	telemetry_receiver_pkg::atb_id_t	Input	[NUM_TELMETRY_RECEIVERS-1:0]	ATB telemetry ID input array.	ATB sources → Telemetry receivers	N/A
telemetry_atready_o	logic	Output	[NUM_TELMETRY_RECEIVERS-1:0]	ATB telemetry ready output.	Telemetry receivers → ATB sources	N/A
telemetry_atvalid_i	logic	Input	[NUM_TELMETRY_RECEIVERS-1:0]	ATB telemetry valid input.	ATB sources → Telemetry receivers	N/A
telemetry_atflushvalid_o	logic	Output	[NUM_TELMETRY_RECEIVERS-1:0]	ATB flush valid output.	Telemetry receivers → ATB sources	N/A
telemetry_atflushready_i	logic	Input	[NUM_TELMETRY_RECEIVERS-1:0]	ATB flush ready input.	ATB sources → Telemetry receivers	N/A
cluster_ded_o	logic	Output	1	Cluster double error detection output.	CPU cluster → External systems	N/A
wdt_first_timeout_o	logic	Output	1	First watchdog timeout signal.	CPU cluster WDT → Reset Unit	N/A
wdt_second_timeout_o	logic	Output	1	Second watchdog timeout signal.	Reset Unit → External systems	N/A
smc_region_size_o	logic	Output	[31:0]	SMC address region size output.	SMC Internal Registers → External systems	N/A
ext_interrupts_i	logic	Input	[NUM_EXT_INTERRUPTS-1:0]	External interrupt inputs. Width is config-dependent: 32 (1-core) or 256 (4-core). Maps directly to <code>cpu_interrupts_o[NUM_EXT_INTERRUPTS-1:0]</code> .	External sources → Interrupt aggregator	N/A, tie to '0 if unused
sep_mailbox_interrupts_i	logic	Input	[7:0]	SEP → SMC mailbox interrupt inputs (8 channels). Routed through <code>peripheral_interrupts[7:0] → cpu_interrupts_o[NUM_EXT_INTERRUPTS+7 : NUM_EXT_INTERRUPTS]</code> (1-core: bits 39–32, 4-core: bits 263–256).	SEP → Interrupt aggregator	N/A, tie to '0 if unused
sep_wdt_reset_n_i	logic	Input	1	SEP watchdog reset (active low). Active-low polarity is inverted before routing to <code>peripheral_interrupts[26] → cpu_interrupts_o[NUM_EXT_INTERRUPTS+26]</code> (1-core: bit 58, 4-core: bit 282).	SEP → Interrupt aggregator	N/A
fuse_sense_done_o	logic	Output	1	Fuse sense completion output.	eFuse controller → Memory repair logic	N/A
fuse_reset_n_delayed_o	logic	Output	1	Delayed fuse reset output (after 16 pipeline stages).	eFuse controller → External systems	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
skip_mem_repair_o	logic	Output	1	Skip memory repair output.	Reset Unit → Memory repair logic	N/A
ext_boot_seq_done_i	logic	Input	1	External boot sequence completion input (memory repair and shadow register override done); gates reset release.	External boot/repair controller → eFuse controller	N/A
temp_interrupt_i	logic	Input	1	PVT temperature sensor interrupt. Routes to <code>peripheral_interrupts[27]</code> → <code>cpu_interrupts_o[NUM_EXT_INTERRUPTS+27]</code> (1-core: bit 59, 4-core: bit 283).	PVT wrap → Interrupt aggregator	N/A, tie to '0 if unused
lc_state_i	logic	Input	[2*LC_STATE_WIDTH-1:0] (8 bits)	Lifecycle state input, differentially encoded.	Lifecycle controller → SMC Internal Registers	From SEP; tie to '0 if unused (encoded TEST_DEV)
ras_bank_chip_o	logic	Output	[3:0]	RAS bank chip selection output.	SMC Internal Registers → RAS system	N/A
ras_bank_instance_o	logic	Output	[3:0]	RAS bank instance selection output.	SMC Internal Registers → RAS system	N/A
ext_mailbox_interrupts_o	logic	Output	[NUM_MAILBOXES-1:0]	External mailbox interrupt outputs (32 interrupts, mapped to IDs 156-187).	Mailbox controller → External targets	N/A
cfg_flr_pf_active_i	logic	Input	1	Function Level Reset (FLR) PF active signal.	PCIe/System → Reset Unit	N/A, tie to '0 if unused
isolate_req_o	logic	Output	[31:0]	Isolation request output for 32 subsystems.	Reset Unit → 32 subsystems	N/A
ss_reset_complete_i	logic	Input	[31:0]	Subsystem reset completion status.	32 subsystems → Reset Unit	32'hFFFFFFFF
ss_config_o	logic	Output	[31:0]	Subsystem configuration output.	Reset Unit → 32 subsystems	N/A
ss_reset_ctrl_o	smc_reset_unit_pkg::reset_ctrl_t	Output	[31:0]	Subsystem reset control output array.	Reset Unit → 32 subsystems	N/A
sync_irq_o	logic	Output	1	Software-controlled global synchronisation bit (<code>SYNC_REG.sync</code> in the reset unit). Not an interrupt aggregate output.	Reset unit → External	N/A
gpio_interrupt_o	logic	Output	[NUM_GPIO_WRAPS-1:0]	Raw GPIO interrupt vector from the padding, one bit per GPIO wrap (bonded + unbonded). Software reads GPIO status registers to identify the interrupt source. Already OR-reduced into <code>peripheral_interrupts_o[30:29]</code> internally.	smc_padding → external consumers	N/A, leave floating if unused
uart_interrupt_o	logic	Output	[NUM_UART-1:0]	Raw UART interrupt outputs, one bit per UART instance, in the peripheral clock domain. Already aggregated into <code>peripheral_interrupts_o[21:18]</code> internally via CDC.	uart_wrap → external consumers	N/A, leave floating if unused
rom_intf_req_o	rom_req_t	Output	-	ROM memory interface request. Width depends on CPU cluster configuration	CPU cluster → Boot ROM	N/A
rom_intf_rsp_i	rom_rsp_t	Input	-	ROM memory interface response. Width depends on CPU cluster configuration	Boot ROM → CPU cluster	N/A
scratch_ram_intf_req_o	scratch_ram_req_t	Output	[NUM_SCRATCH_RAM_BANKS2-1:0]	Scratch RAM interface requests. Width depends on CPU cluster configuration	CPU cluster → Scratch RAM banks	N/A
scratch_ram_intf_rsp_i	scratch_ram_rsp_t	Input	[NUM_SCRATCH_RAM_BANKS2-1:0]	Scratch RAM interface responses. Width depends on CPU cluster configuration	Scratch RAM banks → CPU cluster	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
l1_icache_tag_intf_req_o	l1_icache_tag_req_t	Output	[NUM_ICACHE_TAG_BANKS-1:0]	L1 I-cache tag interface requests. Width depends on CPU cluster configuration	CPU cluster → L1 I-cache tag banks	N/A
l1_icache_tag_intf_rsp_i	l1_icache_tag_rsp_t	Input	[NUM_ICACHE_TAG_BANKS-1:0]	L1 I-cache tag interface responses. Width depends on CPU cluster configuration	L1 I-cache tag banks → CPU cluster	N/A
l1_icache_data_intf_req_o	l1_icache_data_req_t	Output	[NUM_ICACHE_DATA_BANKS-1:0]	L1 I-cache data interface requests. Width depends on CPU cluster configuration	CPU cluster → L1 I-cache data banks	N/A
l1_icache_data_intf_rsp_i	l1_icache_data_rsp_t	Input	[NUM_ICACHE_DATA_BANKS-1:0]	L1 I-cache data interface responses. Width depends on CPU cluster configuration	L1 I-cache data banks → CPU cluster	N/A
l1_dcache_tag_intf_req_o	l1_dcache_tag_req_t	Output	[NUM_DCACHE_TAG_BANKS-1:0]	L1 D-cache tag interface requests. Width depends on CPU cluster configuration	CPU cluster → L1 D-cache tag banks	N/A
l1_dcache_tag_intf_rsp_i	l1_dcache_tag_rsp_t	Input	[NUM_DCACHE_TAG_BANKS-1:0]	L1 D-cache tag interface responses. Width depends on CPU cluster configuration	L1 D-cache tag banks → CPU cluster	N/A
l1_dcache_data_intf_req_o	l1_dcache_data_req_t	Output	[NUM_DCACHE_DATA_BANKS-1:0]	L1 D-cache data interface requests. Width depends on CPU cluster configuration	CPU cluster → L1 D-cache data banks	N/A
l1_dcache_data_intf_rsp_i	l1_dcache_data_rsp_t	Input	[NUM_DCACHE_DATA_BANKS-1:0]	L1 D-cache data interface responses. Width depends on CPU cluster configuration	L1 D-cache data banks → CPU cluster	N/A
disable_sram_auto_init_i	logic	Input	1	Disable SRAM automatic initialization.	Control logic → CPU cluster	1'b0
init_mem_done_o	logic	Output	1	Memory initialization completion signal.	CPU cluster → Control logic	N/A
chiplet_is_primary_i	logic	Input	1	Chiplet is primary signal. Used for System Timer OCTS	GPIO Pading Straps (strap 25) → SMC	N/A
timer_count_o	logic	Output	[63:0]	64-bit system timer count output for Open Chiplet Time Synchronization (OCTS).	System Timer OCTS → External systems	N/A
jtag_reset_ctrl_i	smc_pkg::jtag_smc_reset_ctrl_t	Input	-	Typed reset-override control struct carrying all SMC reset override enables and active-low values. Driven by DTP via JTAG IC_RESET. Tie all enables to 0 when not used.	DTP → SMC reset unit	All enables 0
cla_gpio_enable_o	logic	Output	1	CLA GPIO enable output.	DFD controller → CLA	N/A
tdr_dbg_ctrl_clock_stop_en_i	logic	Input	1	DEBUG_CONTROL CLA clock-stop enable.	DTP → DFD	1'b0
tdr_dbg_ctrl_clocks_stopped_by_cla_o	logic	Output	1	CLA halt-clock status under enable; also drives <code>cpu_interrupts_o[NUM_EXT_INTERRUPTS+64]</code> (internal interrupt 0) — 1-core: bit 96, 4-core: bit 320.	DFD → DTP / interrupt	N/A
xtrigger_ss_i	smc_pkg::xtrigger_t	Input	-	Cross-trigger subsystem input. SMU drives from <code>dtp_xtrig_ctm_src_req[1:0]</code> . Internal SMU connection; not an SMU external port.	SMU internal: DTP <code>xtrig_ctm_src_req_o[1:0]</code> → SMC	N/A
xtrigger_ss_o	smc_pkg::xtrigger_t	Output	-	Cross-trigger subsystem output. SMU routes to <code>dtp_xtrig_ctm_dst_req[1:0]</code> . Internal SMU connection; not an SMU external port.	SMU internal: SMC → DTP <code>xtrig_ctm_dst_req_i[1:0]</code>	N/A
trace_mem_req_o	dfd_trace_mem_pkg::SinkMemPktIn_s	Output	[TRC_RAM_INSTANCES-1:0]	Trace memory interface requests.	DFD controller → Trace memory	N/A, leave floating if unused
trace_mem_resp_i	dfd_trace_mem_pkg::SinkMemPktOut_s	Input	[TRC_RAM_INSTANCES-1:0]	Trace memory interface responses.	Trace memory → DFD controller	N/A, tie to '0 if unused

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
<code>ext_debug_bus_i</code>	logic	Input	[511:0]	External debug bus input (512 bits, 16-bit aligned).	External debug sources → Debug bus aggregator	N/A, tie to '0 if unused
<code>test_en_i</code>	logic	Input	1	DFT/scan test-enable. Drives clock-gater test ports and AXI cell test inputs.	Test controller → All modules	1'b0
<code>scan_rst_ni</code>	logic	Input	1	Scan reset (active-low). Bypasses reset synchronizers during DFT.	Test controller → All modules	1'b1

1.3.12. GPIO Pin Requirements

The SMC provides GPIO functionality with multiplexed peripheral functions and test interfaces. The following table lists the signal and bump location, assigned function, recommended pull, and GPIO-specific notes. It does not define default direction or decode configuration straps; use `smc_padring_pkg::DefaultDirectionMap` and the generated reset-unit `STRAPS_LO` and `STRAPS_HI` register descriptions for those.

Table 8. OCAH GPIO Requirements

Index	Signal	Bump Loc	Function	Recommended Pull	GPIO Notes
0	<code>smc_wrapper.GPIO_PAD[0]</code>	SMC	SPI.DATA[0]	PD	
1	<code>smc_wrapper.GPIO_PAD[1]</code>	SMC	SPI.DATA[1]	PD	
2	<code>smc_wrapper.GPIO_PAD[2]</code>	SMC	SPI.DATA[2]	PD	
3	<code>smc_wrapper.GPIO_PAD[3]</code>	SMC	SPI.DATA[3]	PD	
4	<code>smc_wrapper.GPIO_PAD[4]</code>	SMC	SPI.DATA[4]	PD	
5	<code>smc_wrapper.GPIO_PAD[5]</code>	SMC	SPI.DATA[5]	PD	
6	<code>smc_wrapper.GPIO_PAD[6]</code>	SMC	SPI.DATA[6]	PD	
7	<code>smc_wrapper.GPIO_PAD[7]</code>	SMC	SPI.DATA[7]	PD	
8	<code>smc_wrapper.GPIO_PAD[8]</code>	SMC	SPI.CS	PU	
9	<code>smc_wrapper.GPIO_PAD[9]</code>	SMC	SPI.CLK	PD	
10	<code>smc_wrapper.GPIO_PAD[10]</code>	SMC	SPI.DQS	PD	
11	<code>smc_wrapper.GPIO_PAD[11]</code>	SMC	UART[0].RX	PU	
12	<code>smc_wrapper.GPIO_PAD[12]</code>	SMC	UART[0].TX	PU	
13	<code>smc_wrapper.GPIO_PAD[13]</code>	SMC	UART[0].RTS	PD	
14	<code>smc_wrapper.GPIO_PAD[14]</code>	SMC	UART[0].CTS	PD	

Index	Signal	Bump Loc	Function	Recommended Pull	GPIO Notes
15	smc_wrapper.GPIO_PAD[15]	SMC	UART[1].RX	PU	
16	smc_wrapper.GPIO_PAD[16]	SMC	UART[1].TX	PU	
17	smc_wrapper.GPIO_PAD[17]	SMC	UART[1].RTS	PD	
18	smc_wrapper.GPIO_PAD[18]	SMC	UART[1].CTS	PD	
19	smc_wrapper.GPIO_PAD[19]	SMC	UART[2].RX	PU	
20	smc_wrapper.GPIO_PAD[20]	SMC	UART[2].TX	PU	
21	smc_wrapper.GPIO_PAD[21]	SMC	UART[2].RTS	PD	
22	smc_wrapper.GPIO_PAD[22]	SMC	UART[2].CTS	PD	
23	smc_wrapper.GPIO_PAD[23]	SMC	UART[3].RX	PU	
24	smc_wrapper.GPIO_PAD[24]	SMC	UART[3].TX	PU	
25	smc_wrapper.GPIO_PAD[25]	SMC	UART[3].RTS	PD	
26	smc_wrapper.GPIO_PAD[26]	SMC	UART[3].CTS	PD	
27	smc_wrapper.GPIO_PAD[27]	SMC	I3C[0] SCL	PU	
28	smc_wrapper.GPIO_PAD[28]	SMC	I3C[0] SDA	PU	
29	smc_wrapper.GPIO_PAD[29]	SMC	I3C[2] SCL	PU	
30	smc_wrapper.GPIO_PAD[30]	SMC	I3C[2] SDA	PU	
31	smc_wrapper.GPIO_PAD[31]	SMC	I3C[3] SCL	PU	
32	smc_wrapper.GPIO_PAD[32]	SMC	I3C[3] SDA	PU	
33	smc_wrapper.GPIO_PAD[33]	SMC	I3C[4] SCL	PU	
34	smc_wrapper.GPIO_PAD[34]	SMC	I3C[4] SDA	PU	
35	smc_wrapper.GPIO_PAD[35]	SMC	I3C[5] SCL	PU	
36	smc_wrapper.GPIO_PAD[36]	SMC	I3C[5] SDA	PU	
37	smc_wrapper.GPIO_PAD[37]	SMC	I2C[0] SCL	PU	Every I2C can be programmed to act as I2C, SMBus or PMBus
38	smc_wrapper.GPIO_PAD[38]	SMC	I2C[0] SDA	PU	
39	smc_wrapper.GPIO_PAD[39]	SMC	I2C[0] SMB_A	PU	SMBus Alert

Index	Signal	Bump Loc	Function	Recommended Pull	GPIO Notes
40	smc_wrapper.GPIO_PAD[40]	SMC	I2C[0] SMB_D	PU	SMBus Suspend
41	smc_wrapper.GPIO_PAD[41]	SMC	I2C[1] SCL	PU	
42	smc_wrapper.GPIO_PAD[42]	SMC	I2C[1] SDA	PU	
43	smc_wrapper.GPIO_PAD[43]	SMC	I2C[1] SMB_A	PU	SMBus Alert
44	smc_wrapper.GPIO_PAD[44]	SMC	I2C[1] SMB_D	PU	SMBus Suspend
45	smc_wrapper.GPIO_PAD[45]	SMC	I2C[2] SCL	PU	
46	smc_wrapper.GPIO_PAD[46]	SMC	I2C[2] SDA	PU	
47	smc_wrapper.GPIO_PAD[47]	SMC	I2C[2] SMB_A	PU	SMBus Alert
48	smc_wrapper.GPIO_PAD[48]	SMC	I2C[2] SMB_D	PU	SMBus Suspend
49	smc_wrapper.GPIO_PAD[49]	SMC	AVS.CLOCK	PU	
50	smc_wrapper.GPIO_PAD[50]	SMC	AVS.MDATA	PU	
51	smc_wrapper.GPIO_PAD[51]	SMC	AVS.SDATA	PU	
52	smc_wrapper.GPIO_PAD[52]	SMC	Thermal Emergency N	PU	
53	smc_wrapper.GPIO_PAD[53]	SMC	Isolate Request	PD	
54	smc_wrapper.GPIO_PAD[54]	SMC	SPI DQS Loopback	PD	
55	smc_wrapper.GPIO_PAD[55]	SMC	System Timer Sync Load	PD	
56	smc_wrapper.GPIO_PAD[56]	SMC	System Timer Credit Count	PD	
57	smc_wrapper.GPIO_PAD[57]	SMC	Boot Stall	PD	Stall after fuse sensing to override shadow register values. Boot will resume when this GPIO is released. This value can be overridden by the value specified over JTAG using the boot_stall_jtag_ovrd and boot_stall_jtag_val signals from DTP.
58	smc_wrapper.GPIO_PAD[58]	SMC	OCCP Interface	PD	
59	smc_wrapper.GPIO_PAD[59]	SMC	Reserved	PD	
60	smc_wrapper.GPIO_PAD[60]	SMC	Reserved	PD	
61	smc_wrapper.UNBONDED_GPIO[61]	SMC	Cool Reset In	PU	Not packaged out
62	smc_wrapper.UNBONDED_GPIO[62]	SMC	Cool Reset Out	PU	Not packaged out
63	smc_wrapper.UNBONDED_GPIO[63]	SMC	I3C[1] SCL	PU	Not packaged out

Index	Signal	Bump Loc	Function	Recommended Pull	GPIO Notes
64	smc_wrapper.UNBONDED_GPIO[64]	SMC	I3C[1] SDA	PU	Not packaged out
65	smc_wrapper.PAD_JTAG_IN_TCK	?	PTAP TCK		
66	smc_wrapper.PAD_JTAG_IN_TRSTN	?	PTAP TRST		
67	smc_wrapper.PAD_JTAG_IN_TDI	?	PTAP TDI		
68	smc_wrapper.PAD_JTAG_IN_TMS	?	PTAP TMS		
69	smc_wrapper.PAD_JTAG_IN_TDO	?	PTAP TDO		
70	smc_wrapper.PAD_JTAG_OUT_TCK	?	STAP TCK		
71	smc_wrapper.PAD_JTAG_OUT_TRSTN	?	STAP TRST		
72	smc_wrapper.PAD_JTAG_OUT_TDI	?	STAP TDI		
73	smc_wrapper.PAD_JTAG_OUT_TMS	?	STAP TMS		
74	smc_wrapper.PAD_JTAG_OUT_TDO	?	STAP TDO		
75	smc_wrapper.PAD_REFCLK	SMC	BP_REFCLK_VDD_SYS		
76	smc_wrapper.PAD_PRSTN	SMC	BP_RESETN		
77	smc_wrapper.PAD_POWERGOOD	SMC	BP_POWERGOOD		
78		SEP	OTP[0]		
79		SEP	OTP[1]		
80		SEP	OTP[2]		
81		SEP	OTP[3]		
82		SEP	OTP[4]		
83		SEP	OTP[5]		
84		SMC	PVT[0]		
85		SMC	PVT[1]		
86			VDDIO		
87	smc_wrapper.PLL_CLK_OBS_PAD	SMC	PLL clock OBS	PD	PLL clock observation pin
88	smc_wrapper.PVT_CLK_OBS_PAD	SMC	PVT RO	PD	Process sensor RO output
89					

For multi-chiplet systems, the **PRIMARY_CHIPLET** strap (**STRAPS_LO[25]**) determines whether the chiplet operates as primary or secondary. The primary chiplet's System Timer OCTS generates synchronization pulses (**timer_sync_load_o**, **timer_cnt_credit_o**) that secondary chiplets receive to maintain coherent timestamps across the system.

1.4. Debug & Test Ports (DTP) Integration

1.4.1. Configuration

The cross-trigger topology is fixed at elaboration by `ntp_pkg`, a committed source file at `hw/sys/ntp/rtl/ntp_pkg.sv`. `DEFAULT_NUM_CTP`, `DEFAULT_NUM_INT_CT`, and `DEFAULT_NUM_CLK_STOP_REQ` there set the `XTRIG_NUM_CTP`, `XTRIG_NUM_INT_CT`, and `XTRIG_NUM_CLK_STOP_REQ` localparams of `ntp`, so a different port count is applied by editing that package rather than by overriding parameters at the instantiation. The remaining DTP parameters, including `XTRIG_INT_CT_MODE` and every `JTAG_*` enable, are set at the instantiation as shown below.

Include the DTP in your build using Bender target `ntp`, which pulls in the DTP RTL together with the JTAG and cross-trigger IPs it instantiates. The legacy filelist `hw/sys/ntp/rtl/config/ntp.f` covers only the JTAG portion of the hierarchy and is not a complete DTP recipe.

1.4.2. Instantiation

```
ntp #(
    // JTAG Configuration
    .JTAG_BSR_ENABLE      (1'b1),
    .JTAG_TMP_ENABLE      (1'b1),
    .JTAG_SMC_DBG_ENABLE  (1'b1),
    .JTAG_SEP_DBG_ENABLE  (1'b1),
    .JTAG_IDCODE_MFR_ID   (11'h15A),
    .JTAG_IDCODE_PART_NUM (16'h1234),
    .JTAG_IDCODE_SI_REV   (4'h1),
    // Cross Trigger Configuration
    .XTRIG_INT_CT_MODE    (10'b0000000000)
) u_ntp (
    .clk_i                (clk),
    .rst_n_i              (rst_n),
    .pwr_on_rst_ni        (pwr_on_rst_n),
    // JTAG, Boundary Scan, Cross Trigger interfaces...
);
```

1.4.3. Clock and Reset

- **TCK Domain:** JTAG logic operates on external Test Clock
- **System Clock** (`clk_i`): Cross trigger logic and JTAG2AXI bridges
- **System Reset** (`rst_n_i`): Resets cross trigger and AXI; does not affect TAP controller
- **Power-On Reset** (`pwr_on_rst_ni`): Asynchronous reset for JTAG state machine

1.4.4. Constraints

- JTAG interface: 10-50 MHz TCK frequency
- Cross trigger signals synchronized to system clock

- DTP remains active during clock stop for debug access
- External CTP signals require pad ring integration with ESD protection

1.4.5. STAP Scan Chain Topology

The DTP secondary scan path chains multiple STAPs on a single PTAP-managed scan string. The fixed order on this string is: PTAP, I/O STAP (if `JTAG_STAP_IO_ENABLE=1`), SMC debug STAP (if `JTAG_SMC_DBG_ENABLE=1`), SEP debug STAP (if `SEP=1` in the SMU), extra STAP[N-1:0], and return to PTAP.

Both the SMC and SEP debug STAP host interfaces are internal to the SMU and are not exported on the SMU boundary. The SMU wires the SMC debug STAP host outputs to the SMC CPU/system JTAG ports (`smc_cpu_jtag_*`) when `JTAG_SMC_DBG_ENABLE=1`; when disabled, the DTP bypasses this STAP in the secondary scan chain and drives the internal host outputs low. The SEP debug STAP connects to the SEP JTAG debug port when `SEP=1`; when `SEP=0`, the SMU ties `sep_stap_tdo_to_dtp = 1'b0` to bypass the SEP STAP return path.

Each STAP host interface is gated by lifecycle feature controls, synchronized to TCK inside the DTP. `feat_ctrl_i` is enable-polarity (1 = feature enabled), so a STAP host interface is enabled only when all of its required enables are asserted and is disabled when any one of them is deasserted.

STAP	Enabled When	Host Gating Disable Condition
SMC debug STAP	<code>feat_ctrl_i.soc_debug & feat_ctrl_i.ap_debug</code>	<code>!feat_ctrl_i.soc_debug !feat_ctrl_i.ap_debug</code>
SEP debug STAP	<code>feat_ctrl_i.sep_debug & feat_ctrl_i.soc_debug & feat_ctrl_i.ap_debug</code>	<code>!feat_ctrl_i.sep_debug !feat_ctrl_i.soc_debug !feat_ctrl_i.ap_debug</code>

The full set of lifecycle-gated debug paths, including the iJTAG chains and the JTAG2AXI bridges, is tabulated in the DTP JTAG integration section of the TRM.

1.4.6. DTP Port Interface

The DTP exposes interfaces for JTAG TAP control, boundary scan, multiple STAP host ports, iJTAG scan chains, AXI debug bridges, cross trigger networks, and CTP GPIO pads. The following table defines all DTP top-level ports. Refer to the dedicated DTP documentation in [hw/sys/dtp/doc/](#) for internal design details.

Table 9. DTP Port Declaration

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
<code>clk_i</code>	logic	Input	1	System clock for cross trigger logic and JTAG2AXI bridges.	System clock	N/A
<code>rst_n_i</code>	logic	Input	1	System reset (active low). Resets cross trigger and AXI logic; does not affect the TAP controller.	System reset	N/A
<code>pwr_on_rst_ni</code>	logic	Input	1	Power-on reset for JTAG state machine (active low, asynchronous).	Power-on reset	N/A
<code>feat_ctrl_i</code>	<code>sep_efuse_pkg::sep_efuse_map_lc_disable_reg_t</code>	Input	-	Lifecycle feature control vector gating debug interfaces. Enable-polarity (1 = feature enabled).	Lifecycle controller	N/A
<code>jtag_ptap_client_tap_ctrl_i</code>	<code>jtag_tap_ctrl_t</code>	Input	-	Primary JTAG TAP control (TCK, TMS, TRST_N).	External JTAG pads	N/A
<code>jtag_ptap_client_tdi_i</code>	logic	Input	1	Primary JTAG TDI.	External JTAG pads	N/A
<code>jtag_ptap_client_tdo_o</code>	logic	Output	1	Primary JTAG TDO.	External JTAG pads	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
jtag_ptap_client_tdo_oen_o	logic	Output	1	Primary JTAG TDO output enable.	External JTAG pads	N/A
jtag_bsr_host_scan_ctrl_o	jtag_scan_ctrl_t	Output	-	Boundary scan control. Tie scan_in to scan_out if BSR is disabled.	Boundary scan chain	N/A
jtag_bsr_host_scan_in_i	logic	Input	1	Boundary scan data in.	Boundary scan chain	N/A
jtag_bsr_host_scan_out_o	logic	Output	1	Boundary scan data out.	Boundary scan chain	N/A
jtag_stap_io_host_tap_ctrl_o	jtag_tap_ctrl_t	Output	-	I/O STAP TAP control for chiplet-to-chiplet connectivity.	I/O STAP host	Tie to '0 if JTAG_STAP_IO_ENABLE=0
jtag_stap_io_host_tdi_i	logic	Input	1	I/O STAP TDI.	I/O STAP host	Tie to '0 if unused
jtag_stap_io_host_tdo_o	logic	Output	1	I/O STAP TDO.	I/O STAP host	N/A
jtag_stap_io_host_tdo_oen_o	logic	Output	1	I/O STAP TDO output enable.	I/O STAP host	N/A
jtag_stap_smc_host_tap_ctrl_o	jtag_tap_ctrl_t	Output	-	SMC debug STAP TAP control.	SMC JTAG port	Tie to '0 if JTAG_SMC_DBG_ENABLE=0
jtag_stap_smc_host_tdi_i	logic	Input	1	SMC debug STAP TDI (from SMC TDO).	SMC JTAG port	Tie to '0 if unused
jtag_stap_smc_host_tdo_o	logic	Output	1	SMC debug STAP TDO (to SMC TDI).	SMC JTAG port	N/A
jtag_stap_smc_host_tdo_oen_o	logic	Output	1	SMC debug STAP TDO output enable.	SMC JTAG port	N/A
jtag_stap_sep_host_tap_ctrl_o	jtag_tap_ctrl_t	Output	-	SEP debug STAP TAP control.	SEP JTAG port	Tie to '0 if JTAG_SEP_DBG_ENABLE=0
jtag_stap_sep_host_tdi_i	logic	Input	1	SEP debug STAP TDI (from SEP TDO).	SEP JTAG port	Tie to '0 if unused
jtag_stap_sep_host_tdo_o	logic	Output	1	SEP debug STAP TDO (to SEP TDI).	SEP JTAG port	N/A
jtag_stap_sep_host_tdo_oen_o	logic	Output	1	SEP debug STAP TDO output enable.	SEP JTAG port	N/A
jtag_stap_extra_host_tap_ctrl_o	jtag_tap_ctrl_t	Output	[JTAG_NUM_EXTRA_STAP_PORTS-1:0]	Extra STAP TAP control array.	Extra STAP hosts	Tie to '0 if JTAG_NUM_EXTRA_STAPS=0
jtag_stap_extra_host_tdi_i	logic	Input	[JTAG_NUM_EXTRA_STAP_PORTS-1:0]	Extra STAP TDI array.	Extra STAP hosts	Tie to '0 if unused
jtag_stap_extra_host_tdo_o	logic	Output	[JTAG_NUM_EXTRA_STAP_PORTS-1:0]	Extra STAP TDO array.	Extra STAP hosts	N/A
jtag_stap_extra_host_tdo_oen_o	logic	Output	[JTAG_NUM_EXTRA_STAP_PORTS-1:0]	Extra STAP TDO output enable array.	Extra STAP hosts	N/A
jtag_stap_host_scan_ctrl_o	jtag_scan_ctrl_t	Output	-	Extended STAP scan control. Tie scan_in to scan_out if unused.	STAP scan chain	N/A
jtag_stap_host_scan_in_i	logic	Input	1	Extended STAP scan data in.	STAP scan chain	N/A
jtag_stap_host_scan_out_o	logic	Output	1	Extended STAP scan data out.	STAP scan chain	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
jtag_dfd_host_scan_ctrl_o	jtag_scan_ctrl_t	Output	-	DFD iJTAG scan control. Tie scan_in to scan_out if unused.	DFD iJTAG chain	N/A
jtag_dfd_host_scan_in_i	logic	Input	1	DFD iJTAG scan data in.	DFD iJTAG chain	N/A
jtag_dfd_host_scan_out_o	logic	Output	1	DFD iJTAG scan data out.	DFD iJTAG chain	N/A
jtag_dft_secure_host_scan_ctrl_o	jtag_scan_ctrl_t	Output	-	Secure DFT iJTAG scan control. Tie scan_in to scan_out if unused.	Secure DFT iJTAG chain	N/A
jtag_dft_secure_host_scan_in_i	logic	Input	1	Secure DFT iJTAG scan data in.	Secure DFT iJTAG chain	N/A
jtag_dft_secure_host_scan_out_o	logic	Output	1	Secure DFT iJTAG scan data out.	Secure DFT iJTAG chain	N/A
jtag_dft_host_scan_ctrl_o	jtag_scan_ctrl_t	Output	-	Non-secure DFT iJTAG scan control. Tie scan_in to scan_out if unused.	Non-secure DFT iJTAG chain	N/A
jtag_dft_host_scan_in_i	logic	Input	1	Non-secure DFT iJTAG scan data in.	Non-secure DFT iJTAG chain	N/A
jtag_dft_host_scan_out_o	logic	Output	1	Non-secure DFT iJTAG scan data out.	Non-secure DFT iJTAG chain	N/A
axi_smc_dbg_req_o	smc_jtag_axi_req_t	Output	-	SMC fabric debug AXI request. Active when <code>JTAG_SMC_DBG_ENABLE=1</code> .	SMC <code>jtag_axi_in_req_i</code>	Tie to '0 if disabled
axi_smc_dbg_resp_i	smc_jtag_axi_resp_t	Input	-	SMC fabric debug AXI response.	SMC <code>jtag_axi_in_resp_o</code>	Tie to '0 if disabled
axil_smc_otp_jtag_req_o	smc_otp_axil_req_t	Output	-	SMC OTP debug AXI-Lite request.	SMC <code>axil_smc_otp_jtag_req_i</code>	Tie to '0 if disabled
axil_smc_otp_jtag_resp_i	smc_otp_axil_resp_t	Input	-	SMC OTP debug AXI-Lite response.	SMC <code>axil_smc_otp_jtag_resp_o</code>	Tie to '0 if disabled
axil_sep_otp_jtag_req_o	sep_otp_axil_req_t	Output	-	SEP OTP debug AXI-Lite request.	SEP <code>axil_sep_otp_jtag_req_i</code>	Tie to '0 if disabled
axil_sep_otp_jtag_resp_i	sep_otp_axil_resp_t	Input	-	SEP OTP debug AXI-Lite response.	SEP <code>axil_sep_otp_jtag_resp_o</code>	Tie to '0 if disabled
stop_clks_o	logic	Output	1	Clock stop output from JTAG_DEBUG_CONTROL or CLA clock-stop requests via CTN.	PLL clock gates	N/A
cla_clock_stop_en_o	logic	Output	1	CLA clock stop enable from DEBUG_CONTROL register.	Clock control logic	N/A
jtag_boot_stall_ovrd_o	logic	Output	1	JTAG boot stall override enable.	Boot control	N/A
jtag_boot_stall_o	logic	Output	1	JTAG boot stall value when override is asserted.	Boot control	N/A
jtag_ic_reset_smc_o	ic_reset_smc_t	Output	<code>\$bits(ic_reset_smc_t)</code>	SMC IC_RESET override slice. Struct contains <code>.ovrd</code> (override enables) and <code>.val</code> (active-low override values), each sized to match the corresponding SMC reset control struct. Tied to '0 when <code>JTAG_IC_RESET_SMC_ENABLE=0</code> .	SMC <code>jtag_smc_reset_ctrl_i</code>	N/A
jtag_ic_reset_sep_o	ic_reset_sep_t	Output	<code>\$bits(ic_reset_sep_t)</code>	SEP IC_RESET override slice. Struct contains <code>.ovrd</code> (override enables) and <code>.val</code> (active-low override values), each sized to match the corresponding SEP reset control struct. Tied to '0 when <code>JTAG_IC_RESET_SEP_ENABLE=0</code> .	SEP <code>jtag_sep_reset_ctrl_i</code>	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
jtag_ic_reset_ext_o	ic_reset_ext_t	Output	\$bits(ic_reset_ext_t)	External IC_RESET override slice. Struct contains <code>.ovrd</code> (override enables) and <code>.val</code> (active-low override values), each sized per integrator-defined type. Tied to '0 when <code>JTAG_IC_RESET_EXT_ENABLE=0</code> .	Integrator-defined reset overrides for logic outside the SMU	N/A
jtag_ptap_state_o	tap_state_e	Output	-	Current PTAP TAP state.	Debug infrastructure	N/A
jtag_ptap_inst_decoded_o	jtag_instruction_decoded_e	Output	-	Current decoded PTAP instruction.	Debug infrastructure	N/A
axil_xtrig_req_i	xtrig_axil_req_t	Input	-	Cross trigger CSR AXI-Lite request.	SMC <code>axil_dtp_csr_req_o</code>	N/A
axil_xtrig_resp_o	xtrig_axil_resp_t	Output	-	Cross trigger CSR AXI-Lite response.	SMC <code>axil_dtp_csr_resp_i</code>	N/A
xtrig_ctm_src_req_o	logic	Output	[XTRIG_NUM_IN_T_CT-1:0]	Cross trigger source requests from CTM to destination sinks.	Cross trigger network	N/A
xtrig_ctm_src_ack_i	logic	Input	[XTRIG_NUM_IN_T_CT-1:0]	Cross trigger source acknowledges. Unused in pulse-sync mode (mode bit = 0).	Cross trigger network	Tie to '0 in pulse-sync mode
xtrig_ctm_dst_req_i	logic	Input	[XTRIG_NUM_IN_T_CT-1:0]	Cross trigger destination requests from source generators.	Cross trigger network	Tie to '0 if unused
xtrig_ctm_dst_ack_o	logic	Output	[XTRIG_NUM_IN_T_CT-1:0]	Cross trigger destination acknowledges. Unused in pulse-sync mode (mode bit = 0).	Cross trigger network	N/A
xtrig_clk_stop_req_i	logic	Input	[XTRIG_NUM_CLK_STOP_REQ-1:0]	Clock stop requests from CLAs or other devices in the chiplet.	Clock stop requesters	Tie to '0 if unused
xtrig_ctp_req_out_dout_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP request-out data output (to GPIO pad ring).	GPIO pad ring	N/A
xtrig_ctp_req_out_dout_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP request-out data output enable.	GPIO pad ring	N/A
xtrig_ctp_req_out_din_i	logic	Input	[XTRIG_NUM_CTP-1:0]	CTP request-out data input (from GPIO pad ring).	GPIO pad ring	Tie to '0 if unused
xtrig_ctp_req_out_din_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP request-out data input enable.	GPIO pad ring	N/A
xtrig_ctp_req_in_dout_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP request-in data output.	GPIO pad ring	N/A
xtrig_ctp_req_in_dout_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP request-in data output enable.	GPIO pad ring	N/A
xtrig_ctp_req_in_din_i	logic	Input	[XTRIG_NUM_CTP-1:0]	CTP request-in data input.	GPIO pad ring	Tie to '0 if unused
xtrig_ctp_req_in_din_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP request-in data input enable.	GPIO pad ring	N/A
xtrig_ctp_ack_in_dout_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP ack-in data output.	GPIO pad ring	N/A
xtrig_ctp_ack_in_dout_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP ack-in data output enable.	GPIO pad ring	N/A
xtrig_ctp_ack_in_din_i	logic	Input	[XTRIG_NUM_CTP-1:0]	CTP ack-in data input.	GPIO pad ring	Tie to '0 if unused
xtrig_ctp_ack_in_din_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP ack-in data input enable.	GPIO pad ring	N/A
xtrig_ctp_ack_out_dout_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP ack-out data output.	GPIO pad ring	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
<code>xtrig_ctp_ack_out_dout_en_o</code>	logic	Output	<code>[XTRIG_NUM_CTP-1:0]</code>	CTP ack-out data output enable.	GPIO pad ring	N/A
<code>xtrig_ctp_ack_out_din_i</code>	logic	Input	<code>[XTRIG_NUM_CTP-1:0]</code>	CTP ack-out data input.	GPIO pad ring	Tie to '0 if unused
<code>xtrig_ctp_ack_out_din_en_o</code>	logic	Output	<code>[XTRIG_NUM_CTP-1:0]</code>	CTP ack-out data input enable.	GPIO pad ring	N/A

1.5. Security Processor (SEP) Integration

The Security Processor (SEP) is the Root of Trust for the chiplet, providing a physically isolated execution environment for security-critical services such as secure boot, lifecycle control, device attestation, cryptographic acceleration, and key management. This section provides practical integration procedures for adopters integrating the SEP as part of the SMU or as a standalone out-of-context IP.

For detailed architectural documentation about the SEP internal design, refer to the dedicated SEP documentation located in `hw/sys/sep/doc/`. The SEP hardware overview and interconnect fabric are described in `hw/sys/sep/doc/overview.adoc` and `hw/sys/sep/doc/fabric.adoc` respectively.

1.5.1. Module Overview

The `sep.sv` module provides the complete SEP functionality with logic-level interfaces. Technology-specific components—SRAM, ROM, and eFuse shims—are instantiated by the adopter outside SEP, following the same vendor-neutral philosophy used for the SMC.

Separately, SEP exposes interfaces for supplementary peripherals an adopter may add on rather than substitute: an external TRNG and an additional SPI controller. These are not technology-specific—SEP contains its own entropy source and SPI host—so the interfaces exist to supplement what SEP already provides, and may be left unused.

When integrated inside the SMU, the SEP's external interfaces are passed through the SMU boundary with a `sep_` prefix.

1.5.2. Clock and Reset Requirements

The SEP requires two clock inputs. Connect `clk_i` to the primary system clock driving all SEP internal logic. Connect `clk_wdt_i` to a 200 kHz clock source for the watchdog timer; this clock must be independent of the system clock so that the WDT can detect system clock failures.

The primary reset `rst_ni` follows the standard active-low, asynchronous assertion / synchronous deassertion convention. The `dbg_rstb_i` input provides a separate reset for the EL2 CPU debug module. The `wdt_rst_ni` input receives the watchdog reset generated by the SMC reset unit. The SEP outputs `wdt_timer_rst_req_o` when the WDT triggers a reset request.

1.5.3. AXI Interfaces

The SEP connects to the system through three AXI paths. The `smn_outbound_axi` master interface allows SEP firmware to access system resources outside the SEP domain, such as SMC registers or external memory. The `smn_inbound_axi` slave interface accepts requests from system masters requiring access to SEP-managed resources. The `sep_ext_to_smc_axi` output provides a dedicated AXI path for SEP-to-SMC transactions, connecting to the SMC's `sep_axi_in` port.

An additional AXI extension port (`axi_extension_axi`) enables adopters to attach custom peripherals to the SEP

interconnect fabric without modifying the SEP core. When unused, tie the response input to return DECERR.

1.5.4. Memory Interfaces

All SEP memory macros are instantiated externally and connected through dedicated request/response interfaces. The `sep_cpu_tcm` interface provides tightly-coupled memory (ICCM and DCCM) for the VeeR EL2 CPU. The `sep_sram` interface connects scratch SRAM used for firmware data. The `sep_boot_rom` interface provides the BL0 ROM containing the first-stage bootloader.

The crypto subsystem requires additional memory interfaces. `sep_crypto_pka_imem_sram` and `sep_crypto_pka_dmem_sram` connect to the OTBN public key accelerator instruction and data memories respectively. `km_rom_mem` and `km_sram_mem` connect to the Key Manager's dedicated ROM and SRAM macros.

`abr_mem` connects the Adams Bridge working SRAM. On an idle cycle (`re` low), ABR expects the channel's read data to return 0 the following cycle. `SEP_ABR_SRAM_LATENCY` must equal the macros' native registered-read latency (default 1). Word-write channels are full-word `we` / `re` / address / data. `sig_z` and `pk` carry one-bit-per-byte `wstrobe`; map those to the macros' byte-write enables. `SEP_ABR_MASKING_EN` (default 1) selects 2-share DOM: thirteen 1R1W macros versus nine, plus a second NTT. This changes ABR internal behavior (sequencing, SCA, memory topology), not merely throughput. Give the same value to SEP and to the SRAM instances. When it is 0, do not instantiate the four masked coefficient banks; their response fields read as 0.

1.5.5. Cryptographic Subsystem Connections

The SEP includes an internal entropy source that requires a ring-oscillator sample clock connected to `entropy_rosc_sample_clk_i`. This clock must be asynchronous to `clk_i` and typically runs between 100 and 400 MHz.

For adopters providing an external TRNG, the `ext_trng_axil` AXI-Lite interface connects to the external TRNG's control registers, while the `ext_trng_axis` AXI-Stream interfaces carry entropy data. The `ext_trng_irq_i` input is routed to the PIC; `ext_trng_alarm_i` is reserved for RAS.

1.5.6. eFuse and Lifecycle

The SEP accesses its own eFuse bank through the `efuse_bank_ctrl` AXI-Lite interface and the `efuse_shim_command` custom interface, mirroring the pattern used by the SMC. The SEP must wait for the SMC fuse sense to complete (`smc_fuse_sense_done_i`) before initiating its own fuse operations, and it signals completion through `sep_fuse_sense_done_o`.

The lifecycle controller exports `lc_state_o` (differentially encoded), `lc_sigint_err_o` (signal integrity error), `security_disable_o`, `secure_tm_o`, and `dbg_disable_o` (the typed debug-disable struct that the SMU routes to the DTP). The feature-control vector (`feat_ctrl`) is separately produced by the lifecycle controller and consumed internally within `sep.sv`; it is not an external port. `lcc_demote_state_1_o` and `lcc_demote_state_2_o` are exported at the SMU boundary; at present the SMU does not connect them to the SMC instance.

An OTP debug AXI-Lite port (`axil_sep_otp_jtag`) connects to the DTP for JTAG access to SEP fuses. When SEP is disabled (`SEP=0` at the SMU level), the SMU instantiates an AXI-Lite error slave on this path.

1.5.7. SPI Host Interface

The SEP provides a SPI host controller through the `sep_io_spi` interface, supporting single and quad modes. The controller's own interrupt leaves the module as the `irq` field of `sep_io_spi_req_o`, so no separate interrupt output is needed.

The `spi_irq_i` input is a separate SPI interrupt request routed to the SEP PIC. Adopters that replace or supplement the internal SPI host with an external SPI controller connect that controller's interrupt here; adopters that use only the internal controller may loop back the `irq` field of `sep_io_spi_req_o` or, if a mux over multiple SPI sources is needed, drive the selected interrupt from their own integration logic. Tie the input low when unused.

1.5.8. Bring-Up Considerations

The SEP boot sequence begins after the SMC completes fuse sensing and releases the SEP reset. The SEP BL0 ROM FW verifies the BL1 software image using non-volatile public keys before transferring control. Feature controls derived from secure boot authentication and lifecycle state govern what operations are permitted after boot.

The `sep_straps_i` input provides configuration straps for SEP-specific settings. The `jtag_id` input must be tied to an appropriate constant in the top-level integration. A `rst_vec_i` port exists on `tb_top.sv` for DV reset vector injection; it is not present on `sep.sv` itself and requires no integration tie-off.

1.5.9. SEP Port Interface

The SEP exposes interfaces for clocks, resets, JTAG debug, CPU control, memory macros, AXI bus connections, cryptographic subsystem, eFuse, lifecycle, SPI host, and mailbox. The following table defines all SEP top-level ports.

Table 10. SEP Port Declaration

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
<code>clk_i</code>	logic	Input	1	Primary SEP clock. All SEP internal logic operates synchronously to this clock.	System clock	N/A
<code>clk_wdt_i</code>	logic	Input	1	200 kHz clock for watchdog timer.	Low-frequency clock source	N/A
<code>rst_ni</code>	logic	Input	1	Primary reset (active low). Asynchronous assertion, synchronous deassertion.	System reset	N/A
<code>dbg_rstb_i</code>	logic	Input	1	EL2 debugger reset (active low).	JTAG debug infrastructure	N/A
<code>wdt_rst_ni</code>	logic	Input	1	Watchdog timer reset from SMC (active low).	SMC reset unit	N/A
<code>wdt_timer_rst_req_o</code>	logic	Output	1	WDT reset request output. Triggers memory interface reset in <code>sep_wrapper</code> .	SEP wrapper	N/A
<code>jtag_tck</code>	logic	Input	1	JTAG test clock.	JTAG pads (via DTP STAP)	N/A
<code>jtag_tms</code>	logic	Input	1	JTAG test mode select.	JTAG pads (via DTP STAP)	N/A
<code>jtag_tdi</code>	logic	Input	1	JTAG test data in.	JTAG pads (via DTP STAP)	N/A
<code>jtag_trst_n</code>	logic	Input	1	JTAG test reset (active low).	JTAG pads (via DTP STAP)	N/A
<code>jtag_tdo</code>	logic	Output	1	JTAG test data out.	JTAG pads (via DTP STAP)	N/A
<code>jtag_tdoEn</code>	logic	Output	1	JTAG test data out enable.	JTAG pads (via DTP STAP)	N/A
<code>axil_sep_otp_jtag_req_i</code>	<code>sep_efuse_pkg::efuse_axil_req_t</code>	Input	-	OTP debug AXI-Lite request from JTAG.	DTP <code>axil_sep_otp_jtag_req_o</code>	Tie to '0 if unused
<code>axil_sep_otp_jtag_resp_o</code>	<code>sep_efuse_pkg::efuse_axil_resp_t</code>	Output	-	OTP debug AXI-Lite response to JTAG.	DTP <code>axil_sep_otp_jtag_resp_i</code>	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
mpc_debug_halt_req	logic	Input	1	Async halt request from MPC.	Debug controller	Tie to '0 if unused
mpc_debug_run_req	logic	Input	1	Async run request from MPC.	Debug controller	Tie to '0 if unused
mpc_reset_run_req	logic	Input	1	Run/halt selection after reset.	Debug controller	1'b1
i_cpu_halt_req	logic	Input	1	Async halt request to CPU.	Debug controller	Tie to '0 if unused
i_cpu_run_req	logic	Input	1	Async restart request to CPU.	Debug controller	Tie to '0 if unused
test_en_i	logic	Input	1	DFT/scan test-enable. Drives clock-gater test ports and AXI cell test inputs.	Test controller	1'b0
scan_rst_ni	logic	Input	1	Scan reset (active-low). Bypasses reset synchronizers during DFT.	Test controller	1'b1
ext_boot_seq_done_i	logic	Input	1	External boot sequence completion (memory repair and shadow register override done); gates reset release.	External boot/repair controller	1'b1
dmi_core_enable	logic	Input	1	Enables DMI accesses to the core debug module (addresses 0x00-0x4F). Tied high at SMU; TAP access is already lifecycle-gated.	Debug infrastructure	1'b1
dmi_uncore_enable	logic	Input	1	Enables DMI accesses to an external uncore aperture (addresses 0x50+). Unused: the core debug module AXI master already reaches the SEP fabric.	Debug infrastructure	Tie to '0 if unused
dmi_uncore_en	logic	Output	1	DMI uncore access enable.	Debug infrastructure	N/A
dmi_uncore_wr_en	logic	Output	1	DMI uncore write enable.	Debug infrastructure	N/A
dmi_uncore_addr	logic	Output	[6:0]	DMI uncore address.	Debug infrastructure	N/A
dmi_uncore_wdata	logic	Output	[31:0]	DMI uncore write data.	Debug infrastructure	N/A
dmi_uncore_rdata	logic	Input	[31:0]	DMI uncore read data.	Debug infrastructure	N/A
dmi_active	logic	Output	1	DMI active indicator.	Debug infrastructure	N/A
sep_cpu_trace	sep_cpu_trace_t	Output	-	CPU trace output for debug.	Trace infrastructure	N/A
lockstep_ctrl_i	sep_pkg::sep_lockstep_ctrl_t	Input	-	VeeR lockstep controls: disable corruption detection, and error-injection enable. Inert unless the core is built with RV_LOCKSTEP_ENABLE .	Chip-level RAS control	Tie to '0 if unused
lockstep_status_o	sep_pkg::sep_lockstep_status_t	Output	-	VeeR lockstep status: corruption detected. Reads 0 unless the core is built with RV_LOCKSTEP_ENABLE .	Chip-level RAS aggregation	N/A
jtag_id	logic	Input	[31:1]	JTAG ID value.	Configuration tie-off	N/A
timer_int	logic	Input	1	Timer interrupt.	Internal timer	Tie to '0 if unused
soft_int	logic	Input	1	Software interrupt.	Software IRQ source	Tie to '0 if unused
extintsrc_req	logic	Input	[NUM_EXTERNAL_IRQS-1:0]	External interrupt sources for PIC.	External IRQ sources	Tie to '0 if unused

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
sep_cpu_tcm_req_o	sep_cpu_tcm_req_t	Output	-	CPU tightly-coupled memory (TCM) request (ICCM + DCCM).	SRAM macros	N/A
sep_cpu_tcm_rsp_i	sep_cpu_tcm_rsp_t	Input	-	CPU TCM response.	SRAM macros	N/A
sep_sram_req	sep_sram_req_t	Output	-	Scratch SRAM request.	SRAM macros	N/A
sep_sram_rsp	sep_sram_rsp_t	Input	-	Scratch SRAM response.	SRAM macros	N/A
sep_boot_rom_req	sep_sram_req_t	Output	-	Boot ROM (BL0) request.	ROM macro	N/A
sep_boot_rom_rsp	sep_sram_rsp_t	Input	-	Boot ROM (BL0) response.	ROM macro	N/A
smn_outbound_axi_req_o	sep_system_peripherals_outbound_axi_req_t	Output	-	SMN outbound AXI request (system bus master).	SMN fabric	N/A
smn_outbound_axi_resp_i	sep_system_peripherals_outbound_axi_resp_t	Input	-	SMN outbound AXI response.	SMN fabric	N/A
smn_inbound_axi_req_i	sep_system_peripherals_internal_axi_req_t	Input	-	SMN inbound AXI request (system bus slave).	SMN fabric	Tie to '0 if unused
smn_inbound_axi_resp_o	sep_system_peripherals_internal_axi_resp_t	Output	-	SMN inbound AXI response.	SMN fabric	N/A
sep_ext_to_smc_axi_req_o	sep_system_peripherals_internal_axi_req_t	Output	-	SEP-to-SMC AXI request. Dedicated path for SEP to access SMC resources.	SMC sep_axi_in_req_i	N/A
sep_ext_to_smc_axi_resp_i	sep_system_peripherals_internal_axi_resp_t	Input	-	SEP-to-SMC AXI response.	SMC sep_axi_in_resp_o	N/A
entropy_rosc_sample_clk_i	logic	Input	1	Ring-oscillator sample clock for entropy source (async vs clk_i; ~100-400 MHz typical).	Ring oscillator	N/A
sep_crypto_pka_imem_sram_req	sep_crypto_pka_imem_sram_req_t	Output	-	OTBN PKA instruction memory SRAM request.	SRAM macro	N/A
sep_crypto_pka_imem_sram_rsp	sep_crypto_pka_imem_sram_rsp_t	Input	-	OTBN PKA instruction memory SRAM response.	SRAM macro	N/A
sep_crypto_pka_dmem_sram_req	sep_crypto_pka_dmem_sram_req_t	Output	-	OTBN PKA data memory SRAM request.	SRAM macro	N/A
sep_crypto_pka_dmem_sram_rsp	sep_crypto_pka_dmem_sram_rsp_t	Input	-	OTBN PKA data memory SRAM response.	SRAM macro	N/A
ext_trng_axil_req_o	sep_32_32_axil_req_t	Output	-	External TRNG AXI-Lite passthrough request (to sep_ip_integration).	External TRNG shim	N/A
ext_trng_axil_resp_i	sep_32_32_axil_resp_t	Input	-	External TRNG AXI-Lite passthrough response.	External TRNG shim	N/A
ext_trng_axis_req_i	ext_trng_axis_req_t	Input	[EXT_TRNG_NUM_AXIS-1:0]	External TRNG AXI-Stream request array (from sep_ip_integration).	External TRNG shim	N/A
ext_trng_axis_rsp_o	ext_trng_axis_rsp_t	Output	[EXT_TRNG_NUM_AXIS-1:0]	External TRNG AXI-Stream response array.	External TRNG shim	N/A
ext_trng_irq_i	logic	Input	1	External TRNG interrupt (routed to PIC).	External TRNG shim	Tie to '0 if unused
ext_trng_alarm_i	logic	Input	1	External TRNG alarm (reserved for RAS).	External TRNG shim	Tie to '0 if unused
km_rom_mem_req_o	km_intf_pkg::km_rom_mem_req_t	Output	-	Key Manager ROM memory request.	ROM macro	N/A
km_rom_mem_rsp_i	km_intf_pkg::km_rom_mem_rsp_t	Input	-	Key Manager ROM memory response.	ROM macro	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
km_sram_mem_req_o	km_intf_pkg::km_sram_mem_req_t	Output	-	Key Manager SRAM memory request.	SRAM macro	N/A
km_sram_mem_rsp_i	km_intf_pkg::km_sram_mem_rsp_t	Input	-	Key Manager SRAM memory response.	SRAM macro	N/A
efuse_bank_ctrl_req_o	sep_efuse_pkg::efuse_axil_req_t	Output	-	eFuse bank control AXI-Lite request.	eFuse shim	N/A
efuse_bank_ctrl_resp_i	sep_efuse_pkg::efuse_axil_resp_t	Input	-	eFuse bank control AXI-Lite response.	eFuse shim	N/A
efuse_shim_command_req_o	sep_efuse_pkg::fuse_command_req_t	Output	-	eFuse command interface request.	eFuse shim	N/A
efuse_shim_command_resp_i	sep_efuse_pkg::fuse_command_resp_t	Input	-	eFuse command interface response.	eFuse shim	N/A
lcc_demote_state_1_o	logic	Output	[1:0]	Lifecycle demote state output 1. Exported at SMU boundary; not currently connected to SMC.	SMU external port	N/A
lcc_demote_state_2_o	logic	Output	[1:0]	Lifecycle demote state output 2. Exported at SMU boundary; not currently connected to SMC.	SMU external port	N/A
sep_io_spi_req_o	sep_io_spi_req_t	Output	-	SPI host controller request.	SPI pad interface	N/A
sep_io_spi_rsp_i	sep_io_spi_rsp_t	Input	-	SPI host controller response.	SPI pad interface	N/A
spi_irq_i	logic	Input	1	SPI interrupt request into the SEP PIC, driven by whichever SPI controller the integration selects.	SPI controller	Tie to '0 if unused
lc_state_o	logic	Output	[2*LC_STATE_BIT_WIDTH-1:0]	Lifecycle state output (differentially encoded). Routed to SMC and the SMU external output; debug-disable control to DTP is via <code>dbg_disable_o</code> .	SMC, SMU external	N/A
dbg_disable_o	sep_lifecycle_ctrl_pkg::dbg_disable_t	Output	-	Typed debug-disable struct routed by SMU to DTP debug-disable control.	SMU → DTP	N/A
lc_sigint_err_o	logic	Output	1	Lifecycle signal integrity error.	SMU	N/A
security_disable_o	logic	Output	1	Security disable output.	SMU	N/A
secure_tm_o	logic	Output	1	Secure test mode output.	SMU	N/A
smc_mailbox_interrupt_o	logic	Output	[NUM_MAILBOXES-1:0]	Mailbox interrupt outputs to SMC.	SMC interrupt aggregator	N/A
smc_fuse_sense_done_i	logic	Input	1	SMC fuse sense done indication.	SMC <code>fuse_sense_done_o</code>	N/A
sep_fuse_sense_done_o	logic	Output	1	SEP fuse sense done output.	System integration	N/A
sep_straps_i	sep_pkg::sep_straps_t	Input	-	SEP configuration straps.	GPIO padding / tie-off	N/A
axi_extension_axi_req_o	sep_32_64_6_12_axi_req_t	Output	-	AXI extension port request for adopter peripheral expansion.	Adopter peripherals	N/A
axi_extension_axi_resp_i	sep_32_64_6_12_axi_resp_t	Input	-	AXI extension port response.	Adopter peripherals	Tie to DECERR if unused
smc_global_base_addr_i	logic	Input	[55:0]	SMC global base address for outbound address translation.	System configuration	N/A
smc_region_size_i	logic	Input	[55:0]	SMC region size for outbound address translation.	SMC <code>smc_region_size_o</code>	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
km_unrecoverable_err_o	logic	Output	1	Key Manager unrecoverable error.	RAS / error handling	N/A
km_recoverable_err_o	logic	Output	1	Key Manager recoverable error.	RAS / error handling	N/A
ext_debug_bus_o	logic	Output	[511:0]	External debug observation bus.	Debug infrastructure	N/A

1.6. SMU Composition and Internal Wiring

The System Management Unit (SMU) combines the SMC, DTP, and optionally the SEP into a unified subsystem. The interface descriptions in the preceding sections apply directly when integrating the SMU, since the SMU inherits all three subsystems' ports. This section covers the module variants available to adopters, how the SMU wires its subsystems together internally, and what additional configuration it exposes.

1.6.1. Module Variants and IP Integration

OCAH core modules (`smc.sv`, `sep.sv`, `dtp.sv`, and the composite `smu.sv`) expose logic-level interfaces only. All technology-specific components — PLLs, PVT sensors, eFuse shims, memory macros, GPIO padrings, SPI controllers, and external TRNG — are instantiated outside the core modules by the adopter. This separation keeps the OCAH deliverable technology-neutral.

The repository provides reference wrappers that demonstrate how to bridge the gap between OCAH core interfaces and adopter-specific IP:

- `smu_wrapper.sv` wraps `smu.sv` with pad-level signal handling for clocks, resets, GPIO, eFuse, and JTAG interfaces. It internally instantiates `smc_ip_integration.sv` for SMC-related vendor IP and (when `SEP=1`) `sep_ip_integration.sv` for SEP-related vendor IP.
- `smc_wrapper.sv` provides an equivalent reference wrapping `smc.sv` alone, useful when integrating the SMC without the DTP/SEP subsystems.

Neither the wrappers nor the IP integration modules are part of the OCAH deliverable; they exist as integration guides and for Tenstorrent internal usage. Adopters must develop their own equivalents following the interface patterns demonstrated in these modules. The interface contracts at the module boundaries ensure technology substitution without modifying the verified OCAH logic.

The figure below illustrates the relationship between the OCAH core modules, the wrapper layer, and the IP integration modules.

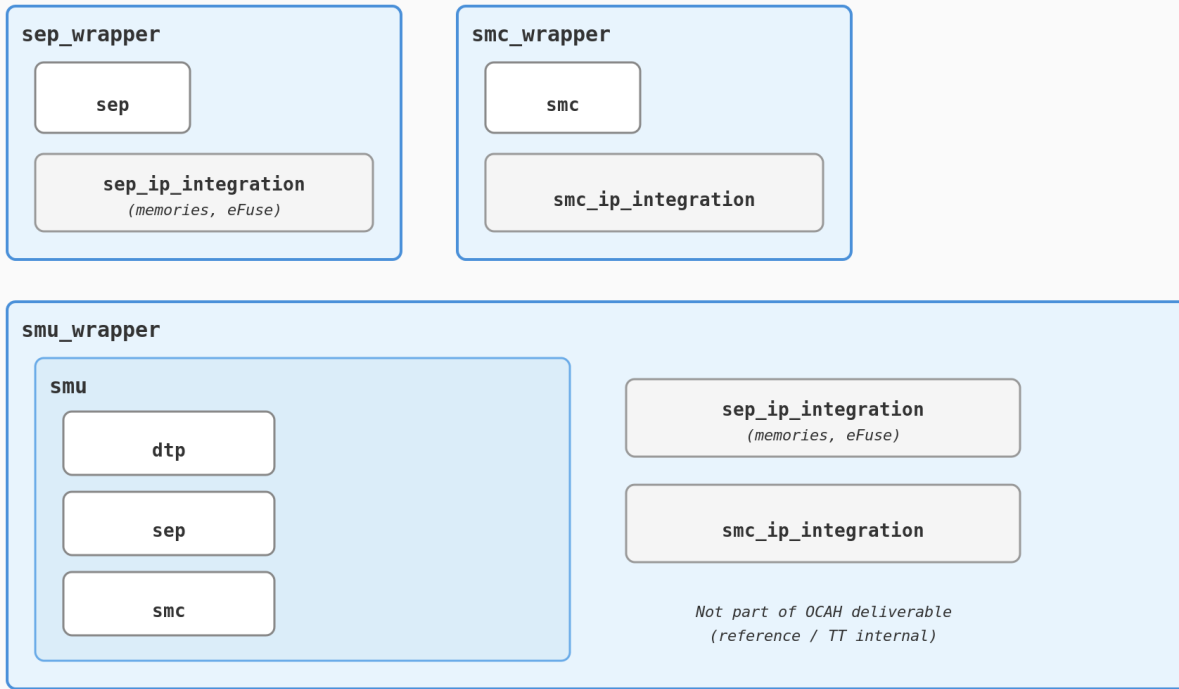


Figure 1. SMU Integration Components

IP Integration Modules

`smc_ip_integration.sv` demonstrates integration of SMC-side technology components: PLL wrappers (generating `pll_cgm_clk_o` and `pll_awm_clk_o`), PVT sensor interfaces, eFuse shims, GPIO padding logic, memory macros (ROM, scratchpad RAM, cache tag and data arrays), and trace memory instances.

`sep_ip_integration.sv` demonstrates integration of SEP-side technology components: SEP memory macros (TCM, scratch SRAM, boot ROM, OTBN IMEM/DMEM, Key Manager ROM/SRAM), eFuse shim, external TRNG, and SPI controller selection.

Wrapper Clock Architecture

The `smu_wrapper.sv` reference wrapper accepts pre-generated clocks directly: `clk_smu_i` (primary), `clk_ref_i` (reference), and `clk_periph_i` (peripheral). The `smc_wrapper.sv` uses `clk_smc_i` for the primary clock. Clock generation is entirely the adopter's responsibility. The `pll_wrap` model included in the DV environment only terminates the SMC's AXI-Lite PLL control interface; it does not generate functional clocks and must not be used as a synthesis model.

1.6.2. Clock and Reset Architecture

Connect `clk_smu_i` to the primary system clock and `clk_ref_i` to a stable reference source. The DTP operates on TCK for JTAG logic and on the system clock for cross-trigger and AXI bridges. Reset sequencing follows the SMC flow; the DTP JTAG state machine uses `pwr_on_rst_ni`, which the SMU connects to `rst_cold_ni`. The SMU internally connects DTP debug paths to the SMC and optionally instantiates the SEP when the `SEP` parameter is set to 1. The full interconnection details are described in [SMC-DTP Internal Interconnection](#) and [SEP-Conditional Wiring](#) below.

1.6.3. SMU AXI Crossbar Address Map

Inside the SMU, the `smu_axi_xbar` module routes traffic between three AXI4 manager/subordinate pairs: the SEP

outbound master (**sep_out**), the SMC outbound master (**smc_out**), and the external SMU AXI input (**ext_in** / **smu_axi_in**). The corresponding targets are the SEP and SMC inbound subordinates, and the external SMU AXI output (**ext_out** / **smu_axi_out**).

See the *SMU AXI Crossbar* and *OCAH Memory Map* sections of the Technical Reference Manual for connectivity and the responsibility split between hardware-fixed and firmware-programmable apertures.

The crossbar uses a **runtime** address map. Only the SEP and SMC apertures are programmable at runtime; the **ext_out** rule is static:

- **sep_global_base_addr_i** / **sep_region_size_i** come from the **sep_cpu_ctrl** CSRs **SEP_GLOBAL_BASE_ADDR** / **SEP_REGION_SIZE**, exposed at both the **sep.sv** and **smu.sv** boundaries as **sep_global_base_addr_o** / **sep_region_size_o** (both 56-bit). **SEP_REGION_SIZE** is a 32-bit CSR field, zero-extended to 56-bit at the SEP system CSR; the xbar input takes only the lower 32 bits. **SEP_REGION_SIZE** defines the SEP global/chiplet aperture for SMU routing and SEP inbound global-to-local remap. It is distinct from the SEP CPU's architectural local-alias span. **SEP_REGION_SIZE** currently resets to **0x0100_0000** (16 MiB). Tied to '0' at the **smu.sv** boundary when the **SEP** parameter is 0.
- **smc_global_base_addr_i** / **smc_region_size_i** come from the SMC CPU control CSRs **GLOBAL_BASE** / **REGION_SIZE**, exposed at both the **smc.sv** and **smu.sv** boundaries as **smc_global_base_o** (56-bit) / **smc_region_size_o** (32-bit). **REGION_SIZE** resets to **0x0100_0000** (16 MiB).

SEP, SMC, and **smu_axi_xbar** all run on **clk_smu_i**, so these quasi-static CSR fields need no clock-domain crossing. The crossbar's address arithmetic is done on 57-bit widened sums. **ext_out** is a catch-all port that carries any outbound SEP/SMC manager request not matched by the programmed apertures; it has no fixed lower-bound address. Unmatched inbound (**ext_in**) requests return a decode error and do not reach **ext_out**.

Firmware programming order

Firmware must program **SEP_GLOBAL_BASE_ADDR** / **SEP_REGION_SIZE** and SMC **GLOBAL_BASE** / **REGION_SIZE** during early boot, **before any cross-subsystem AXI traffic is issued**, and must ensure the two apertures do not overlap and both fall within the chiplet-visible SMU envelope assigned by the integrator. The SMU crossbar does not itself constrain envelope size.

See the *OCAH Memory Map* section of the Technical Reference Manual for that responsibility split.

1.6.4. SMU Default Parameters

The SMU exposes a set of configurable parameters that control DTP JTAG features, identification, cross trigger, and pipeline depths. The table below lists each parameter with its default value.

Parameter	Value	Description
JTAG_BSR_ENABLE	1	Boundary scan enabled
JTAG_EXTEST_TRAIN_ENABLE	1	EXTEST_TRAIN instruction enabled
JTAG_EXTEST_PULSE_ENABLE	1	EXTEST_PULSE instruction enabled
JTAG_INTEST_ENABLE	1	INTEST instruction enabled
JTAG_CLAMP_ENABLE	1	CLAMP instruction enabled
JTAG_HIGHZ_ENABLE	1	HIGHZ instruction enabled
JTAG_RUNBIST_ENABLE	1	RUNBIST instruction enabled
JTAG_TMP_ENABLE	1	TMP controller enabled

Parameter	Value	Description
JTAG_IC_RESET_ENABLE	1	Enables the external IC_RESET TDR slice (SMU drives this into the DTP's JTAG_IC_RESET_EXT_ENABLE). SMC and SEP slices are enabled unconditionally by the SMU.
JTAG_SMC_DBG_ENABLE	1	SMC debug interface enabled
JTAG_STAP_IO_ENABLE	1	I/O STAP enabled
JTAG_NUM_EXTRA_STAPS	1	Number of additional STAPs
JTAG_IDCODE_MFR_ID	110h000	Manufacturer ID
JTAG_IDCODE_PART_NUM	160h0000	Part number
JTAG_IDCODE_SI_REV	40h0	Silicon revision
JTAG_OCH_VER	80h00	OCAH IP version
XTRIG_INT_CT_MODE	80h00	Internal CT protocol mode bits (0=pulse sync, 1=req/ack)
SMC_OTP_RD_PL_DEPTH	20h3	SMC OTP read pipeline depth
SMC_OTP_WR_PL_DEPTH	20h3	SMC OTP write pipeline depth
SMC_RD_PL_DEPTH	20h3	SMC fabric read pipeline depth
SMC_WR_PL_DEPTH	20h3	SMC fabric write pipeline depth
SEP_ABR_SRAM_LATENCY	1	ABR 1R1W registered-read latency, in clocks
SEP_ABR_MASKING_EN	1	Adams Bridge 2-share DOM masking

SEP-related DTP configuration is not exposed as an SMU config-struct field. The SMU derives SEP debug/STAP enablement from the module parameter `SEP` and wires the corresponding DTP interfaces internally to the SEP block when `SEP=1`.

1.6.5. IC_RESET TDR Structure

The DTP primary TAP implements the IEEE 1149.1-2013 §17 IC_RESET TDR as three independently enabled slices plus a single `reset_hold` bit nearest TDO. At the SMU the scan order from TDI to TDO is:

TDI → SMC slice → SEP slice → external slice → reset_hold → TDO

The SMU enables the SMC slice unconditionally, the SEP slice when `SEP=1` (otherwise that slice has width 0 and is omitted), and the external slice when `JTAG_IC_RESET_ENABLE` is set. Each slice is a packed struct with `.ovrd` and `.val` halves of equal width; `jtag_ptap` sizes the slice from $\$bits(type)/2$ and interleaves the halves into the `{reset_enable, reset_control}` pair IEEE §17 requires per port. Declaration order within each half is the TDI-to-TDO scan order: the first field is nearest TDI.

Slice	Ports	Defined by	Consumed by
SMC	68	OCAH (<code>smc_pkg::jtag_smc_reset_ctrl_t</code>)	SMC reset unit; SMU-internal, no SMU port
SEP	7 when <code>SEP=1</code> , else 0	OCAH (<code>sep_pkg::jtag_sep_reset_ctrl_t</code>)	SEP reset controller; SMU-internal, no SMU port
External	$\$bits(ic_reset_ext_t)/2$	Adopter (<code>ic_reset_ext_t</code>)	SMU output <code>jtag_ic_reset_ext_o</code>
<code>reset_hold</code>	1 bit	IEEE 1149.1 §17	Nearest TDO; not a reset port

The TDR length is $2 * (\text{SMC ports} + \text{SEP ports} + \text{external ports}) + 1$. Programming order and `reset_enable` polarity are in the DTP JTAG chapter of the Technical Reference Manual.

SMC slice (TDI to TDO)

The SMC slice covers the SMC reset domains and the 32 managed-subsystem (SS) cold and warm resets. Vector index [31] of each SS field is nearer TDI than [0].

Port	Reset domain
ss_warm_reset_n[31:0]	Per-subsystem warm reset
ss_cold_reset_n[31:0]	Per-subsystem cold reset
cold_reset_n	SMC cold reset
cool_reset_n	SMC cool reset
warm_reset_n	SMC warm reset
fuse_reset_n	Fuse reset (nearest the SEP slice)

SEP slice (TDI to TDO)

When SEP=1 the SEP slice covers the SEP main reset and the crypto, Key Manager, and TRNG resets consumed by sep_reset_ctrl. trng_jtag_rst_n is nearest TDI.

Port	Reset domain
trng_jtag_rst_n	TRNG
sep_reset_n	SEP main reset
kmac_jtag_rst_n	KMAC
hmac_jtag_rst_n	HMAC
aes_jtag_rst_n	AES
otbn_jtag_rst_n	OTBN
km_jtag_rst_n	Key Manager (nearest the external slice)

External slice (adopter-defined)

The external slice is reserved for reset overrides on logic instantiated outside the SMU. OCAH does not name its ports. Set the SMU type parameter ic_reset_ext_t to a packed struct with .ovrd and .val halves of equal width, and route jtag_ic_reset_ext_o to that logic. The parameter default (jtag_tap_pkg::jtag_ic_reset_default_t) is a single-override stub. Clear JTAG_IC_RESET_ENABLE when no external overrides are required.

Declare a new external override at each half's MSB so already-assigned TDR positions do not shift. Extra subsystem reset controls beyond the SMC slice use this typed port.

1.6.6. DTP Parameter Pass-Through

The SMU passes through most DTP parameters to allow integrator customization. Type parameters are set internally based on SMC package types.

JTAG Configuration Parameters

Parameter	SMU Default	Pass-through	Notes
JTAG_BSR_ENABLE	1	Yes	Boundary scan enabled
JTAG_EXTTEST_TRAIN_ENABLE	1	Yes	

Parameter	SMU Default	Pass-through	Notes
JTAG_EXTTEST_PULSE_ENABLE	1	Yes	
JTAG_INTTEST_ENABLE	1	Yes	
JTAG_CLAMP_ENABLE	1	Yes	
JTAG_HIGHZ_ENABLE	1	Yes	
JTAG_RUNBIST_ENABLE	1	Yes	
JTAG_TMP_ENABLE	1	Yes	
JTAG_IC_RESET_ENABLE	1	No	SMU routes this to the DTP's JTAG_IC_RESET_EXT_ENABLE; SMC and SEP slices are enabled inside the SMU instantiation (SMC unconditionally, SEP when SEP=1)
JTAG_SMC_DBG_ENABLE	1	Yes	
JTAG_SEP_DBG_ENABLE	1 (SEP=1)	No	Derived internally from the SMU module parameter SEP; not exposed in smu_cfg_t
JTAG_STAP_IO_ENABLE	1	Yes	
JTAG_NUM_EXTRA_STAPS	1	Yes	Port array size equals parameter value; when set to 0, array size is 1 (tie-off required)

JTAG ID Parameters

Parameter	SMU Default	Pass-through	Notes
JTAG_IDCODE_MFR_ID	110h000	Yes	Manufacturer ID
JTAG_IDCODE_PART_NUM	160h0000	Yes	Part number
JTAG_IDCODE_SI_REV	40h0	Yes	Silicon revision
JTAG_OCH_VER	80h00	Yes	OCAH IP version

Cross Trigger Parameters

Parameter	SMU Default	DTP Internal	Pass-through	Notes
XTRIG_INT_CT_MODE	[7:0]	[9:0]	Yes	8-bit on SMU; DTP bits [1:0]=0 for SMC

The SMU parameter width is two less than DTP's for cross trigger (8 vs 10) because internal cross trigger ports [1:0] and clock stop request #0 are connected to the SMC internally. The SMU remaps DTP ports [9:2] to SMU external ports [7:0].

Pipeline Depth Parameters

Parameter	SMU Default	Pass-through	Notes
SMC_OTP_RD_PL_DEPTH	20h3	Yes	SMC OTP read pipeline depth
SMC_OTP_WR_PL_DEPTH	20h3	Yes	SMC OTP write pipeline depth
SEP_OTP_RD_PL_DEPTH	20h3	No	Fixed to 20h3 in SMU; used when SEP=1, ignored when SEP=0
SEP_OTP_WR_PL_DEPTH	20h3	No	Fixed to 20h3 in SMU; used when SEP=1, ignored when SEP=0
SMC_RD_PL_DEPTH	20h3	Yes	SMC fabric read pipeline depth
SMC_WR_PL_DEPTH	20h3	Yes	SMC fabric write pipeline depth

Type Parameters

The following type parameters are set internally by the SMU and are not exposed for integrator configuration.

Parameter	Set To	Notes
jtag_tap_ctrl_t	prim_jtag_pkg::jtag_tap_ctrl_t	JTAG TAP control struct
jtag_scan_ctrl_t	prim_jtag_pkg::jtag_scan_ctrl_t	JTAG scan control struct
smc_jtag_axi_req_t	smc_pkg::smc_jtag_56_64_2_12_axi_req_t	SMC debug AXI request
smc_jtag_axi_resp_t	smc_pkg::smc_jtag_56_64_2_12_axi_resp_t	SMC debug AXI response
smc_otp_axil_req_t	smc_efuse_pkg::efuse_axil_req_t	SMC OTP AXI-Lite request
smc_otp_axil_resp_t	smc_efuse_pkg::efuse_axil_resp_t	SMC OTP AXI-Lite response
sep_otp_axil_req_t	smc_efuse_pkg::efuse_axil_req_t	SEP OTP AXI-Lite request (error slave when SEP=0)
sep_otp_axil_resp_t	smc_efuse_pkg::efuse_axil_resp_t	SEP OTP AXI-Lite response (error slave when SEP=0)
xtrig_axil_req_t	smc_pkg::smc_axil_32_32_req_t	Cross trigger CSR request
xtrig_axil_resp_t	smc_pkg::smc_axil_32_32_resp_t	Cross trigger CSR response

1.6.7. DTP Port Mapping

The following tables define the complete DTP port list as connected within the SMU. Ports marked "SMU external port" are exposed on the SMU boundary for integrator connection. Ports marked with a subsystem name (e.g. SMC, SEP) are wired internally within the SMU.

Clock and Reset

Port	Direction	Width	SMU Connection
clk_i	input	1	SMU clk_smu_i (shared with SMC clk_smc_i)
rst_n_i	input	1	SMC rst_primary_smc_clk_no
pwr_on_rst_ni	input	1	SMU rst_cold_ni (power-on reset for JTAG logic)

Primary JTAG TAP Interface

Port	Direction	Width	SMU Connection
jtag_ptap_client_tap_ctrl_i	input	jtag_tap_ctrl_t	SMU external port
jtag_ptap_client_tdi_i	input	1	SMU external port
jtag_ptap_client_tdo_o	output	1	SMU external port
jtag_ptap_client_tdo_oen_o	output	1	SMU external port

Boundary Scan Interface

Port	Direction	Width	SMU Connection
jtag_bsr_host_scan_ctrl_o	output	jtag_scan_ctrl_t	SMU external port
jtag_bsr_host_scan_in_i	input	1	SMU external port
jtag_bsr_host_scan_out_o	output	1	SMU external port

I/O STAP Interface

Port	Direction	Width	SMU Connection
jtag_stap_io_host_tap_ctrl_o	output	jtag_tap_ctrl_t	SMU external port
jtag_stap_io_host_tdi_i	input	1	SMU external port

Port	Direction	Width	SMU Connection
jtag_stap_io_host_tdo_o	output	1	SMU external port
jtag_stap_io_host_tdo_oen_o	output	1	SMU external port

SMC Debug STAP Interface

Port	Direction	Width	SMU Connection
jtag_stap_smc_host_tap_ctrl_o	output	jtag_tap_ctrl_t	SMC smc_cpu_jtag_{TCK,TMS}_i; trst_n inverted drives SMC smc_cpu_jtag_reset_i when JTAG_SMC_DBG_ENABLE=1, else tie-off
jtag_stap_smc_host_tdi_i	input	1	SMC smc_cpu_jtag_TDO_data_o when JTAG_SMC_DBG_ENABLE=1, else 1'b0
jtag_stap_smc_host_tdo_o	output	1	SMC smc_cpu_jtag_TDI_i when JTAG_SMC_DBG_ENABLE=1, else unused
jtag_stap_smc_host_tdo_oen_o	output	1	Internal only; not consumed by SMC

SEP Debug STAP Interface

Port	Direction	Width	SMU Connection
jtag_stap_sep_host_tap_ctrl_o	output	jtag_tap_ctrl_t	SEP jtag_{tck,tms,trst_n} when SEP=1, else unused
jtag_stap_sep_host_tdi_i	input	1	SEP jtag_tdo when SEP=1, else 1'b0
jtag_stap_sep_host_tdo_o	output	1	SEP jtag_tdi when SEP=1, else unused
jtag_stap_sep_host_tdo_oen_o	output	1	Internal only; not consumed by SEP at SMU level

Extra STAP Interfaces

Port	Direction	Width	SMU Connection
jtag_stap_extra_host_tap_ctrl_o	output	jtag_tap_ctrl_t[JTAG_NUM_EXTRA_STAP_PORTS-1:0]	SMU external port
jtag_stap_extra_host_tdi_i	input	[JTAG_NUM_EXTRA_STAP_PORTS-1:0]	SMU external port
jtag_stap_extra_host_tdo_o	output	[JTAG_NUM_EXTRA_STAP_PORTS-1:0]	SMU external port
jtag_stap_extra_host_tdo_oen_o	output	[JTAG_NUM_EXTRA_STAP_PORTS-1:0]	SMU external port

Extended STAP Scan Interface

Port	Direction	Width	SMU Connection
jtag_stap_host_scan_ctrl_o	output	jtag_scan_ctrl_t	SMU external port
jtag_stap_host_scan_in_i	input	1	SMU external port
jtag_stap_host_scan_out_o	output	1	SMU external port

DFD iJTAG Interface

Port	Direction	Width	SMU Connection
jtag_dfd_host_scan_ctrl_o	output	jtag_scan_ctrl_t	SMU external port
jtag_dfd_host_scan_in_i	input	1	SMU external port
jtag_dfd_host_scan_out_o	output	1	SMU external port

DFT iJTAG Interface

Port	Direction	Width	SMU Connection
jtag_dft_host_scan_ctrl_o	output	jtag_scan_ctrl_t	SMU external port
jtag_dft_host_scan_in_i	input	1	SMU external port
jtag_dft_host_scan_out_o	output	1	SMU external port

SMC Fabric Debug AXI Interface

Port	Direction	Width	SMU Connection
axi_smc_dbg_req_o	output	smc_jtag_axi_req_t	SMC jtag_axi_in_req_i
axi_smc_dbg_resp_i	input	smc_jtag_axi_resp_t	SMC jtag_axi_in_resp_o

SMC OTP Debug AXI-Lite Interface

Port	Direction	Width	SMU Connection
axil_smc_otp_jtag_req_o	output	smc_otp_axil_req_t	SMC axil_smc_otp_jtag_req_i
axil_smc_otp_jtag_resp_i	input	smc_otp_axil_resp_t	SMC axil_smc_otp_jtag_resp_o

SEP OTP Debug AXI-Lite Interface

When `SEP=1`, this interface connects to the SEP `axil_sep_otp_jtag_*` ports. When `SEP=0`, the SMU instantiates `u_sep_otp_axil_err_slv`, which responds with DECERR to all transactions.

Port	Direction	Width	SMU Connection
axil_sep_otp_jtag_req_o	output	sep_otp_axil_req_t	SEP <code>axil_sep_otp_jtag_req_i</code> when <code>SEP=1</code> , else AXI-Lite error slave (<code>u_sep_otp_axil_err_slv</code>)
axil_sep_otp_jtag_resp_i	input	sep_otp_axil_resp_t	SEP <code>axil_sep_otp_jtag_resp_o</code> when <code>SEP=1</code> , else AXI-Lite error slave (<code>u_sep_otp_axil_err_slv</code>)

Clock Control

Port	Direction	Width	SMU Connection
stop_clks_o	output	1	SMU external port: <code>dtb_stop_clks_o</code> (to PLL clock gates)

Boot Stall Control

Port	Direction	Width	SMU Connection
jtag_boot_stall_ovrd_o	output	1	SMC <code>boot_stall_jtag_ovrd_i</code>
jtag_boot_stall_o	output	1	SMC <code>boot_stall_jtag_val_i</code>

JTAG Reset Control

Port	Direction	Width	SMU Connection
jtag_ic_reset_smc_o	output	smc_pkg::jtag_smc_reset_ctrl_t	SMC <code>jtag_smc_reset_ctrl_i</code> (packed struct with <code>.ovrd</code> override enables and <code>.val</code> active-low override values)

Port	Direction	Width	SMU Connection
jtag_ic_reset_sep_o	output	sep_pkg::jtag_sep_reset_ctrl_t	SEP jtag_sep_reset_ctrl_i (packed struct with .ovrd override enables and .val active-low override values); SMU-internal, no SMU port
jtag_ic_reset_ext_o	output	ic_reset_ext_t	SMU jtag_ic_reset_ext_o external port (integrator-defined packed struct; carries reset overrides for logic outside the SMU)

JTAG State Outputs

Port	Direction	Width	SMU Connection
jtag_ptap_state_o	output	tap_state_e	SMU external port
jtag_ptap_inst_decoded_o	output	jtag_instruction_decoded_e	SMU external port

Cross Trigger CSR AXI-Lite Interface

Port	Direction	Width	SMU Connection
axil_xtrig_req_i	input	xtrig_axil_req_t	SMC axil_dtp_csr_req_o
axil_xtrig_resp_o	output	xtrig_axil_resp_t	SMC axil_dtp_csr_resp_i

Cross Trigger Matrix Interface

Port	Direction	Width	SMU Connection
xtrig_ctm_src_req_o	output	[XTRIG_NUM_INT_CT-1:0]	[1:0]: SMU internal (SMC), [9:2]: SMU external xtrig_ctm_src_req_o[7:0]
xtrig_ctm_src_ack_i	input	[XTRIG_NUM_INT_CT-1:0]	[1:0]: Tie-off (SMC), [9:2]: SMU external xtrig_ctm_src_ack_i[7:0]
xtrig_ctm_dst_req_i	input	[XTRIG_NUM_INT_CT-1:0]	[1:0]: SMC xtrigger_ss_o[1:0], [9:2]: SMU external xtrig_ctm_dst_req_i[7:0]
xtrig_ctm_dst_ack_o	output	[XTRIG_NUM_INT_CT-1:0]	[1:0]: Unused (SMC connection), [9:2]: SMU external xtrig_ctm_dst_ack_o[7:0]

Internal cross trigger ports [1:0] are reserved for SMC (both bits of `xtrigger_ss_o`). The SMU exposes ports [9:2] as external ports [7:0], so the SMU `XTRIG_NUM_INT_CT` parameter is two less than DTP's (8 vs 10). The SMC uses pulse-synchronization mode, so ack signals for ports [1:0] are unused: `xtrig_ctm_src_ack_i[1:0]` should be tied to `2'b00` and `xtrig_ctm_dst_ack_o[1:0]` are left unconnected. The usage of ack signals depends on the `XTRIG_INT_CT_MODE` parameter: when a mode bit is 0 (pulse synchronization), the corresponding ack is unused; when 1 (req/ack handshaking), the ack must be connected to the cross trigger endpoint.

Clock Stop Request Interface

Port	Direction	Width	SMU Connection
xtrig_clk_stop_req_i	input	[XTRIG_NUM_CLK_STOP_REQ-1:0]	[0]: SMC tdr_dbg_ctrl_clocks_stopped_by_cla_o, [8:1]: SMU external xtrig_clk_stop_req_i[7:0]

Clock stop request port #0 is reserved for SMC. The SMU exposes ports [8:1] as external ports [7:0], so the SMU `XTRIG_NUM_CLK_STOP_REQ` parameter should be one less than DTP's (8 vs 9).

Cross Trigger Port GPIO Interface

The 16 CTP signals from DTP are ported through the SMU and are direct `smu_wrapper` external ports (connected through the `.*` SMU instantiation). The adopter must connect them to GPIO pad logic; no tie-off or dangling is present in the wrapper itself.

Port	Direction	Width	SMU Connection
<code>xtrig_ctp_req_out_dout_o</code>	output	[15:0]	SMU external port; connect to GPIO pad logic
<code>xtrig_ctp_req_out_dout_en_o</code>	output	[15:0]	SMU external port; connect to GPIO pad logic
<code>xtrig_ctp_req_out_din_i</code>	input	[15:0]	SMU external port; connect to GPIO pad logic
<code>xtrig_ctp_req_out_din_en_o</code>	output	[15:0]	SMU external port; connect to GPIO pad logic
<code>xtrig_ctp_req_in_dout_o</code>	output	[15:0]	SMU external port; connect to GPIO pad logic
<code>xtrig_ctp_req_in_dout_en_o</code>	output	[15:0]	SMU external port; connect to GPIO pad logic
<code>xtrig_ctp_req_in_din_i</code>	input	[15:0]	SMU external port; connect to GPIO pad logic
<code>xtrig_ctp_req_in_din_en_o</code>	output	[15:0]	SMU external port; connect to GPIO pad logic
<code>xtrig_ctp_ack_in_dout_o</code>	output	[15:0]	SMU external port; connect to GPIO pad logic
<code>xtrig_ctp_ack_in_dout_en_o</code>	output	[15:0]	SMU external port; connect to GPIO pad logic
<code>xtrig_ctp_ack_in_din_i</code>	input	[15:0]	SMU external port; connect to GPIO pad logic
<code>xtrig_ctp_ack_in_din_en_o</code>	output	[15:0]	SMU external port; connect to GPIO pad logic
<code>xtrig_ctp_ack_out_dout_o</code>	output	[15:0]	SMU external port; connect to GPIO pad logic
<code>xtrig_ctp_ack_out_dout_en_o</code>	output	[15:0]	SMU external port; connect to GPIO pad logic
<code>xtrig_ctp_ack_out_din_i</code>	input	[15:0]	SMU external port; connect to GPIO pad logic
<code>xtrig_ctp_ack_out_din_en_o</code>	output	[15:0]	SMU external port; connect to GPIO pad logic

1.6.8. SMC Port Mapping

The following tables define the complete SMC port list as connected within the SMU. Ports marked "SMU external port" are exposed on the SMU boundary. Ports marked with "DTP" or "SEP" are wired internally.

Clocks

Port	Direction	Width	SMU Connection
<code>clk_smc_i</code>	input	1	SMU <code>clk_smu_i</code> (shared with DTP <code>clk_i</code>)
<code>clk_periph_i</code>	input	1	SMU external port
<code>clk_ref_i</code>	input	1	SMU external port

Resets

Port	Direction	Width	SMU Connection
<code>rst_cold_ni</code>	input	1	SMU external port
<code>rst_primary_ref_clk_no</code>	output	1	SMU external port
<code>rst_primary_smc_clk_no</code>	output	1	SMU external port + DTP <code>rst_n_i</code>

System AXI Input

Port	Direction	Width	SMU Connection
sys_axi_in_req_i	input	smc_sys_in_56_64_6_12_axi_req_t	SMU external port
sys_axi_in_resp_o	output	smc_sys_in_56_64_6_12_axi_resp_t	SMU external port

JTAG AXI Input

Port	Direction	Width	SMU Connection
jtag_axi_in_req_i	input	smc_jtag_56_64_2_12_axi_req_t	DTP axi_smc_dbg_req_o
jtag_axi_in_resp_o	output	smc_jtag_56_64_2_12_axi_resp_t	DTP axi_smc_dbg_resp_i

SMC OTP JTAG AXI-Lite

Port	Direction	Width	SMU Connection
axil_smc_otp_jtag_req_i	input	efuse_axil_req_t	DTP axil_smc_otp_jtag_req_o
axil_smc_otp_jtag_resp_o	output	efuse_axil_resp_t	DTP axil_smc_otp_jtag_resp_i

SEP AXI Input

When **SEP=0**, the SEP AXI subordinate input is tied off to '0' and the response output is left unconnected. When **SEP=1**, the SMU drives this path from the SEP-to-SMC dedicated AXI connection.

Port	Direction	Width	SMU Connection
sep_axi_in_req_i	input	smc_sep_in_56_64_6_12_axi_req_t	Tied off to '0' (no incoming transactions)
sep_axi_in_resp_o	output	smc_sep_in_56_64_6_12_axi_resp_t	Unconnected (no transactions to respond to)

AXI Output

Port	Direction	Width	SMU Connection
output_axi_req_o	output	smc_sys_out_56_64_8_12_axi_req_t	SMU external port
output_axi_resp_i	input	smc_sys_out_56_64_8_12_axi_resp_t	SMU external port

DTP CSR Interface

Port	Direction	Width	SMU Connection
axil_dtp_csr_req_o	output	smc_axil_32_32_req_t	DTP axil_xtrig_req_i
axil_dtp_csr_resp_i	input	smc_axil_32_32_resp_t	DTP axil_xtrig_resp_o

Adopter Peripheral Window AXI-Lite

Port	Direction	Width	SMU Connection
smc_external_req_o	output	smc_axil_32_32_req_t	SMU external port (to adopter peripherals)
smc_external_resp_i	input	smc_axil_32_32_resp_t	SMU external port (from adopter peripherals)

eFuse Interface

Port	Direction	Width	SMU Connection
efuse_bank_ctrl_req_o	output	efuse_axil_req_t	SMU external port
efuse_bank_ctrl_resp_i	input	efuse_axil_resp_t	SMU external port
efuse_shim_command_req_o	output	efuse_command_req_t	SMU external port
efuse_shim_command_resp_i	input	efuse_command_resp_t	SMU external port
shadow_regs_o	output	efuse_map_t	SMU external port

GPIO Data Signals

Port	Direction	Width	SMU Connection
lsio_interface_select_o	output	[NUM_GPIO_WRAPS-1:0]	SMU external port
pad2core_i	input	[NUM_GPIO_WRAPS-1:0]	SMU external port
core2pad_o	output	[NUM_GPIO_WRAPS-1:0]	SMU external port
pad2core_en_o	output	[NUM_GPIO_WRAPS-1:0]	SMU external port
core2pad_en_o	output	[NUM_GPIO_WRAPS-1:0]	SMU external port

GPIO External Pins

Port	Direction	Width	SMU Connection
rst_cool_n_from_pin_i	input	1	SMU external port

ATB Telemetry

Port	Direction	Width	SMU Connection
clk_telemetry_i	input	1	SMU external port
rst_telemetry_ni	input	1	SMU external port
telemetry_atdata_i	input	telemetry_data_t[NUM_TELEMETRY_RECEIVERS-1:0]	SMU external port
telemetry_atid_i	input	atb_id_t[NUM_TELEMETRY_RECEIVERS-1:0]	SMU external port
telemetry_atready_o	output	[NUM_TELEMETRY_RECEIVERS-1:0]	SMU external port
telemetry_atvalid_i	input	[NUM_TELEMETRY_RECEIVERS-1:0]	SMU external port
telemetry_afvalid_o	output	[NUM_TELEMETRY_RECEIVERS-1:0]	SMU external port
telemetry_afready_i	input	[NUM_TELEMETRY_RECEIVERS-1:0]	SMU external port

DED/WDT

Port	Direction	Width	SMU Connection
cluster_ded_o	output	1	SMU external port
wdt_first_timeout_o	output	1	SMU external port
wdt_second_timeout_o	output	1	SMU external port

Region Size

Port	Direction	Width	SMU Connection
smc_region_size_o	output	[31:0]	SMU external port

External Interrupts

Port	Direction	Width	SMU Connection
<code>ext_interrupts_i</code>	input	[NUM_EXT_INTERRUPTS-1:0]	SMU external port
<code>sep_mailbox_interrupts_i</code>	input	[7:0]	Internal: from SEP <code>smc_mailbox_interrupt_o</code> when <code>SEP=1</code> , else tied to '0
<code>sep_wdt_reset_n_i</code>	input	1	Internal: <code>~sep_wdt_timer_rst_req</code> when <code>SEP=1</code> , else tied to 1b1

Fuse Signals

Port	Direction	Width	SMU Connection
<code>fuse_sense_done_o</code>	output	1	SMU external port
<code>fuse_reset_n_delayed_o</code>	output	1	SMU external port

DFT Signals

Port	Direction	Width	SMU Connection
<code>skip_mem_repair_o</code>	output	1	SMU external port
<code>ext_boot_seq_done_i</code>	input	1	SMU external port

PVT

Port	Direction	Width	SMU Connection
<code>temp_interrupt_i</code>	input	1	SMU external port

Lifecycle State

Port	Direction	Width	SMU Connection
<code>lc_state_o</code>	output	[2*LC_STATE_WIDTH-1:0]	SEP lifecycle state when <code>SEP=1</code> , else 80hf0; ported through SMU
<code>lc_sigint_err_o</code>	output	1	SEP lifecycle integrity error when <code>SEP=1</code> , else SMC <code>efuse_lc_sigint_err</code>

RAS Bank Settings

Port	Direction	Width	SMU Connection
<code>ras_bank_chip_o</code>	output	[3:0]	SMU external port
<code>ras_bank_instance_o</code>	output	[3:0]	SMU external port

Mailbox Interrupts

Port	Direction	Width	SMU Connection
<code>ext_mailbox_interrupts_o</code>	output	[NUM_MAILBOXES-1:0]	SMU external port

Reset Unit Signals

Port	Direction	Width	SMU Connection
cfg_flr_pf_active_i	input	1	SMU external port
isolate_req_o	output	[31:0]	SMU external port
ss_reset_complete_i	input	[31:0]	SMU external port
ss_config_o	output	[31:0]	SMU external port
ss_reset_ctrl_o	output	reset_ctrl_t[31:0]	SMU external port
sync_irq_o	output	1	SMU external port

CPU Memory Signals

Port	Direction	Width	SMU Connection
rom_intf_req_o	output	rom_req_t	SMU external port
rom_intf_rsp_i	input	rom_rsp_t	SMU external port
scratch_ram_intf_req_o	output	scratch_ram_req_t[31:0]	SMU external port
scratch_ram_intf_rsp_i	input	scratch_ram_rsp_t[31:0]	SMU external port
l1_icache_tag_intf_req_o	output	l1_icache_tag_req_t[3:0]	SMU external port
l1_icache_tag_intf_rsp_i	input	l1_icache_tag_rsp_t[3:0]	SMU external port
l1_icache_data_intf_req_o	output	l1_icache_data_req_t[7:0]	SMU external port
l1_icache_data_intf_rsp_i	input	l1_icache_data_rsp_t[7:0]	SMU external port
l1_dcache_tag_intf_req_o	output	l1_dcache_tag_req_t[3:0]	SMU external port
l1_dcache_tag_intf_rsp_i	input	l1_dcache_tag_rsp_t[3:0]	SMU external port
l1_dcache_data_intf_req_o	output	l1_dcache_data_req_t[3:0]	SMU external port
l1_dcache_data_intf_rsp_i	input	l1_dcache_data_rsp_t[3:0]	SMU external port

Memory Init

Port	Direction	Width	SMU Connection
disable_sram_auto_init_i	input	1	SMU external port
init_mem_done_o	output	1	SMU external port

System Timer OCTS Interface

Port	Direction	Width	SMU Connection
timer_count_o	output	[63:0]	SMU external port

Reset Overrides (from DTP)

Port	Direction	Width	SMU Connection
jtag_reset_ctrl_i	input	smc_pkg::jtag_smc_reset_ctrl_t	DTP jtag_ic_reset_smc_o (packed struct with .ovrd override enables and .val active-low override values)

DFD Signals

Port	Direction	Width	SMU Connection
cla_gpio_enable_o	output	1	SMU external port

Port	Direction	Width	SMU Connection
<code>tdr_dbg_ctrl_clock_stop_en_i</code>	input	1	DTP <code>cla_clock_stop_en_o</code>
<code>tdr_dbg_ctrl_clocks_stopped_by_cla_o</code>	output	1	DTP <code>xtrig_clk_stop_req_i[0]</code> (SMU connects SMC output to DTP input)

Cross Trigger Interface

Port	Direction	Width	SMU Connection
<code>xtrigger_ss_i</code>	input	<code>smc_pkg::xtrigger_t</code>	SMU internal: DTP <code>xtrig_ctm_src_req_o[1:0]</code> → SMC. Not an SMU external port.
<code>xtrigger_ss_o</code>	output	<code>smc_pkg::xtrigger_t</code>	SMU internal: SMC → DTP <code>xtrig_ctm_dst_req_i[1:0]</code> . Not an SMU external port.

Trace Memory

Port	Direction	Width	SMU Connection
<code>trace_mem_req_o</code>	output	<code>SinkMemPktIn_s[TRC_RAM_INSTANCES-1:0]</code>	SMU external port
<code>trace_mem_resp_i</code>	input	<code>SinkMemPktOut_s[TRC_RAM_INSTANCES-1:0]</code>	SMU external port

External Debug Bus

Port	Direction	Width	SMU Connection
<code>ext_debug_bus_i</code>	input	[127:0]	SMU external port

Test Mode

Port	Direction	Width	SMU Connection
<code>test_en_i</code>	input	1	SMU external port
<code>scan_rst_i</code>	input	1	SMU external port

Captured Straps

Port	Direction	Width	SMU Connection
<code>captured_straps_i</code>	input	[63:0]	SMU external port

Please refer to the `reset_unit` STRAPS_LO and STRAPS_HI register descriptions for the decoded value of the straps.

1.6.9. SMC-DTP Internal Interconnection

The following table summarizes the key internal connections between the DTP and SMC within the SMU. These connections are not visible on the SMU boundary.

Connection	DTP Port	SMC Port	Direction	Notes
AXI Debug Bus	<code>axi_smc_dbg_req_o</code>	<code>jtag_axi_in_req_i</code>	DTP → SMC	
	<code>axi_smc_dbg_resp_i</code>	<code>jtag_axi_in_resp_o</code>	SMC → DTP	
AXI OTP Bus	<code>axil_smc_otp_jtag_req_o</code>	<code>axil_smc_otp_jtag_req_i</code>	DTP → SMC	

Connection	DTP Port	SMC Port	Direction	Notes
	<code>axil_smc_otp_jtag_resp_i</code>	<code>axil_smc_otp_jtag_resp_o</code>	SMC → DTP	
DTP CSR Bus	<code>axil_xtrig_req_i</code>	<code>axil_dtp_csr_req_o</code>	SMC → DTP	
	<code>axil_xtrig_resp_o</code>	<code>axil_dtp_csr_resp_i</code>	DTP → SMC	
JTAG SMC Reset Overrides	<code>jtag_ic_reset_smc_o</code>	<code>jtag_reset_ctrl_i</code>	DTP → SMC	Packed <code>smc_pkg::jtag_smc_reset_ctrl_t</code> struct carrying all SMC reset override enables and active-low values
DTP Reset	<code>rst_n_i</code>	<code>rst_primary_smc_clk_no</code>	SMC → DTP	
Cross Trigger [1:0]	<code>xtrig_ctm_dst_req_i[1:0]</code>	<code>xtrigger_ss_o[1:0]</code>	SMC → DTP	Both bits
Clock Stop #0	<code>xtrig_clk_stop_req_i[0]</code>	<code>tdr_dbg_ctrl_clocks_stopped_by_cla_o</code>	SMC → DTP	
Boot Stall	<code>jtag_boot_stall_ovrd_o</code>	<code>boot_stall_jtag_ovrd_i</code>	DTP → SMC	
	<code>jtag_boot_stall_o</code>	<code>boot_stall_jtag_val_i</code>	DTP → SMC	

1.6.10. SEP-Conditional Wiring

SEP-related interfaces are handled internally within the SMU. Their behavior depends on the module parameter `SEP`.

Module/Signal	Implementation	Description
SMC SEP AXI Input	<code>sep_axi_in_req_i</code> = SEP traffic when <code>SEP=1</code> , else '0	Dedicated SEP-to-SMC subordinate path
DTP SEP OTP JTAG	SEP <code>axil_sep_otp_jtag_*</code> when <code>SEP=1</code> , else <code>u_sep_otp_axil_err_slv</code>	SEP OTP debug AXI-Lite path
DTP SEP STAP return	SEP <code>jtag_tdo</code> when <code>SEP=1</code> , else <code>sep_stap_tdo_to_dtp = 10b0</code>	SEP JTAG STAP return path
SMU	<code>sep_mailbox_interrupts</code>	Internal SEP-to-SMC mailbox interrupts when <code>SEP=1</code> , else '0
SMC	<code>sep_wdt_reset_n_i</code> = <code>~sep_wdt_timer_rst_req</code> when <code>SEP=1</code> , else 10b1	SEP watchdog reset input to SMC

Other Internal Tie-offs

Module	Port	Tie-off Value	Reason
DTP	<code>xtrig_ctm_src_ack_i[1:0]</code>	20b00	SMC uses pulse-sync mode

`xtrig_ctm_src_ack_i[9:2]` (exposed as SMU `xtrig_ctm_src_ack_i[7:0]`) is ported out through the SMU. The integrator should tie off to '0 when using pulse-sync mode (`XTRIG_INT_CT_MODE` bit = 0), or connect to actual ack signals when using req/ack mode (bit = 1).

1.6.11. SMU Port Interface

The SMU exposes a single consolidated boundary that an adopter connects to at the chip or chiplet level. The following table defines all SMU top-level ports including clock, reset, JTAG, cross trigger, SMC-originated interfaces (AXI, GPIO, eFuse, PLL, PVT, memory), and SEP passthrough interfaces (memory macros, eFuse, crypto SRAM, SPI, AXI extension). Ports prefixed with `sep_` are only active when the module parameter `SEP=1`.

Table 11. SMU Port Declaration

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
clk_smu_i	logic	Input	1	Primary SMU clock domain. Shared by SMC and DTP system logic.	PLL output	N/A
clk_ref_i	logic	Input	1	Reference clock domain for timing-sensitive operations.	PLL reference clock	N/A
rst_cold_ni	logic	Input	1	Cold reset (active low). Asynchronous assertion, synchronous deassertion. Also serves as DTP power-on reset.	System cold reset	N/A
rst_cold_stable_ref_clk_no	logic	Output	1	Stable cold reset synchronized to reference clock domain (active low).	Downstream subsystems	N/A
powergood_i	logic	Input	1	Power-good indication from supply supervisor.	Power supply supervisor	N/A
jtag_ptap_client_tap_ctrl_i	prim_jtag_pkg::jtag_tap_ctrl_t	Input	-	Primary JTAG TAP control (TCK, TMS, TRST_N).	External JTAG pads	N/A
jtag_ptap_client_tdi_i	logic	Input	1	Primary JTAG TDI.	External JTAG pads	N/A
jtag_ptap_client_tdo_o	logic	Output	1	Primary JTAG TDO.	External JTAG pads	N/A
jtag_ptap_client_tdo_oen_o	logic	Output	1	Primary JTAG TDO output enable.	External JTAG pads	N/A
jtag_bsr_host_scan_ctrl_o	prim_jtag_pkg::jtag_scan_ctrl_t	Output	-	Boundary scan control.	Boundary scan chain	N/A
jtag_bsr_host_scan_in_i	logic	Input	1	Boundary scan data in.	Boundary scan chain	N/A
jtag_bsr_host_scan_out_o	logic	Output	1	Boundary scan data out.	Boundary scan chain	N/A
jtag_stap_io_host_tap_ctrl_o	prim_jtag_pkg::jtag_tap_ctrl_t	Output	-	I/O STAP TAP control.	I/O STAP host	N/A
jtag_stap_io_host_tdi_i	logic	Input	1	I/O STAP TDI.	I/O STAP host	Tie to '0 if unused
jtag_stap_io_host_tdo_o	logic	Output	1	I/O STAP TDO.	I/O STAP host	N/A
jtag_stap_io_host_tdo_oen_o	logic	Output	1	I/O STAP TDO output enable.	I/O STAP host	N/A
jtag_stap_extra_host_tap_ctrl_o	prim_jtag_pkg::jtag_tap_ctrl_t	Output	[JTAG_NUM_EXTRA_STAP_PORTS-1:0]	Extra STAP TAP control array.	Extra STAP hosts	Tie to '0 if unused
jtag_stap_extra_host_tdi_i	logic	Input	[JTAG_NUM_EXTRA_STAP_PORTS-1:0]	Extra STAP TDI array.	Extra STAP hosts	Tie to '0 if unused
jtag_stap_extra_host_tdo_o	logic	Output	[JTAG_NUM_EXTRA_STAP_PORTS-1:0]	Extra STAP TDO array.	Extra STAP hosts	N/A
jtag_stap_extra_host_tdo_oen_o	logic	Output	[JTAG_NUM_EXTRA_STAP_PORTS-1:0]	Extra STAP TDO output enable array.	Extra STAP hosts	N/A
jtag_stap_host_scan_ctrl_o	prim_jtag_pkg::jtag_scan_ctrl_t	Output	-	Extended STAP scan control.	STAP scan chain	N/A
jtag_stap_host_scan_in_i	logic	Input	1	Extended STAP scan data in.	STAP scan chain	N/A
jtag_stap_host_scan_out_o	logic	Output	1	Extended STAP scan data out.	STAP scan chain	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
jtag_dfd_host_scan_ctrl_o	prim_jtag_pkg::jtag_scan_ctrl_t	Output	-	DFD iJTAG scan control.	DFD iJTAG chain	N/A
jtag_dfd_host_scan_in_i	logic	Input	1	DFD iJTAG scan data in.	DFD iJTAG chain	N/A
jtag_dfd_host_scan_out_o	logic	Output	1	DFD iJTAG scan data out.	DFD iJTAG chain	N/A
jtag_dft_secure_host_scan_ctrl_o	prim_jtag_pkg::jtag_scan_ctrl_t	Output	-	Secure DFT iJTAG scan control.	Secure DFT iJTAG chain	N/A
jtag_dft_secure_host_scan_in_i	logic	Input	1	Secure DFT iJTAG scan data in.	Secure DFT iJTAG chain	N/A
jtag_dft_secure_host_scan_out_o	logic	Output	1	Secure DFT iJTAG scan data out.	Secure DFT iJTAG chain	N/A
jtag_dft_host_scan_ctrl_o	prim_jtag_pkg::jtag_scan_ctrl_t	Output	-	Non-secure DFT iJTAG scan control.	Non-secure DFT iJTAG chain	N/A
jtag_dft_host_scan_in_i	logic	Input	1	Non-secure DFT iJTAG scan data in.	Non-secure DFT iJTAG chain	N/A
jtag_dft_host_scan_out_o	logic	Output	1	Non-secure DFT iJTAG scan data out.	Non-secure DFT iJTAG chain	N/A
dtp_stop_clks_o	logic	Output	1	Clock stop output from DTP (JTAG_DEBUG_CONTROL or CLA clock-stop).	PLL clock gates	N/A
jtag_ptap_state_o	jtag_tap_pkg::tap_state_e	Output	-	Current PTAP TAP state.	Debug infrastructure	N/A
jtag_ptap_inst_decoded_o	jtag_inst_reg_pkg::jtag_instruction_decoded_e	Output	-	Current decoded PTAP instruction.	Debug infrastructure	N/A
jtag_ic_reset_ext_o	ic_reset_ext_t	Output	$\$bits(ic_reset_ext_t)$	External JTAG IC_RESET override slice. Integrator-defined packed struct with <code>.ovrd</code> override enables and <code>.val</code> active-low override values. Driven by the DTP's <code>jtag_ic_reset_ext_o</code> when <code>Cfg.JTAG_IC_RESET_ENABLE=1</code> , otherwise tied to '0. An integrator needing JTAG reset overrides for logic outside the SMU sets <code>ic_reset_ext_t</code> to their own override struct and routes this port to it; the SMU does not define that type.	Integrator-defined reset overrides	N/A
xtrig_ctm_src_req_o	logic	Output	$[XTRIG_NUM_INT_CT-1:0]$	Cross trigger source requests (SMU external ports [7:0]; DTP [1:0] reserved for SMC).	Cross trigger network	N/A
xtrig_ctm_src_ack_i	logic	Input	$[XTRIG_NUM_INT_CT-1:0]$	Cross trigger source acknowledges. Unused in pulse-sync mode (mode bit = 0).	Cross trigger network	Tie to '0 in pulse-sync mode
xtrig_ctm_dst_req_i	logic	Input	$[XTRIG_NUM_INT_CT-1:0]$	Cross trigger destination requests.	Cross trigger network	Tie to '0 if unused
xtrig_ctm_dst_ack_o	logic	Output	$[XTRIG_NUM_INT_CT-1:0]$	Cross trigger destination acknowledges. Unused in pulse-sync mode.	Cross trigger network	N/A
xtrig_clk_stop_req_i	logic	Input	$[XTRIG_NUM_CLK_STOP_REQ-1:0]$	Clock stop requests (SMU external ports [7:0]; port [0] reserved for SMC internally).	Clock stop requesters	Tie to '0 if unused
xtrig_ctp_req_out_dout_o	logic	Output	$[XTRIG_NUM_CTP-1:0]$	CTP request-out data output.	GPIO pad ring	N/A
xtrig_ctp_req_out_dout_en_o	logic	Output	$[XTRIG_NUM_CTP-1:0]$	CTP request-out data output enable.	GPIO pad ring	N/A
xtrig_ctp_req_out_din_i	logic	Input	$[XTRIG_NUM_CTP-1:0]$	CTP request-out data input.	GPIO pad ring	Tie to '0 if unused

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
xtrig_ctp_req_out_din_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP request-out data input enable.	GPIO pad ring	N/A
xtrig_ctp_req_in_dout_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP request-in data output.	GPIO pad ring	N/A
xtrig_ctp_req_in_dout_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP request-in data output enable.	GPIO pad ring	N/A
xtrig_ctp_req_in_din_i	logic	Input	[XTRIG_NUM_CTP-1:0]	CTP request-in data input.	GPIO pad ring	Tie to '0 if unused
xtrig_ctp_req_in_din_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP request-in data input enable.	GPIO pad ring	N/A
xtrig_ctp_ack_in_dout_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP ack-in data output.	GPIO pad ring	N/A
xtrig_ctp_ack_in_dout_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP ack-in data output enable.	GPIO pad ring	N/A
xtrig_ctp_ack_in_din_i	logic	Input	[XTRIG_NUM_CTP-1:0]	CTP ack-in data input.	GPIO pad ring	Tie to '0 if unused
xtrig_ctp_ack_in_din_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP ack-in data input enable.	GPIO pad ring	N/A
xtrig_ctp_ack_out_dout_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP ack-out data output.	GPIO pad ring	N/A
xtrig_ctp_ack_out_dout_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP ack-out data output enable.	GPIO pad ring	N/A
xtrig_ctp_ack_out_din_i	logic	Input	[XTRIG_NUM_CTP-1:0]	CTP ack-out data input.	GPIO pad ring	Tie to '0 if unused
xtrig_ctp_ack_out_din_en_o	logic	Output	[XTRIG_NUM_CTP-1:0]	CTP ack-out data input enable.	GPIO pad ring	N/A
rst_primary_ref_clk_no	logic	Output	1	Primary reset synchronized to reference clock domain (active low).	Downstream subsystems	N/A
rst_primary_smc_clk_no	logic	Output	1	Primary reset synchronized to SMC clock domain (active low).	Downstream subsystems	N/A
smu_axi_in_req_i	smu_axi_xbar_pkg::axi_56_64_req_t	Input	-	SMU AXI crossbar inbound request (from SMN).	SMN fabric	Tie to '0 if unused
smu_axi_in_resp_o	smu_axi_xbar_pkg::axi_56_64_resp_t	Output	-	SMU AXI crossbar inbound response.	SMN fabric	N/A
smu_axi_out_req_o	smu_axi_xbar_pkg::axi_out_req_t	Output	-	SMU AXI crossbar outbound request (to SMN).	SMN fabric	N/A
smu_axi_out_resp_i	smu_axi_xbar_pkg::axi_out_resp_t	Input	-	SMU AXI crossbar outbound response.	SMN fabric	Tie to '0 if unused
smc_external_req_o	smc_pkg::smc_axil_32_req_t	Output	-	SMC AXI-Lite external port request for adopter peripheral expansion.	Adopter peripherals	N/A
smc_external_resp_i	smc_pkg::smc_axil_32_resp_t	Input	-	SMC AXI-Lite external port response.	Adopter peripherals	Tie to DECERR if unused
smc_efuse_bank_ctrl_req_o	smc_pkg::smc_axil_32_req_t	Output	-	SMC eFuse bank control AXI-Lite request.	eFuse shim	N/A
smc_efuse_bank_ctrl_resp_i	smc_pkg::smc_axil_32_resp_t	Input	-	SMC eFuse bank control AXI-Lite response.	eFuse shim	N/A
smc_efuse_shim_command_req_o	smc_efuse_pkg::fuse_command_req_t	Output	-	SMC eFuse command interface request.	eFuse shim	N/A
smc_efuse_shim_command_resp_i	smc_efuse_pkg::fuse_command_resp_t	Input	-	SMC eFuse command interface response.	eFuse shim	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
smc_shadow_regs_o	smc_efuse_pkg::efuse_map_t	Output	-	SMC eFuse shadow register output.	External systems	N/A
pll_clk_obs_i	logic	Input	1	PLL clock observation input.	PLL shim	N/A
pll_clk_obs_en_i	logic	Input	1	PLL clock observation enable.	Control logic	N/A
pvt_process_clk_obs_i	logic	Input	1	PVT process monitor clock observation input.	PVT shim	N/A
pvt_process_clk_obs_en_i	logic	Input	1	PVT process monitor clock observation enable.	Control logic	N/A
cat_therm_i	logic	Input	1	CAT thermal sensor input.	CAT thermal sensor	Tie to '0 if unused
lsio_interface_select_o	logic	Output	[NUM_GPIO_WRAP S-1:0]	LSIO interface selection.	GPIO padring	N/A
pad2core_i	logic	Input	[NUM_GPIO_WRAP S-1:0]	GPIO input data from pad ring to core.	GPIO padring → SMU	N/A
core2pad_o	logic	Output	[NUM_GPIO_WRAP S-1:0]	GPIO output data from core to pad ring.	SMU → GPIO padring	N/A
pad2core_en_o	logic	Output	[NUM_GPIO_WRAP S-1:0]	GPIO pad-to-core path enable.	SMU → GPIO padring	N/A
core2pad_en_o	logic	Output	[NUM_GPIO_WRAP S-1:0]	GPIO core-to-pad path enable.	SMU → GPIO padring	N/A
rst_cool_n_from_pin_i	logic	Input	1	Cool reset input from GPIO pin.	GPIO pin	N/A
clk_telemetry_i	logic	Input	1	Telemetry clock domain for ATB data collection.	Telemetry clock source	N/A
rst_telemetry_n_i	logic	Input	1	Telemetry reset (active low).	Telemetry reset source	N/A
telemetry_atdata_i	telemetry_receiver_pkg::telemetry_data_t	Input	[NUM_TELEMETRY_RECEIVERS-1:0]	ATB telemetry data input array.	ATB sources	N/A
telemetry_atid_i	telemetry_receiver_pkg::atb_id_t	Input	[NUM_TELEMETRY_RECEIVERS-1:0]	ATB telemetry ID input array.	ATB sources	N/A
telemetry_atready_o	logic	Output	[NUM_TELEMETRY_RECEIVERS-1:0]	ATB telemetry ready output.	ATB sources	N/A
telemetry_atvalid_i	logic	Input	[NUM_TELEMETRY_RECEIVERS-1:0]	ATB telemetry valid input.	ATB sources	N/A
telemetry_afvalid_o	logic	Output	[NUM_TELEMETRY_RECEIVERS-1:0]	ATB flush valid output.	ATB sources	N/A
telemetry_afready_i	logic	Input	[NUM_TELEMETRY_RECEIVERS-1:0]	ATB flush ready input.	ATB sources	N/A
cluster_ded_o	logic	Output	1	Cluster double error detection output.	External systems	N/A
wdt_first_timeout_o	logic	Output	1	First watchdog timeout signal.	Reset unit	N/A
wdt_second_timeout_o	logic	Output	1	Second watchdog timeout signal.	External systems	N/A
smc_region_size_o	logic	Output	[31:0]	SMC address region size output.	External systems	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
ext_interrupts_i	logic	Input	[Cfg.NUM_INT_T O_SMC-1:0]	External interrupt inputs to SMC.	External interrupt sources	Tie to '0 if unused
fuse_sense_done_o	logic	Output	1	SMC fuse sense completion output.	Memory repair logic	N/A
fuse_reset_n_delayed_o	logic	Output	1	Delayed fuse reset output.	External systems	N/A
skip_mem_repair_o	logic	Output	1	Skip memory repair output.	Memory repair logic	N/A
ext_boot_seq_done_i	logic	Input	1	External boot sequence completion input (memory repair and shadow register override done); gates reset release.	External boot/repair controller	N/A
temp_interrupt_i	logic	Input	1	Temperature interrupt input.	PVT wrap	Tie to '0 if unused
lc_state_o	logic	Output	[2*LC_STATE_WIDTH-1:0]	Lifecycle state output (driven by SEP when SEP=1, else 8'hf0).	External systems	N/A
lc_sigint_err_o	logic	Output	1	Lifecycle signal integrity error.	External systems	N/A
ras_bank_chip_o	logic	Output	[3:0]	RAS bank chip selection output.	RAS system	N/A
ras_bank_instance_o	logic	Output	[3:0]	RAS bank instance selection output.	RAS system	N/A
ndmreset_request_i	logic	Input	[CPU_CLUSTER_COUNT-1:0]	Non-debug-module reset requests.	Debug infrastructure	Tie to '0 if unused
ndmreset_process_o	logic	Output	[CPU_CLUSTER_COUNT-1:0]	Non-debug-module reset process indicators.	Debug infrastructure	N/A
ext_mailbox_interrupts_o	logic	Output	[NUM_MAILBOXES-1:0]	External mailbox interrupt outputs.	External targets	N/A
cfg_flr_pf_active_i	logic	Input	1	Function Level Reset (FLR) PF active signal.	PCIe / system	Tie to '0 if unused
isolate_req_o	logic	Output	[31:0]	Isolation request output for 32 subsystems.	Subsystems	N/A
ss_reset_completion_i	logic	Input	[31:0]	Subsystem reset completion status.	Subsystems	32'hFFFFFFFF
ss_config_o	logic	Output	[31:0]	Subsystem configuration output.	Subsystems	N/A
ss_reset_ctrl_o	smc_reset_unit_pkg::reset_ctrl_t	Output	[31:0]	Subsystem reset control output array.	Subsystems	N/A
sync_irq_o	logic	Output	1	Software-controlled global synchronisation bit (SYNC_REG.sync in the SMC reset unit). Not an interrupt aggregate output.	External systems	N/A
gpio_interrupt_o	logic	Output	[NUM_GPIO_WRAPS-1:0]	Raw GPIO interrupt vector from the padding, one bit per GPIO wrap (bonded + unbonded). Passed through from SMC. Software reads GPIO status registers to identify the interrupt source.	SMC → external consumers	N/A, leave floating if unused
uart_interrupt_o	logic	Output	[NUM_UART-1:0]	Raw UART interrupt outputs, one bit per UART instance, in the peripheral clock domain. Passed through from SMC.	SMC → external consumers	N/A, leave floating if unused
rom_intf_req_o	rom_req_t	Output	-	SMC ROM memory interface request.	Boot ROM	N/A

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
rom_intf_rsp_i	rom_rsp_t	Input	-	SMC ROM memory interface response.	Boot ROM	N/A
scratch_ram_intf_req_o	scratch_ram_req_t	Output	[31:0]	SMC scratch RAM interface requests.	Scratch RAM banks	N/A
scratch_ram_intf_rsp_i	scratch_ram_rsp_t	Input	[31:0]	SMC scratch RAM interface responses.	Scratch RAM banks	N/A
l1_icache_tag_intf_req_o	l1_icache_tag_req_t	Output	[3:0]	SMC L1 I-cache tag interface requests.	I-cache tag banks	N/A
l1_icache_tag_intf_rsp_i	l1_icache_tag_rsp_t	Input	[3:0]	SMC L1 I-cache tag interface responses.	I-cache tag banks	N/A
l1_icache_data_intf_req_o	l1_icache_data_req_t	Output	[7:0]	SMC L1 I-cache data interface requests.	I-cache data banks	N/A
l1_icache_data_intf_rsp_i	l1_icache_data_rsp_t	Input	[7:0]	SMC L1 I-cache data interface responses.	I-cache data banks	N/A
l1_dcache_tag_intf_req_o	l1_dcache_tag_req_t	Output	[3:0]	SMC L1 D-cache tag interface requests.	D-cache tag banks	N/A
l1_dcache_tag_intf_rsp_i	l1_dcache_tag_rsp_t	Input	[3:0]	SMC L1 D-cache tag interface responses.	D-cache tag banks	N/A
l1_dcache_data_intf_req_o	l1_dcache_data_req_t	Output	[3:0]	SMC L1 D-cache data interface requests.	D-cache data banks	N/A
l1_dcache_data_intf_rsp_i	l1_dcache_data_rsp_t	Input	[3:0]	SMC L1 D-cache data interface responses.	D-cache data banks	N/A
disable_sram_auto_init_i	logic	Input	1	Disable SRAM automatic initialization.	Control logic	1'b0
init_mem_done_o	logic	Output	1	Memory initialization completion signal.	Control logic	N/A
chiplet_is_primary_i	logic	Input	1	Chiplet is primary (for System Timer OCTS).	GPIO padring straps	N/A
timer_count_o	logic	Output	[63:0]	64-bit system timer count output (OCTS).	External systems	N/A
trace_mem_req_o	dfd_trace_mem_pkg::SinkMemPktIn_s	Output	[TRC_RAM_INSTANCES-1:0]	Trace memory interface requests.	Trace memory	N/A
trace_mem_resp_i	dfd_trace_mem_pkg::SinkMemPktOut_s	Input	[TRC_RAM_INSTANCES-1:0]	Trace memory interface responses.	Trace memory	N/A
test_en_i	logic	Input	1	DFT/scan test-enable. Drives clock-gater test ports and AXI cell test inputs.	Test controller	1'b0
scan_rst_ni	logic	Input	1	Scan reset (active-low). Bypasses reset synchronizers during DFT.	Test controller	1'b1
captured_straps_i	logic	Input	[63:0]	Captured strap configuration. Please refer to the reset_unit STRAPS_LO and STRAPS_HI register descriptions for the decoded value of the straps.	GPIO padring	N/A
mem_repair_done_i	logic	Input	1	Memory repair done indicator.	DFT controller	1'b0
mem_repair_success_i	logic	Input	1	Memory repair success indicator.	DFT controller	1'b0
mem_repair_abort_i	logic	Input	1	Memory repair abort indicator.	DFT controller	1'b0
mbist_done_i	logic	Input	1	MBIST done indicator.	DFT controller	1'b0
mbist_pass_i	logic	Input	1	MBIST pass indicator.	DFT controller	1'b0
mbist_abort_i	logic	Input	1	MBIST abort indicator.	DFT controller	1'b0

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
sep_sram_req_o	sep_pkg::sep_sram_req_t	Output	-	SEP scratch SRAM request passthrough.	SEP SRAM macros	N/A
sep_sram_rsp_i	sep_pkg::sep_sram_rsp_t	Input	-	SEP scratch SRAM response passthrough.	SEP SRAM macros	N/A
sep_boot_rom_req_o	sep_pkg::sep_sram_req_t	Output	-	SEP boot ROM request passthrough.	SEP ROM macro	N/A
sep_boot_rom_rsp_i	sep_pkg::sep_sram_rsp_t	Input	-	SEP boot ROM response passthrough.	SEP ROM macro	N/A
sep_cpu_tcm_req_o	sep_pkg::sep_cpu_tcm_req_t	Output	-	SEP CPU TCM request passthrough (ICCM + DCCM).	SEP SRAM macros	N/A
sep_cpu_tcm_rsp_i	sep_pkg::sep_cpu_tcm_rsp_t	Input	-	SEP CPU TCM response passthrough.	SEP SRAM macros	N/A
sep_efuse_bank_ctrl_req_o	sep_efuse_pkg::efuse_axil_req_t	Output	-	SEP eFuse bank control AXI-Lite request passthrough.	SEP eFuse shim	N/A
sep_efuse_bank_ctrl_rsp_i	sep_efuse_pkg::efuse_axil_rsp_t	Input	-	SEP eFuse bank control AXI-Lite response passthrough.	SEP eFuse shim	N/A
sep_efuse_shim_command_req_o	sep_efuse_pkg::fuse_command_req_t	Output	-	SEP eFuse command interface request passthrough.	SEP eFuse shim	N/A
sep_efuse_shim_command_rsp_i	sep_efuse_pkg::fuse_command_rsp_t	Input	-	SEP eFuse command interface response passthrough.	SEP eFuse shim	N/A
sep_crypto_pka_imem_sram_req_o	sep_crypto_pkg::sep_crypto_pka_imem_sram_req_t	Output	-	SEP OTBN PKA instruction memory SRAM request passthrough.	SRAM macro	N/A
sep_crypto_pka_imem_sram_rsp_i	sep_crypto_pkg::sep_crypto_pka_imem_sram_rsp_t	Input	-	SEP OTBN PKA instruction memory SRAM response passthrough.	SRAM macro	N/A
sep_crypto_pka_dmem_sram_req_o	sep_crypto_pkg::sep_crypto_pka_dmem_sram_req_t	Output	-	SEP OTBN PKA data memory SRAM request passthrough.	SRAM macro	N/A
sep_crypto_pka_dmem_sram_rsp_i	sep_crypto_pkg::sep_crypto_pka_dmem_sram_rsp_t	Input	-	SEP OTBN PKA data memory SRAM response passthrough.	SRAM macro	N/A
sep_km_rom_mem_req_o	km_intf_pkg::km_rom_mem_req_t	Output	-	SEP Key Manager ROM memory request passthrough.	ROM macro	N/A
sep_km_rom_mem_rsp_i	km_intf_pkg::km_rom_mem_rsp_t	Input	-	SEP Key Manager ROM memory response passthrough.	ROM macro	N/A
sep_km_sram_mem_req_o	km_intf_pkg::km_sram_mem_req_t	Output	-	SEP Key Manager SRAM memory request passthrough.	SRAM macro	N/A
sep_km_sram_mem_rsp_i	km_intf_pkg::km_sram_mem_rsp_t	Input	-	SEP Key Manager SRAM memory response passthrough.	SRAM macro	N/A
spi_irq_i	logic	Input	1	SPI interrupt request into the SEP PIC, driven by whichever SPI controller the integration selects.	SPI controller	Tie to '0 if unused
sep_external_req_o	sep_pkg::sep_32_64_12_axi_req_t	Output	-	SEP AXI extension port request passthrough.	Adopter peripherals	N/A
sep_external_rsp_i	sep_pkg::sep_32_64_12_axi_rsp_t	Input	-	SEP AXI extension port response passthrough.	Adopter peripherals	Tie to DECERR if unused
sep_cpu_trace_o	sep_pkg::sep_cpu_trace_t	Output	-	SEP CPU trace output.	Trace infrastructure	N/A
sep_lockstep_ctrl_i	sep_pkg::sep_lockstep_ctrl_t	Input	-	SEP CPU lockstep controls, passed through to the SEP. Inert unless the SEP core is built with RV_LOCKSTEP_ENABLE .	Chip-level RAS control	Tie to '0 if unused

Interface Name	Type	Direction	Width	Description	Connection	Default value (if applicable)
sep_lockstep_status_o	sep_pkg::sep_lockstep_status_t	Output	-	SEP CPU lockstep status. Reads 0 when the SEP is absent (SEP=0) or the core is built without RV_LOCKSTEP_ENABLE.	Chip-level RAS aggregation	N/A
lcc_demote_state_1_o	logic	Output	[1:0]	SEP lifecycle demote state 1.	External lifecycle logic	N/A
lcc_demote_state_2_o	logic	Output	[1:0]	SEP lifecycle demote state 2.	External lifecycle logic	N/A
sep_fuse_sense_done_o	logic	Output	1	SEP fuse sense done output.	System integration	N/A
clk_sep_wdt_i	logic	Input	1	SEP watchdog timer clock passthrough.	Low-frequency clock source	N/A
sep_straps_i	sep_pkg::sep_straps_t	Input	-	SEP configuration straps passthrough.	GPIO padding / tie-off	N/A

Chapter 2. System Management Network (SMN) Integration

WARNING

This section is currently under development. Content will be added to describe SMN integration procedures, network topology, and management interfaces.

Chapter 3. User IP Integration

This chapter describes how to add adopter-owned IP to a chiplet containing the Open Chiplet Atlas Harness (OCAH). It covers three distinct attachment classes: management peripherals, security extensions, and application logic in the user subsystem. Choosing the correct class matters because connecting an IP to an OCAH bus does not automatically make it trusted, securely booted, lifecycle-controlled, or protected from other initiators. [REQ-UIP-MOD-1] The integrator shall define those properties and provide the fabric, firewalls, IOMMU, secure-boot handoff, and physical implementation needed to enforce them.

OCAH Phase 1 does not contain the general-purpose System Management Network (SMN), die-to-die fabric, or user-subsystem interconnect shown in the system block diagram. The application subsystem remains an adopter-owned black box. Consequently, the application path in this chapter is an integration contract, not a drop-in RTL port on `smu.sv`. The adopter owns the application fabric, address decode, clock and power domains, and application software stack; OCAH provides management and security services around that subsystem.

3.1. Select the trust and attachment model

First decide whether SMC firmware owns the IP as part of the management domain, SEP firmware owns it inside or in direct support of the security boundary, or application firmware or an OS owns it in the adopter application domain. An application-domain IP remains outside the OCAH trusted computing base unless a threat-model review explicitly places it inside. This classification determines the correct CSR, interrupt, data, boot, and debug paths; the classes are not interchangeable.

Table 12. User IP attachment classes

Class	Runtime and trust owner	CSR connection	DMA or data connection	Interrupt destination
SMC-managed peripheral	SMC firmware; management-plane policy	The supplementary portion of the SMC adopter peripheral window, <code>smc_external_req_o</code> / <code>smc_external_resp_i</code>	Usually none; attach any independent manager to an adopter-owned fabric	An assigned bit of SMU <code>ext_interrupts_i</code>
SEP security extension	SEP firmware; inside or directly supporting the security boundary	The SEP external region through <code>sep_external_req_o</code> / <code>sep_external_resp_i</code>	The SEP extension port is a manager-to-subordinate path. Any extension DMA manager uses a separately contained adopter-fabric connection	An assigned bit of <code>sep_extintsrc_req_i</code>
Application-owned accelerator	Application firmware or OS; outside the OCAH trusted computing base unless a threat-model review explicitly places it inside	An adopter SoC CSR fabric	The IP's AXI manager connects through an IOMMU or equivalent firewall to the adopter memory fabric	The application interrupt controller; optionally also an assigned bit of SMU <code>ext_interrupts_i</code> for management or fault notification

Before selecting a class, record:

- the code, configuration, keys, model weights, and user data handled by the IP;
- which of those assets require integrity, confidentiality, or anti-rollback;
- whether the IP and its supply chain are trusted;
- which boot stage authenticates and releases the IP;
- which initiators may access its CSRs and memory;

- the least-privilege AXI security attributes and source identity it receives;
- its debug, scan, test, and production-lifecycle policy; and
- where security and fatal faults are reported.

The OCAH SMC and SEP inbound/outbound filters protect their own traffic paths. They do not automatically protect an IP attached to the adopter fabric. [REQ-UIP-MOD-2] The integrator shall use default-deny hardware policy at every new trust boundary and prevent less trusted software or the IP itself from rewriting or spoofing that policy.

[REQ-UIP-MOD-3] The integrator shall not connect an adopter DMA manager to `smu_axi_out_req_o`. That port is an OCAH **manager output**: it carries transactions initiated by the SMC or SEP to the adopter fabric. A user IP's DMA interface is another manager on that fabric.



The SMU inbound AXI port is also not a route from an external manager to an arbitrary user-subsystem target. Its internal crossbar routes inbound traffic only to the programmed SMC and SEP apertures. Unmatched inbound requests return a decode error.

3.2. Integrate an SMC-managed peripheral

[REQ-UIP-SMC-1] The integrator shall use this model only for product-specific control and status logic that SMC firmware configures or monitors, such as a sensor, pad controller, or small management engine.

3.2.1. Allocate the SMC supplementary window

`smc_external_req_o` and `smc_external_resp_i` form a 32-bit AXI4-Lite request/response pair. The complete adopter peripheral window starts at SMC local `BASE + 0x040_0000` and is 4 MiB. The mandatory region contains the technology-specific GPIO, PLL, PVT, and eFuse SHIM blocks. Optional adopter IP belongs in the supplementary region beginning at local `BASE + 0x040_4000`.

The as-shipped `smc_ip_integration.sv` decoder applies modulo-0x4000 address folding (`addr % ExtWindowSize`) before comparing against the mandatory-region device offsets (GPIO at `0x1100`, PLL at `0x2000`, PVT at `0x3000`). A supplementary-region address with a low-12-bit offset that falls within a mandatory device range will alias to that device. [REQ-UIP-SMC-2] The adopter shall replace or extend the decoder in `hw/top/smc_ip_integration.sv` so that supplementary-region addresses are not subject to modulo folding before mandatory-device comparison. The modulo decoder shall not remain in place in front of the supplementary region.

[REQ-UIP-SMC-3] The downstream decoder shall reserve a naturally aligned range large enough for the IP register map. [REQ-UIP-SMC-4] The downstream decoder shall reject overlaps with mandatory blocks and other adopter allocations. [REQ-UIP-SMC-5] The downstream decoder shall route both read and write channels to the same subordinate. [REQ-UIP-SMC-6] The downstream decoder shall return `DECERR` for an unallocated address and propagate an IP error as `SLVERR`. [REQ-UIP-SMC-7] The integrator shall keep the allocation in the chiplet address-map source of truth used by RTL, firmware, verification, and documentation.

The SMC `GLOBAL_BASE` and `REGION_SIZE` CSRs describe the complete SMC aperture; they are not per-IP aperture registers. Adding a peripheral to the supplementary window does not create another OCAH aperture CSR.

3.2.2. Adapt the register protocol

No adapter is needed for an IP with a compatible 32-bit AXI4-Lite subordinate. For a standard OCAH/OpenTitan-style TL-UL subordinate, the repository provides `hw/common/axi/axi_lite_to_tlul.sv`. It accepts packed AXI request/response types and produces packed `tlul_pkg::tl_h2d_t` / `tlul_pkg::tl_d2h_t` values, including TL-UL command and data integrity.

That converter is not a drop-in adapter for an IP exposing a flattened subset of TL-UL. In that case, use an IP-supplied wrapper or add an explicit shim and verify every field, response, byte-enable, and error mapping. The [NPU worked example](#) uses the NPU's own AXI4-Lite wrapper for this reason.

3.2.3. Route the interrupt

Assign each SMC-visible source a bit of SMU `ext_interrupts_i`, whose integration width is set by the `NUM_INT_TO_SMC` configuration parameter. The SMC synchronizes this vector into its clock domain. Masking the interrupt does not clear its source, and an un-stretched one-cycle pulse may be lost across this boundary.

[REQ-UIP-SMC-8] A source asynchronous to the SMC clock shall hold the interrupt asserted until SMC firmware reads and clears the status register in the peripheral using the peripheral-defined clear operation such as W1C.

[REQ-UIP-SMC-9] The integrator shall record the bit assignment in the platform interrupt map and use the same value in RTL, SMC firmware, and the verification environment.

[REQ-UIP-SMC-10] `sep_extintsrc_req_i` is a separate SEP interrupt input and shall not be substituted for `ext_interrupts_i`.

3.3. Integrate a SEP security extension

Use this model only for logic that SEP firmware must own as part of, or in direct support of, the chiplet security boundary. Examples include a security-policy engine or an adopter cryptographic service. [REQ-UIP-SEP-1] The integrator shall not place an ordinary application accelerator behind the SEP extension port merely to make it appear trusted.

3.3.1. Allocate the SEP external region

`sep_external_req_o` and `sep_external_resp_i` expose the SEP local crossbar's 32-bit-address, 64-bit-data AXI4 extension target at local addresses `0x2000_0000` through `0x3fff_ffff`. The first four bytes at `0x2000_0000` are intercepted by `sep.sv` for the eFuse SHIM. [REQ-UIP-SEP-2] The adopter decoder shall not allocate any block overlapping `0x2000_0000--0x2000_0003`. Use `hw/sys/sep/dv/models/regs/sep_external.rdl` and the generated `OCH_SEP_TOP_SEP_EXTERNAL_EFUSE_SHIM_CTRL_BASE_ADDR` and `OCH_SEP_TOP_SEP_EXTERNAL_EFUSE_SHIM_CTRL_SIZE` constants in `hw/sys/sep/regs/gen/sv/sep_addrmap_pkg.sv` as the machine-readable source of truth; the map may evolve independently of this guide. [REQ-UIP-SEP-3] The adopter decoder shall reserve naturally aligned, non-overlapping ranges in the remainder and return `DECERR` for holes.

The as-shipped `sep_ip_integration.sv` terminates `axi_extension_axi_req_i` / `axi_extension_axi_resp_o` with an `axi_err_slv` instance that returns `DECERR` for all transactions. These signals are module-level ports. [REQ-UIP-SEP-4] To use the SEP extension region, the adopter shall modify `hw/top/sep_ip_integration.sv` to replace the `u_axi_extension_err_slv` instance with their own decoder. An outer wrapper module cannot override an internal

instance connection; the adopter must edit or replace the integration file directly.

The port is a transaction path from SEP-local initiators to adopter subordinates. It is not a connection for an adopter IP's DMA manager. The SEP CPU LSU/debug paths, SEP DMA, and the filtered system-interface path have crossbar connectivity to the extension target. External system access remains subject to the SEP inbound filter, which is block-by-default after reset. [REQ-UIP-SEP-4a] SEP firmware shall open the inbound filter deliberately; it is not unlocked by hardware default.

[REQ-UIP-SEP-5] The integrator shall preserve the AXI ID, **PROT**, and user/source information carried by the packed SEP request type. [REQ-UIP-SEP-6] When adapting to another protocol or a narrower data path, the integrator shall verify the adapter explicitly. [REQ-UIP-SEP-7] The integrator shall cover burst conversion, byte strobes, ordering, response IDs, and error propagation.

3.3.2. Apply SEP security policy

Placing a block behind the SEP extension port only changes reachability; it does not prove that the block is trustworthy. [REQ-UIP-SEP-8] The security design shall include executable or mutable extension state in the appropriate verified-boot and anti-rollback policy. [REQ-UIP-SEP-9] The security design shall prevent non-secure or incorrectly identified initiators from reaching protected registers. [REQ-UIP-SEP-10] The security design shall exclude roots, keys, and integrity-critical state from scan, and classify any secret-consuming state under the SEP scan policy. [REQ-UIP-SEP-11] The security design shall define zeroization and reset behavior for sensitive state. [REQ-UIP-SEP-12] The integrator shall constrain any extension DMA manager independently at the adopter fabric. [REQ-UIP-SEP-13] The security design shall route integrity, tamper, and fatal errors to a security or safety handler that cannot be masked by application software.

[REQ-UIP-SEP-14] The integrator shall assign each SEP-visible interrupt an allocated bit of **sep_extintsrc_req_i**. [REQ-UIP-SEP-15] The integrator shall synchronize every asynchronous **sep_extintsrc_req_i** source to the SEP clock domain before the SEP boundary. **sep_sv** concatenates **extintsrc_req** into the interrupt vector combinationally with no internal synchronizer; metastability is the integrator's responsibility. [REQ-UIP-SEP-15a] A pulse interrupt source shall additionally use level retention, a toggle handshake, or a pulse stretcher to prevent event loss across the synchronizer. [REQ-UIP-SEP-16] The integrator shall not route a SEP interrupt source to SMU **ext_interrupts_i** unless SMC also has an independent management reason to receive the event.

3.4. Integrate an application-owned accelerator

Use this model when an application processor owns the command queue, buffers, and normal interrupt service for an accelerator.

NOTE

Validation status: this section is integration guidance. OCAH Phase 1 provides neither the application fabric/SMN nor an application-owned accelerator attachment, so the complete path described here has not been implemented or exercised in OCAH simulation. [REQ-UIP-APP-1] Adopters shall record product-level RTL, CDC/RDC, simulation, firmware, and security-review evidence.

3.4.1. Connect control and DMA independently

Place the CSR subordinate on the adopter's application CSR fabric. In a future full-harness integration this fabric may be the SMN shown in the OCAH system architecture; Phase 1 supplies no SMN RTL. [REQ-UIP-APP-2] The integrator shall connect the DMA manager through a trusted ingress point to the adopter memory fabric as a peer of the application CPU and the OCAH outbound manager. [REQ-UIP-APP-3] The fabric shall supply any required address-width, data-width, ID-width, clock-domain, security, and coherency adapters.

For a manager with an address width narrower than the fabric, zero-extend its addresses. Zero-extension restricts the manager to the low part of the physical address space; it is not address translation or security containment.

[REQ-UIP-APP-4] The integrator shall treat a third-party or application-owned DMA engine as untrusted by default. [REQ-UIP-APP-5] The integrator shall place an IOMMU or equivalent address-and-permission firewall between the DMA engine and memory. [REQ-UIP-APP-5a] The integrator shall assign a non-spoofable stream/source identity to the DMA engine at the trusted fabric ingress. [REQ-UIP-APP-5b] The integrator shall configure the firewall to permit only the buffers and operations required for the current workload. [REQ-UIP-APP-5c] An integration may omit containment only when a threat-model review explicitly accepts the IP inside the relevant trust boundary or proves the hardware connection cannot reach another asset.

[REQ-UIP-APP-6] If the IP does not generate AXI **AxPROT** or fabric user/source fields, the trusted adapter shall supply least-privilege values. Application IP normally originates non-secure, unprivileged data transactions (**AxPROT** = **3'b010**). [REQ-UIP-APP-7] The integrator shall not let an untrusted IP assert secure privilege or choose its own authenticated source ID. [REQ-UIP-APP-8] The integrator shall validate unsupported transaction types, including atomics and exclusive accesses, at the ingress.

[REQ-UIP-APP-9] The system address map shall define the CPU-visible CSR aperture independently. [REQ-UIP-APP-10] The system address map shall define the DMA-visible memory aperture independently. [REQ-UIP-APP-11] The system address map shall state whether the SMC or SEP can also reach the CSR aperture through **smu_axi_out_req_o**. [REQ-UIP-APP-12] The system address map shall state which initiators and security states are permitted by the adopter fabric. [REQ-UIP-APP-13] The system address map shall state the IOMMU/firewall context and non-spoofable source identity. [REQ-UIP-APP-14] The system address map shall state which security policy registers become locked before the IP is released.

3.4.2. Route application and management interrupts

Route completion and recoverable work errors to the application interrupt controller. A separate fault signal, or an intentional duplicate of a retained level interrupt, may also be assigned to SMU **ext_interrupts_i** for SMC management or fault recovery.

[REQ-UIP-APP-15] The integrator shall not join two independently cleared interrupt controllers to a pulse. [REQ-UIP-APP-16] The integrator shall define the ownership of status acknowledgement and masking so one handler cannot hide an event from the other.

3.5. Integrate secure boot, lifecycle, debug, and test

3.5.1. Establish a trusted release sequence

[REQ-UIP-BOOT-1] The integrator shall hold application IP in reset while the chiplet root of trust is establishing policy. [REQ-UIP-BOOT-1a] The integrator shall keep the IP's bus-master capability blocked while the chiplet root of trust is establishing policy.

Before enabling transactions:

1. [REQ-UIP-BOOT-2] SEP shall authenticate every executable or mutable configuration item required by the product threat model, check its security version, and extend the measurement into attestation when required.
2. [REQ-UIP-BOOT-3] Trusted firmware shall configure the CSR firewall, DMA stream identity, IOMMU/firewall ranges, interrupt ownership, and fault routing.
3. [REQ-UIP-BOOT-4] Trusted firmware shall lock policy that application software and the IP shall not change.

4. [REQ-UIP-BOOT-5] SMC shall complete the power, clock, reset, and isolation sequence.
5. [REQ-UIP-BOOT-6] The secure-management handoff shall record success before the application driver enables queues or DMA.

OCAH does not provide a generic manifest loader or release API for arbitrary user IP. [REQ-UIP-BOOT-7] The adopter shall extend the verified-boot policy and trusted firmware for the selected IP, or integrate an equivalent platform root-of-trust service, then verify that implementation as part of the product security case.

The authenticated set is product-specific. For an accelerator it may include microcode, firmware, register presets, bitstreams, or integrity-sensitive model metadata. Bulk user data need not be part of verified boot, but it still needs runtime access control. A hardwired accelerator with no mutable executable state still requires identity/version checks and containment before release.

[REQ-UIP-BOOT-8] The release sequence shall fail closed. [REQ-UIP-BOOT-9] An authentication, rollback, policy-lock, or firewall setup failure shall leave the IP reset or bus-blocked. [REQ-UIP-BOOT-10] An authentication, rollback, policy-lock, or firewall setup failure shall produce a diagnosable security event. [REQ-UIP-BOOT-11] Application software shall not be able to convert that failure into a release.

3.5.2. Apply lifecycle and scan policy

Classify adopter IP state using the SEP lifecycle asset model:

- Class 1 confidentiality or integrity assets are excluded from scan;
- Class 2 secret-consuming state is placed on a dedicated chain that is never scan-dumpable and observes the secure-test rules; and
- Class 3 general logic is accessible only when the applicable owner and lifecycle debug policy permit it.

The SEP lifecycle architecture reserves adopter bits in its feature-control vector for additional debug, test, and functional gates. The bit-allocation convention is currently open. The current SMU boundary exposes `lc_state_o`, an 8-bit differential encoding of the four-bit lifecycle state, and `secure_tm_o`, a single-bit secure-test-mode indication, but not the complete feature-control vector. An IP that needs lifecycle-controlled enables therefore requires an explicitly reviewed extension to the integration wrapper and lifecycle/RDL allocation. [REQ-UIP-BOOT-12] An IP that needs lifecycle-controlled enables shall not infer an ad-hoc debug authorization from `lc_state_o` alone.

`secure_tm_o` indicates secure test mode; it is not a general functional-access authorization. [REQ-UIP-BOOT-13] The integrator shall document how the IP's DTP/scan path, test enables, JTAG-visible registers, SRAM test, and secret zeroization behave in every production, development, RMA, and secure-test condition used by the product.

3.6. Integrate clocks, reset, and power

3.6.1. Single-clock IP

An IP may use an existing adopter functional clock when its timing target and power state match that domain. [REQ-UIP-CLK-1] The integrator shall generate an asynchronous-assert, synchronous-deassert reset for that clock. [REQ-UIP-CLK-2] The integrator shall not use an SMC-domain reset output as a synchronous reset in an unrelated clock domain without a reset-domain adapter.

3.6.2. Independent-clock IP

When the user IP runs at a different frequency, the integrator shall place CDC logic at every boundary. [REQ-UIP-CLK-3] The integrator shall use an asynchronous AXI bridge or correctly sized channel FIFOs for the CSR or DMA interface. [REQ-UIP-CLK-4] The integrator shall synchronize retained status levels across the clock boundary. [REQ-UIP-CLK-5] The integrator shall transfer pulses with a toggle, handshake, or pulse stretcher. [REQ-UIP-CLK-6] The integrator shall constrain every asynchronous clock relationship in the SoC timing-constraint file. [REQ-UIP-CLK-7] The integrator shall run static CDC and reset-domain-crossing analysis on the assembled SoC.

An AXI register slice is not a CDC bridge. [REQ-UIP-CLK-8] All five AXI channels shall obey the same clocking contract.

3.6.3. Power and isolation sequence

OCAH exposes the 32-bit vectors `ss_config_o`, `isolate_req_o`, and `ss_reset_complete_i`, plus `smc_reset_unit_pkg::reset_ctrl_t ss_reset_ctrl_o[31:0]`. The latter is a 32-element unpacked array of reset-control structures, not a plain 32-bit vector; use the field definitions in `smc_reset_unit_pkg` when connecting it. These interfaces provide adopter-defined subsystem lifecycle control.

[REQ-UIP-CLK-9] The adopter shall assign a subsystem index from the `ss_reset_ctrl_o` array to the user IP. [REQ-UIP-CLK-9a] The adopter shall translate the `smc_reset_unit_pkg::reset_ctrl_t` fields for that index into technology-specific reset and power-switch controls. [REQ-UIP-CLK-9b] The adopter shall insert isolation cells on all outputs and retention cells on state that must survive power removal. [REQ-UIP-CLK-9c] The adopter shall define the completion condition that drives `ss_reset_complete_i` for that index.

OCAH does not instantiate the user IP's power controller.

Use this ordering for a switchable user domain:

1. [REQ-UIP-CLK-10] The integrator shall hold the IP in reset, isolate its outputs, and block its manager ports.
2. [REQ-UIP-CLK-11] The integrator shall enable the domain supply and wait for the adopter power-good indication.
3. [REQ-UIP-CLK-12] The integrator shall start and qualify the functional clock. Clock qualification includes confirming PLL lock or equivalent clock-gating stability before the clock is considered clean.
4. [REQ-UIP-CLK-13] The integrator shall configure and lock the trusted fabric ingress, IOMMU/firewall, and fault routing while the IP remains blocked.
5. [REQ-UIP-CLK-14] The integrator shall complete the required authentication and secure-management release checks.
6. [REQ-UIP-CLK-15] The integrator shall deassert the IP reset synchronously to its clock only after release is authorized.
7. [REQ-UIP-CLK-16] The integrator shall complete any IP-local memory initialization.
8. [REQ-UIP-CLK-17] The adopter logic shall assert the assigned `ss_reset_complete_i` bit only when the IP is internally initialized and ready to accept transactions once isolation is removed. SMC firmware polls the `SS_RESET_COMPLETE` CSR (which synchronises `ss_reset_complete_i` through a 3-stage synchronizer) and, when set, clears the applicable isolation request source(s) for that subsystem index. `isolate_req_o[i]` is the OR of three sources: `isolate_req_reg[i]` (per-subsystem software isolation, the normal sequencing path), the pin source when `isolate_req_pinen_reg[i]` is set, and the FLR source when `isolate_req_smcen_reg[i]` and the global `isolate_req_smc_reg` are both set. [REQ-UIP-CLK-17a] All contributing isolation sources for that index shall be inactive before `isolate_req_o[i]` de-asserts. [REQ-UIP-CLK-17b] The adopter logic shall deassert `ss_reset_complete_i` for each subsystem index before that index enters a new reset or power cycle. SMC firmware shall confirm that the `SS_RESET_COMPLETE` CSR reads zero for that index before beginning or accepting completion of a new sequence,

ensuring the three-stage synchronizer has propagated the deassertion low before firmware acts on any new assertion.

9. [REQ-UIP-CLK-18] The adopter logic shall remove isolation clamps only after observing `isolate_req_o[i]` de-asserted for that subsystem index.
10. [REQ-UIP-CLK-19] The integrator shall enable DMA transactions only after isolation clamps are removed and the manager path is explicitly unblocked.

NOTE	The ordering between <code>ss_reset_complete_i</code> assertion, SMC clearing <code>isolate_req_o</code> , and removal of isolation clamps is enforced by firmware convention, not by RTL interlock. REQ-UIP-CLK-18 requires the adopter integration to observe <code>isolate_req_o</code> de-asserted before removing clamps.
-------------	--

Reverse the dependencies when powering down: stop new work, drain or abort DMA, mask interrupts, isolate outputs, assert reset, stop the functional clock, and then remove power. [REQ-UIP-CLK-20] The integrator shall not remove power while an AXI transaction can still return a response.

3.7. Integrate software and firmware

3.7.1. Maintain one platform contract

[REQ-UIP-SW-1] Software-visible allocations shall be generated from, or checked against, the hardware address and interrupt maps. [REQ-UIP-SW-2] The platform contract shall record the CSR base and span. [REQ-UIP-SW-3] The platform contract shall record the interrupt number, trigger type, polarity, and owner. [REQ-UIP-SW-4] The platform contract shall record the DMA address width and accessible ranges. [REQ-UIP-SW-5] The platform contract shall record the DMA stream/source identity, IOMMU context, and permission owner. [REQ-UIP-SW-6] The platform contract shall record the AXI security/privilege attributes and allowed CSR initiators. [REQ-UIP-SW-7] The platform contract shall record the coherency and cache-maintenance policy. [REQ-UIP-SW-8] The platform contract shall record the reset and power-domain index. [REQ-UIP-SW-9] The platform contract shall record the IP configuration or ABI version, authenticated components, minimum security versions, and secure-release indication.

Use the generated `*_addr.h` and register headers under a block's `regs/gen/c/` directory when its register map is supplied as SystemRDL. [REQ-UIP-SW-10] An external IP without SystemRDL shall supply an equivalent reviewed header. [REQ-UIP-SW-11] The integrator shall not infer register offsets from generated SystemVerilog or duplicate unexplained numeric addresses in multiple drivers.

3.7.2. Place initialization in the boot flow

[REQ-UIP-SW-12] For an SMC-managed peripheral, the integrator shall initialize it after the SMC has completed fuse sensing and required memory repair, programmed the SMC/SEP global apertures, and made the peripheral's clock and power domain available, then configure and clear the peripheral before enabling its SMC interrupt source.

[REQ-UIP-SW-13] For a SEP security extension, SEP firmware shall validate the extension identity and version, establish access policy, initialize sensitive state, clear pending status, and enable its SEP interrupt only after the relevant verified-boot stage succeeds.

For an application-owned accelerator, trusted SEP/SMC firmware first performs the authenticated release and containment sequence. [REQ-UIP-SW-14] Application firmware or the OS driver shall verify that secure release succeeded before enabling queues or DMA. [REQ-UIP-SW-15] The driver shall validate the hardware/ABI version against the authenticated minimum before use. [REQ-UIP-SW-16] The driver shall allocate DMA-safe buffers through the assigned IOMMU context. [REQ-UIP-SW-17] The driver shall clear stale interrupt status and register the application

interrupt handler before enabling work. [REQ-UIP-SW-18] A driver shall not substitute its own version check for the secure boot decision.

3.7.3. Handle DMA coherency

[REQ-UIP-SW-19] If the user IP is not coherent with CPU caches, software shall use uncached buffers or perform cache maintenance. [REQ-UIP-SW-20] Before starting device reads, the driver shall clean CPU dirty lines and execute the platform's required write barrier. [REQ-UIP-SW-21] After device writes complete, the driver shall execute the completion/read barrier and invalidate affected CPU cache lines before consuming results. [REQ-UIP-SW-22] The driver shall prevent buffers from being unmapped or reused while DMA is active.

3.7.4. Bound failures and recovery

[REQ-UIP-SW-23] Every wait shall have a timeout. [REQ-UIP-SW-24] On failure, the driver shall capture the IP error registers and fabric response before clearing them. [OPT-UIP-SW-1] Recovery should progress from least to most disruptive. . [OPT-UIP-SW-2] The driver should stop new submissions and mask the IP interrupt first. . [OPT-UIP-SW-3] The driver should acknowledge retained interrupt status. . [OPT-UIP-SW-4] The driver should request an IP-local soft reset, if supported. . [OPT-UIP-SW-5] The integrator should assert the adopter subsystem reset after DMA has quiesced. . [OPT-UIP-SW-6] The integrator should power-cycle the domain only after isolation is asserted.

Document which state survives each reset. A soft reset that leaves a datapath or DMA engine in an error state is not a complete recovery mechanism.

3.8. Worked example: application-owned NPU

This example uses the external NPU as a representative DMA-capable accelerator. The OCAH tree does not contain or redistribute NPU source. The example adapter instantiates the external `npu_axi_csr_wrap`, not the raw `npu_top`. It is an illustrative boundary adapter, not a self-contained public regression: an open-source integrator cannot compile it without separately obtaining the NPU package and wrapper. [REQ-UIP-EX-0] A release-quality reproducible example shall pin an open-source dependency or replace this block with an in-tree reference model.

The example classifies the NPU as application-owned and outside the OCAH trusted computing base. It receives no device roots or secret keys, originates non-secure unprivileged transactions, and remains reset/bus-blocked until trusted firmware has configured its IOMMU/firewall context and authorized release. A product that makes the NPU part of its trusted computing base needs a separate threat-model and verification argument; changing an AXI attribute is not sufficient.

The reference NPU wrapper exposes:

Interface	Role	Integration
<code>s_axil_*</code>	32-bit AXI4-Lite subordinate	[REQ-UIP-EX-1] Connect to the application CSR fabric through a 4 KiB aperture. Rebase the fabric address to a zero-based offset, even though the current NPU register decoder consumes only address bits <code>[11:0]</code> .
<code>m_axi_*</code>	64-bit AXI4 manager	[REQ-UIP-EX-2] Connect through a trusted IOMMU/firewall ingress. Zero-extend <code>m_axi_awaddr_o[31:0]</code> and <code>m_axi_araddr_o[31:0]</code> on a wider fabric, assign a non-spoofable stream ID, and constrain all untranslated addresses below 4 GiB. The adapter supplies <code>AxPROT=3'b010</code> because the NPU wrapper has no protection outputs.
<code>intr_done_o, intr_error_o</code>	Retained level interrupts	[REQ-UIP-EX-3] Route to separate application interrupt sources. Optionally route the error level to an assigned SMU <code>ext_interrupts_i</code> bit for SMC fault handling; a fatal security violation from the trusted ingress shall not depend on the NPU's own interrupt.

Interface	Role	Integration
<code>clk_i, rst_ni</code>	Single NPU clock and asynchronous active-low reset	[REQ-UIP-EX-4] Use an adopter clock/reset domain. Before the adapter input, condition <code>rst_ni</code> to assert asynchronously and deassert synchronously to <code>clk_i</code> . Meet the implementation technology's minimum reset pulse width; the NPU architecture does not prescribe a clock-cycle count. The NPU contains no internal CDC.
<code>security_release_i</code>	Trusted release into the example adapter	[REQ-UIP-EX-5] Keep low until authentication and containment setup succeed. Before the adapter input, synchronize its rising edge to <code>clk_i</code> ; a falling edge may asynchronously force the NPU back into reset. This signal does not replace the external DMA firewall.

The example adapter combines `rst_ni` and `security_release_i` with a combinational AND gate. It contains no synchronizer: the caller is responsible for presenting both rising edges synchronously to `clk_i` so the combined NPU reset deasserts synchronously.

The `security_release_i` signal in the example adapter is an active-high persistent level. It holds the NPU in reset while low; a rising edge deasserts reset and a falling edge re-asserts it. The adapter contains no synchronizer: the caller is responsible for synchronizing the rising edge to `clk_i` before asserting. A falling edge may be asynchronous and will immediately re-assert reset.

The software-visible counterpart is `platform_npu_security_release_confirmed()`, which the adopter shall implement as a read of a firmware-written, application-read-only status word. Trusted SEP or SMC firmware writes this word as part of the authenticated release sequence and application software shall not be able to manufacture or overwrite it. The hardware `security_release_i` and the software `platform_npu_security_release_confirmed()` shall share a common source: the same firmware release step that asserts `security_release_i` shall also write the software-readable status word, so the two signals are coherent.

[REQ-UIP-EX-6] The integrator shall implement `platform_npu_security_release_confirmed()` as a read of a firmware-written application-read-only status word. [REQ-UIP-EX-7] The hardware `security_release_i` and the software release-confirmed indication shall be written by the same firmware release step. [REQ-UIP-EX-8] `security_release_i` shall remain asserted for the lifetime of the NPU operating session; de-assertion shall re-assert reset and the software release indication shall be cleared atomically or before `security_release_i` de-asserts. [REQ-UIP-EX-9] The integrator shall ensure that every AXI transaction to the NPU CSR aperture while `npu_rst_ni` is asserted receives a complete, bounded error response. Specifically:

- One accepted AR shall produce exactly one R response with `RRESP = SLVERR` or `RRESP = DECERR`. Response latency is measured from the AR handshake to assertion of RVALID and does not include the manager's RREADY de-assertion time.
- One accepted AW combined with its accepted W (collected in either legal order) shall produce exactly one B response with `BRESP = SLVERR` or `BRESP = DECERR`. A subordinate shall not issue B after accepting only AW if W has not yet arrived. Response latency is measured from the later of the AW and W handshakes to assertion of BVALID.
- If NPU reset asserts after only one half of a write handshake has completed, the integration shall use one of two designs: (a) a boundary element outside `npu_rst_ni` that remains operational while the NPU resets, accepts the missing AW or W channel, and returns B independently of the NPU state; or (b) a coherent AXI interface reset that legally discards the partial transaction at both ends. The boundary shall not delay assertion of NPU reset waiting for the missing AXI channel; security reset shall be immediate.
- The maximum response latency from the defining handshake to RVALID or BVALID assertion shall be defined as a platform-contract parameter and enforced; an unbounded wait is not permitted.

[REQ-UIP-EX-10] The NPU's mutable executable/configuration surface shall be established for the selected product configuration. [REQ-UIP-EX-11] If it loads microcode or other executable state, the integrator shall authenticate it and enforce a minimum security version before release. [REQ-UIP-EX-12] If descriptors or model metadata influence a security decision, the integrator shall authenticate those items under the corresponding policy. Ordinary application

tensors can remain runtime data, protected by the IOMMU/firewall rather than measured boot.

The complete adapter is in `doc/integrator/examples/npu/ocah_npu_fabric_adapter.sv`. The significant address connection is:

```
logic [31:0] npu_dma_awaddr;
logic [31:0] npu_dma_araddr;

// Sized casts zero-extend these unsigned 32-bit addresses.
assign m_axi_awaddr_o = FabricAddrWidth'(npu_dma_awaddr);
assign m_axi_araddr_o = FabricAddrWidth'(npu_dma_araddr);

// The external wrapper supplies no AxPROT; use least privilege at this ingress.
assign m_axi_awprot_o = 3'b010; // data, non-secure, unprivileged
assign m_axi_arprot_o = 3'b010;
```

The example allocation in `npu_platform.h` is illustrative, not an OCAH fixed map:

Resource	Example assignment	Requirement
NPU CSRs	<code>0x3000_0000-0x3000_0fff</code>	4 KiB aligned application CSR aperture
DMA buffer memory	<code>0x4000_0000-0x4007_ffff</code>	Uncached 512 KiB range below 4 GiB; the IOMMU/firewall permits this range and rejects adjacent addresses
Done interrupt	Application IRQ 42	Retained level; W1C at NPU <code>INTR_STATE[0]</code>
Error interrupt	Application IRQ 43	Retained level; W1C at NPU <code>INTR_STATE[1]</code>

[REQ-UIP-EX-13] The platform integration shall review and either replace or explicitly adopt the example address assignments. The integration shall establish a correct DMA address contract that distinguishes the following address domains and mapping state, even when their numeric values happen to coincide:

- the CPU virtual address used by software to access the buffer;
- the backing physical pages to which that virtual address resolves;
- the device-visible DMA address programmed into `NPU_DMA_EXT_BASE` (an IOVA when an IOMMU translates device transactions, or a physical address in identity-mapped systems); and
- for dynamic IOMMU mappings, a platform-specific mapping token (a DMA address, mapping handle, or equivalent opaque record, depending on the OS or firmware API) that the platform shall keep valid for the entire duration of the DMA operation. A static bare-metal platform with no runtime IOMMU mapping may substitute a persistent platform configuration record that serves the same role.

When an IOMMU is present, the mapping shall ensure the device-visible DMA address resolves to the exact same byte range within the same backing physical pages as the CPU virtual address, with transfer direction and permissions that are no more permissive than the operation requires.

NOTE

The example `npu_smoke.c` passes the same `buffer` value to `platform_dma_prepare()` (CPU-side cache maintenance) and to `npu_write(NPU_DMA_EXT_BASE, (uint32_t)buffer)` (device DMA register). This is only valid for an identity-mapped bare-metal system where the CPU virtual address, backing physical address, and device-visible DMA address are numerically equal. On a system with an IOMMU these are distinct address domains. REQ-UIP-EX-13 therefore requires production integrations to treat the CPU virtual address and device-visible DMA address as separate interface parameters and to use the

platform mapping mechanism to establish that they resolve to the exact same byte range within the same backing physical pages where an IOMMU is present.

3.8.1. Polling smoke path

[REQ-UIP-EX-14] The integrator shall run the example only after trusted firmware reports successful authentication, containment setup, policy lock, and NPU release. [REQ-UIP-EX-15] The application shall not be able to manufacture that indication. The example `npu_smoke.c` accepts a valid NPU descriptor supplied by the test application. It clears stale state, programs the 32-bit DMA base, queues the descriptor, starts the NPU, and polls `STATUS.done` / `STATUS.error` with a bounded timeout. The example's `timeout` values count polling-loop iterations, not units of elapsed time. [REQ-UIP-EX-16] The integrator shall replace `timeout` values with a platform timer when the integration requires a wall-clock deadline. Passing the test requires:

- successful CSR read and write transactions;
- DMA reads and writes only inside the assigned buffer range;
- a completed descriptor before timeout;
- no AXI `SLVERR` or `DECERR`; and
- the expected output data after the required cache maintenance.

Security negative tests shall also prove the following. [REQ-UIP-EX-17] The integrator shall verify that the NPU remains reset or bus-blocked after a failed authentication or policy setup. [REQ-UIP-EX-18] The integrator shall verify that DMA immediately below and above the permitted range is rejected and reported. [REQ-UIP-EX-19] The integrator shall verify that the NPU cannot select a secure/privileged attribute or another stream ID. [REQ-UIP-EX-20] The integrator shall verify that application software cannot rewrite locked firewall policy. [REQ-UIP-EX-21] The integrator shall verify that production lifecycle/debug policy prevents unauthorized register, JTAG, and scan access.

3.8.2. Interrupt smoke path

[REQ-UIP-EX-22] Before enabling the CPU interrupt, the integrator shall clear NPU `INTR_STATE` with a write-one-to-clear value and install the handler. The handler records the state and clears the asserted bits. [REQ-UIP-EX-23] The test shall retain a timeout source independent of the NPU interrupt so a missing interrupt cannot sleep forever.

Exercise both the real completion path and `INTR_TEST`: the former verifies work-to-interrupt integration, while the latter isolates interrupt routing from the compute and DMA paths. [REQ-UIP-EX-24] The integrator shall not count an `INTR_TEST` result as proof of DMA or descriptor execution.

3.9. User IP integration checklist

[REQ-UIP-CHK-1] The integrator shall record evidence beside every item. [REQ-UIP-CHK-2] A schematic review or prose assertion is not a substitute for the named RTL, static-analysis, simulation, or firmware result.

Done	ID	Gate	Required evidence	NPU example evidence
[]	[REQ-UIP-CHK-3]	Trust and attachment class selected	Approved data-flow/trust-boundary diagram naming runtime owner, TCB status, CSR, DMA, interrupt, clock, reset, power, and security-policy owners	Application-owned, outside the OCAH TCB; CSR and DMA are independent fabric ports
[]	[REQ-UIP-CHK-4]	Assets and threats classified	Reviewed list of executable/configuration state, secrets, model/data assets, attackers, security objectives, and accepted residual risks	Microcode/configuration policy decided; tensors treated as contained runtime data unless product policy says otherwise

Done	ID	Gate	Required evidence	NPU example evidence
[]	[REQ-UIP-CHK-5]	CSR aperture allocated without overlap	Machine-readable address-map check and positive/unmapped access test	4 KiB aperture; <code>npu_reg_top</code> decodes offsets [11:0]
[]	[REQ-UIP-CHK-6]	Protocol and width adapters reviewed	Interface assertions and tests for byte strobes, independent AXI channels, backpressure, and error responses	<code>npu_axi_csr_wrap</code> ; OCAH packed <code>axi_lite_to_tlul</code> is not used
[]	[REQ-UIP-CHK-7]	Transaction identity and privilege fixed	RTL review and assertions proving the IP cannot spoof secure/privileged <code>AxPROT</code> , authenticated source/stream ID, or unsupported transaction types	Trusted adapter fixes <code>AxPROT=3'b010</code> ; adopter ingress assigns the stream ID
[]	[REQ-UIP-CHK-8]	DMA containment and ranges proven	IOMMU/firewall configuration and lock evidence plus tests at the lowest/highest permitted address and rejected adjacent/protected addresses	64-bit data, 4-bit ID, 32-bit address; AW/AR zero-extended and then contained by an adopter IOMMU/firewall
[]	[REQ-UIP-CHK-9]	Secure boot and release fail closed	Positive and negative tests for authentication, minimum security version, policy setup/lock, reset or bus block on failure, and secure handoff	NPU remains reset/bus-blocked until trusted firmware authorizes release
[]	[REQ-UIP-CHK-10]	Lifecycle, debug, and scan policy closed	Asset-class manifest, feature-control allocation, DTP/scan connectivity, and tests in production, development, RMA, and secure-test conditions	No roots or keys; adopter needs reviewed lifecycle gate allocation before adding NPU debug/scan access
[]	[REQ-UIP-CHK-11]	Clock and reset crossings closed	Clock/reset specification, CDC/RDC reports, and reset assertions	One NPU clock; external CDC required if either fabric uses another clock
[]	[REQ-UIP-CHK-12]	Power and isolation sequence closed	UPF or equivalent intent, power-state simulation, and completion handshake test	Adopter-assigned subsystem index; quiesce DMA before reset or power-off
[]	[REQ-UIP-CHK-13]	Interrupt routing and ownership proven	Assigned interrupt/fault map; mask, trigger, acknowledgement, lost-interrupt, and unmaskable security-fault tests	Done/error retained levels to application IRQs; optional operational error to <code>ext_interrupts_i</code> ; trusted ingress reports containment violations
[]	[REQ-UIP-CHK-14]	Software platform contract generated or reviewed	Base/span, IRQ, DMA width/range, source identity, IOMMU context, security attributes, release state, cache policy, and ABI/security versions agree with RTL	<code>npu_platform.h</code> replaced and extended with adopter security resources
[]	[REQ-UIP-CHK-15]	Polling and interrupt smoke tests pass	Bounded positive tests plus timeout, IP error, fabric-error, containment, and unauthorized-access negative tests	Valid descriptor completes in both modes only after secure release; <code>INTR_TEST</code> separately verifies routing
[]	[REQ-UIP-CHK-16]	DMA coherency policy verified	Uncached mapping or cache clean/invalidate test on real platform	Reference range is uncached; production mapping requires platform confirmation
[]	[REQ-UIP-CHK-17]	Recovery is repeatable	Error capture followed by soft-reset and subsystem-reset tests, then successful resubmission	Escalate to <code>rst_ni</code> if NPU soft reset cannot recover the PE state
[]	[REQ-UIP-CHK-18]	Example and dependency are reproducible	Pinned public dependency or in-tree model with an automated build and test	Current adapter is illustrative and does not pass this gate until the external NPU dependency is made publicly reproducible
[]	[REQ-UIP-CHK-19]	Integration sign-off complete	Lint, CDC/RDC, synthesis, constraints, simulation, firmware, threat-model, security, DFT, and documentation reviews archived against one RTL revision	External NPU revision/configuration recorded; containment and secure-release evidence linked to the sign-off record

Chapter 4. Project Defines

Compile-time defines that apply across OCAH rather than to a single block.

4.1. SIMULATION

SIMULATION selects the verification view of OCAH RTL. Set it on every simulation compile and on no other kind of build:

```
+define+SIMULATION    # VCS, Xcelium
-DSIMULATION          # Verilator, Icarus
```

OCAH's own DV flows set it; a compile driven by an adopter's testbench does not.

WARNING

Do not let **SIMULATION** reach synthesis, formal, or a gate-level flow. It enables **initial** blocks, **\$readmemh**, **\$random** and DPI exports; a tool that skips those rather than rejecting them yields a netlist that does not correspond to the RTL you verified.

Without the define, these testbench hooks are not compiled in and the plusargs are silently ignored:

Plusarg	Effect
+skip_fuse_sense	efuse_shadow_regs reports fuse sense complete immediately rather than sensing the bank through the adopter's SHIM.
+smc_shadow_reg_preload=<file>, +sep_shadow_reg_preload=<file>	Loads the shadow registers from a hex image. Takes effect only alongside +skip_fuse_sense.
+rom_bin64=<file>, +rom_hex=<file>	Preloads OCAH4CORECluster_rom_ext , the SMC CPU ROM simulation model. Applies only to integrations that use the reference memory swaps.

The define also zeroes the RMA and security-disable token and digest bits in **efuse_token_processing** at time 0, which have no reset and otherwise power up X, and routes OpenTitan assertion failures in the SEP **prim_assert** shim to **\$error** on non-UVM runs.

Two model families it enables make simulation deliberately non-deterministic: **prim_flop_2sync** and its Caliptra and Adams Bridge equivalents insert **prim_cdc_rand_delay** ahead of the synchronizer, and **prim_lfsr** and **prim_double_lfsr** randomize their default seed on simulators other than Verilator (+**prim_lfsr_use_default_seed**=1 pins it). Two runs of one test are therefore not identical unless the simulator seed is fixed; record the seed with any failure you intend to reproduce.

Chapter 5. Prim Cell Replacement and Don't-Touch Constraints

OCAH primitives (`prim_stdbuf`, `prim_and2`, `prim_clkgater`, the `prim_flop_*sync*` family, and so on) ship as behavioral models with a reference technology mapping. Adopters replace them with cells from their own library.

Two rules govern those replacements. Both are the adopter's responsibility — OCAH provides the instance names, but the names have no effect until the adopter's synthesis and place-and-route flows constrain them.

5.1. Naming Requirement

Every library cell instantiated inside a replacement primitive must have `d0nt_` in its instance name.

The marker is deliberately misspelled with a zero. `dont_touch` is a reserved keyword in several tools and a common substring in ordinary comments and signal names; `d0nt_` occurs only where the protection is intended, so a name-based match cannot pick up unrelated cells.

Clock-path cells take the extended prefix `d0nt_clk`, which selects the stricter rule below. The `d0nt_*` naming convention appears in the eFuse IP (`efuse_token_digest_sha256`, `efuse_triple_redundant_comparator`) as the primary example. The adopter is expected to carry the same convention into any technology-mapped primitive wrapper they deliver. The intended name assignments are:

Instance name	Rule	Primitive
<code>d0nt_clkgate</code>	Don't touch	<code>prim_clkgater</code>
<code>d0nt_clkbuf</code>	Don't touch	<code>prim_clock_buf</code>
<code>d0nt_clkinv</code>	Don't touch	<code>prim_clock_inv</code>
<code>d0nt_clkmux</code>	Don't touch	<code>prim_clock_mux2</code>
<code>d0nt_clknand</code>	Don't touch	<code>prim_clock_nand2</code>
<code>d0nt_clkxor</code>	Don't touch	<code>prim_xor2</code>
<code>d0nt_stdbuf</code>	Size only	<code>prim_stdbuf</code>
<code>d0nt_stdinv</code>	Size only	<code>prim_flop_*sync_*</code> reset inverter
<code>d0nt_an</code>	Size only	<code>prim_and2</code>
<code>d0nt_ao222</code>	Size only	<code>prim_ao222</code>
<code>d0nt_wrap_sync</code>	Size only	<code>prim_flop_*sync_*</code> synchronizer flops

Keep the same names when substituting your own cells. Adopters who rename must apply the equivalent constraints to whatever names they use.

5.2. Constraint Requirement

In synthesis, apply `set_size_only` to every cell matching `d0nt_`, then `set_dont_touch` to every cell matching `d0nt_clk`. Every step after synthesis needs `size_only` alone — on all `d0nt_` cells, clock cells included.

```
# Order matters: d0nt_clk* also matches d0nt_*, so the clock rule is applied
# second and overrides size-only for those cells.
```

```
set_size_only [get_cells -hierarchical -filter "full_name =~ *d0nt_*"]
set_dont_touch [get_cells -hierarchical -filter "full_name =~ *d0nt_clk*"]
```

size_only keeps the instance in the netlist—the tool may not delete, merge, absorb, restructure, or buffer through it—while still allowing it to be swapped for another library cell of the same logic function, typically to change drive strength or threshold-voltage flavor for timing, area, and EM closure. That is the intent for every replacement cell: these primitives exist because the structure matters, not because the logic function could not be inferred. Synthesis will otherwise remove them—an inverter pair that cancels, redundant AND gates that always agree, a buffer driving a single load are all dead weight to an optimizer.

Clock cells need the stronger **dont_touch** because that swap is exactly the problem. Under **size_only** the tool treats the logic function as the contract and every other property of the cell as free to change. On the clock path the function is the least of the reasons the cell was chosen. Clock buffers, inverters, muxes, and gates are picked for electrical characteristics the function does not capture—matched rise and fall times above all, and with them duty-cycle preservation, balanced input-to-output delay, and low delay variation across PVT. A functionally identical cell pulled from the general-purpose library guarantees none of that. The netlist still says "inverter", functional simulation still passes, and the duty-cycle distortion shows up only in silicon or in a timing run late enough to be expensive.

dont_touch removes the tool's discretion: the cell the designer selected is the cell that reaches the netlist. Sizing and insertion-delay decisions on the clock path belong to clock tree synthesis, which owns them explicitly, rather than to a logic-optimization pass making a substitution it believes is free.

5.2.1. After Synthesis

The **dont_touch** on **d0nt_clk*** is a synthesis-only constraint. Once synthesis has handed off the netlist, release the clock cells to **size_only** and carry that—and nothing stronger—through placement, clock tree synthesis, routing, and optimization:

```
# Post-synthesis: uniform size_only, clock cells included.
set_dont_touch [get_cells -hierarchical -filter "full_name =~ *d0nt_clk*"] false
set_size_only [get_cells -hierarchical -filter "full_name =~ *d0nt_*"]
```

The stricter rule exists to stop logic optimization from choosing the cell. Downstream, clock cell sizing is CTS's job: it cannot balance skew or close transition violations with the cells it needs to resize frozen, and freezing them there buys nothing. Keeping **size_only** in place still prevents deletion, merging, and restructuring for the rest of the flow.

Restrict the physical flow to the library's clock-cell list so a resize stays inside the same clock cell family. Sizing a **CLKBUF_D4** up to a **CLKBUF_D8** keeps the matched rise and fall behavior the cell was chosen for; substituting a general-purpose buffer of equivalent drive does not.

Additional requirements:

- **Carry size_only through PnR.** The **dont_touch** is synthesis-only, but **size_only** is not. A cell protected in synthesis and left entirely unprotected downstream is not protected.
- **Verify the match is non-zero.** A typo in the filter, or a flow that renames instances before the constraint is read, silently protects nothing. Check the collection size and error out if it is empty.
- **Preserve the names.** Uniquification, flattening, and name mapping must not strip the **d0nt_** substring. If your flow renames instances, derive the collections before flattening and carry them forward.

- **Restore protection after ECOs.** Flows that relax preservation constraints for clock-tree or filler ECOs — clearing `eco_honor_dont_touch`, for example — must restore `size_only` before any subsequent optimization pass.
- **Check the final netlist**, not just the post-synthesis one.

Flows other than Design Compiler and Fusion Compiler must apply the equivalent `size_only` and `dont_touch` attributes to the same collections.

5.3. Simulation

These constraints have no functional effect and no simulation impact. RTL simulation compiles the technology-independent primitive models (`not(synth)`). Library cells are not on that filelist, so a simulator never elaborates them.

Confirming that the protected cells survived is therefore a gate-level netlist check. RTL simulation cannot detect their loss.

Chapter 6. CDC Timing Constraints

OCAH ships synthesis constraints that bound the enumerated synchronization structures in the SMC, SEP, DTP and SMU. Independent CDC sign-off is still required for the assembled SoC. This section explains what those constraints do, how they are produced, and what an adopter has to change when integrating OCAH into their own flow and cell library.

The collateral ships with the harness. Each block carries its own constraints in `hw/sys/<block>/synth/`:

- `constraints.sdc` — the block's full timing intent, including the clock definitions and the asynchronous clock groups. It sources the two files below.
- `<block>_cdc_max_delay.tcl` — generated-clock periods and any hand-written overrides.
- `<block>_cdc_max_delay_generated.tcl` — the per-instance calls.

The two pieces shared by every block live in `flows/synth/constraints/`: `async_clock_groups.tcl` and `cdc_max_delay_procs.tcl`.

6.1. Why crossings are bounded rather than timed

A clock-domain crossing has no fixed phase relationship, so it cannot be timed against a capture edge. Its correctness still depends on delay, though: every CDC element assumes the data reaches the receiver within a bounded time.

OCAH therefore keeps the crossings visible to the tool and gives each one an explicit upper bound, sized from the clock that captures it. This asserts no phase relationship — it states only how much delay the crossing tolerates while its CDC element's guarantee still holds.

6.2. What is constrained

Each enumerated CDC element in each block is given its own `set_max_delay`, sized from the clock that actually captures it as read from the netlist. Muxed domains resolve to several clocks; the tightest period wins.

The elements covered are `prim_sync2/3/4` and their reset and set variants, the pulp `sync` shim, `prim_sync_reset`, `prim_sync3_pulse`, `prim_fifo_async`, `avsbus_async_fifo`, `cdc_fifo_gray`, `axi_cdc`, `prim_reg_cdc`, `prim_sync_reqack_data`, `prim_sync_data_autohs`, `cdc_2phase`, `dmi_wrapper` and Rocket `AsyncQueue`.

The bound depends on what protects the crossing:

Kind of path	Bound	Reason
Synchronizer data input	$0.5 * T_{dst}$	Leaves half a destination period of settling margin before the capture edge the synchronizer relies on.
Protocol-protected data	$1.0 * T_{dst}$	Conservative. The requirement is set by the pointer synchronizer's depth — see below.
Two-phase handshake nets	$\min(T_{src}, T_{dst})$	<code>cdc_2phase</code> runs the handshake in both directions, so the tighter domain governs. This is the module's own stated requirement.

T_{dst} rather than $\min(T_{src}, T_{dst})$ in the general case because the gating synchronizer is clocked by the destination, so its settling window is $S * T_{dst}$ and T_{src} does not enter the guarantee. This makes slow-to-fast and fast-to-slow equally safe.

6.2.1. Why protocol-protected data still needs a bound

The payload of a gray-pointer FIFO or a req/ack handshake never passes through a synchronizer — only the pointer or request does. So this is not a metastability constraint, and it is reasonable to ask why such a path needs a bound at all, or why a single blanket bound covering every crossing in the design would not be enough.

Because the protocol guarantees **ordering**, not **arrival**. The read side will not look at an entry until the write pointer has crossed, which takes `SYNC_STAGES` destination cycles, where `SYNC_STAGES` is the depth of the pointer synchronizer. Nothing in the protocol ensures the data physically arrived within that window; that is a timing property, and only a `max_delay` states it. Miss it and the consumer samples the entry's previous contents. There is no violation report and no metastability event, just wrong data.

So the requirement is `data path < SYNC_STAGES * T_dst`. It is not a fixed number of nanoseconds: it scales with the clock that captures the crossing, and the destination periods within one design span an order of magnitude.

Destination period	Guarantee at 3 stages
1 ns	3 ns
5 ns	15 ns
10 ns	30 ns

That is why the bound is applied per instance rather than as one global value. A single blanket `max_delay` has to be chosen once for the whole design: loose enough to be feasible on the slowest domains, it permits several times more delay than a fast crossing's pointer scheme tolerates; tight enough for the fastest domain, it over-constrains every slow crossing and burns effort closing paths that were never at risk. Sizing each crossing from its own capturing clock is the only bound that tracks the requirement.

The blanket default that the clock-group layer applies (see [The two layers](#)) is deliberately on the loose side, and for slow destination domains it is already sufficient — there the explicit per-instance constraint is redundant. It is the fast crossings that need it.

`1.0 * T_dst` is therefore conservative by a factor of two to three. It is kept simple deliberately, but if wide payloads such as AXI data will not close, raising `cdc_data_delay` to `2.0 * T_dst` stays inside a three-stage guarantee with a cycle to spare. An adopter whose synchronizer cells use a different depth should recompute from their own `SYNC_STAGES` rather than inheriting this factor.

Enumeration and clock resolution run once, offline, against an elaborated design, and their output is an explicit per-block file of constraint calls that is reviewed and checked in. The flow itself does no discovery. This matters to an integrator because those files are written against the OCAH hierarchy and OCAH's clock names. If you instantiate a block as delivered, you do not need to regenerate or hand-edit them: set `::cdc_hier_prefix` to where the block sits in your hierarchy, and optionally `::cdc_clock_alias` to map OCAH clock names onto your own. Regeneration is needed only when the block is reconfigured in a way that changes which CDC elements exist, since no prefix can conjure constraints for elements the file never listed (see [Integration checklist](#) item 2).

6.3. The two layers

The per-instance bounds take effect only on top of a correctly configured clock-group layer. That layer decides whether the tool considers inter-domain paths at all, so it must be established first; the per-instance exceptions then refine it.

Layer 1 is `set_async_clock_groups`, defined in `flows/synth/constraints/async_clock_groups.tcl` and called from each block's `constraints.sdc`. It declares the asynchronous groups with `-allow_paths` and applies a loose 20 ns / -20 ns default to every inter-group clock pair.

IMPORTANT

`-allow_paths` is a prerequisite, not a refinement. In DC and PrimeTime exception priority, `set_false_path` outranks `set_max_delay` regardless of which is more specific, so a bare `set_clock_groups -asynchronous` silently masks **every** per-instance constraint. There is no warning; the exceptions simply have no effect.

The 20 ns default is what makes `-allow_paths` safe to enable. Once paths are allowed, every crossing without a specific constraint would otherwise be timed against a meaningless launch-capture relationship and fail. The default bounds them loosely so that only deliberately tightened paths appear as real requirements. It is emitted per ordered clock pair, never group-wide, because a 20 ns bound inside a group would mask genuine setup failures on synchronous paths.

Layer 2 is the per-instance `set_max_delay` output of the three steps above, with two deliberate carve-outs:

- **Paths from primary inputs revert to the default.** A synchronizer input reachable from a top-level port carries that port's `set_input_delay`, which is a budget figure rather than real logic and can alone exceed the tightened bound. Only the parent block sees both ends, so there is nothing to tighten at block level.
- **Same-clock instances are skipped.** A two-clock element wired to one clock is not a crossing, and applying `set_max_delay -ignore_clock_latency` there would replace a proper setup check with a flat budget that discards skew and uncertainty. These are emitted commented out. OCAH has five, where OpenTitan AES, KMAC and OTBN instantiate `prim_sync_reqack_data` on EDN and OTP interfaces that are separate domains upstream but share a clock in SEP.

6.4. What is covered

Block	CDC elements constrained	Exceptions emitted
SMC	365	412
SEP	18	21
DTP	250	270
SMU	758	838

SMU is a flat build containing the other three, so its instance paths carry `u_smc/`, `gen_sep.u_sep/` and `u_dtp/` prefixes. It is not the sum of the parts: DTP contributes 375 elements inside SMU against 250 standalone, because it instantiates three JTAG2AXI bridges in the SMU configuration and two on its own.

6.5. Integration checklist

Five things must change, three should be checked. None of them requires editing per-instance data by hand; every change is either a one-line policy edit or a regeneration.

6.5.1. 1. Repoint the constraints at the pin of the cell inside the prim wrapper

This is the most important change.

As shipped, the procs target the prim wrapper's own port, `prim_sync3/i_d`. That port exists only as long as the wrapper survives as a hierarchical boundary, and a prim cell is ordinary RTL: nothing stops synthesis from ungrouping it and dissolving its ports.

WARNING

Do not rely on the wrapper port surviving. Anchor the constraint to the pin of the cell instantiated **inside** the wrapper. When you swap in your own synchronizer, that cell is a real library cell or

macro—a leaf that cannot be flattened—so its pins persist through optimisation regardless of what happens to the RTL hierarchy around it.

```
proc set_cdc_max_delay_prim_sync3 { inst dst_clk {delay {}} } {  
    cdc_emit "prim_sync3 $inst" [cdc_sync_delay $dst_clk $delay] to \  
        {} [get_pins "$inst/u_my_sync_cell/D" -quiet]  
}
```

`$inst` stays the wrapper instance path the generator recorded; you extend it down to the leaf. If your cell is instantiated in a generate loop or per bit, the path needs a glob. Note that generate scopes are separated by `.`, not `/`:

```
[get_pins "$inst/g_bits*.u_my_sync_cell/D" -quiet]
```

Anchoring this way makes hierarchy preservation a safety margin rather than a load-bearing dependency.

If you keep the prim wrappers unchanged, the shipped port names work as-is, but your flow must then preserve those wrapper boundaries. A `destination empty` warning appearing after `compile` where there was none at constraint-load time means a boundary was dissolved.

If you go further and replace a wrapper with a module of your own name, the enumeration that produced `<block>_cdc_max_delay_generated.tcl` knows nothing about it and those crossings end up with no per-instance bound at all. Nothing warns—an element that was never listed cannot report itself missing. Add the calls for it to `<block>_cdc_max_delay.tcl` by hand; the procs take an instance path and the capturing clock regardless of the module they were named after.

6.5.2. 2. Point the files at your hierarchy, or regenerate

The checked-in `<block>_cdc_max_delay_generated.tcl` files are written against the OCAH hierarchy, with paths relative to the block's own top and clocks named as OCAH names them. Two ways to make them apply to your design.

If you instantiate the block as delivered, reuse the files and set two variables. Both default to no-ops, so leaving them alone reproduces the shipped behaviour exactly.

```
# where the block sits in your hierarchy, with a trailing slash  
set ::cdc_hier_prefix "u_chiplet/u_mgmt/u_smc/"  
  
# optional: map an OCAH clock name onto the clock you actually drive it with  
set ::cdc_clock_alias(SMCCLK) MY_SYS_CLK
```

The prefix is prepended to every instance path. The alias is applied wherever a clock name reaches the tool, so OCAH's names become labels in your design. You can also skip the alias and supply the period directly with `set ::cdc_clock_period(SMCCLK) <ps>`, which is enough for the synchronizer constraints; the alias additionally matters for the data-path constraints, which name the destination clock as a timing endpoint and so need it to exist.

If you reconfigure the block, regenerate instead. Changing parameters that add or remove CDC elements makes the shipped file stale in the silent direction—the constraints that should exist for new elements simply will not, and nothing warns. The prefix does not help with that.

A wrong prefix fails loudly: every path resolves to nothing, so you get a warning per constraint against a zero exception count, and `SUMMARY 0 exceptions` is itself reported as a setup failure.

Regeneration is also the right answer whenever a parent build contains the block. Instance counts there can differ from the standalone build, so a parent's file is not the child's file with a prefix.

6.5.3. 3. Set the default bound

The 20 ns / -20 ns figures are a starting point. Both are `info exists-guarded` in `flows/synth/constraints/async_clock_groups.tcl`, so setting them before that file is sourced overrides without editing it. `set_async_clock_groups` also accepts per-call overrides, which is the cleaner way to give one block a different default:

```
set_async_clock_groups { {CLK_A} {CLK_B} } -max_delay 8000 -min_delay -8000
```

Two constraints on the choice. The default must exceed every per-instance bound you tighten, because it also governs paths from primary inputs into synchronizers; if it is tighter than $0.5 * T_{dst}$ it becomes the binding constraint there. And tighter is not automatically better: this is a net for crossings you have not enumerated, so set it loose enough that those do not fail and rely on layer 2 for the paths that matter.

The minimum is negative deliberately. With `-allow_paths` a crossing has no meaningful hold relationship, and a zero floor would report hold violations where hold is not a real requirement.

6.5.4. 4. Declare your own clock groups

`set_async_clock_groups` is called from each block's `constraints.sdc` with that block's group list. Each inner list is one asynchronous group and globs are allowed. Globs resolve when the call runs, so it has to come after every `create_clock` and `create_generated_clock` it is meant to cover.

```
set_async_clock_groups {  
    {CLK_A CLK_A_DERIVED_*}  
    {CLK_B}  
    {JTAG_TCK}  
} -exclude {{PATTERN_X* PATTERN_Y*}}
```

`-exclude` takes pairs of patterns and suppresses that part of the cross product, for clock pairs that already have a `-logically_exclusive` or `-physically_exclusive` relationship that `-allow_paths` would contradict. Omit it unless you hit a `TIM-184` conflict.

6.5.5. 5. Declare periods for generated clocks

A generated clock exposes no readable `period` attribute in Design Compiler, so any generated clock that clocks a CDC element needs an explicit entry in `<block>_cdc_max_delay.tcl`:

```
set ::cdc_clock_period(MY_DIV_CLK) [expr {4 * $clock_periods(MY_MASTER_PERIOD)}]
```

Omitting one is loud — a `no period for clock` warning per affected element — so a clean run means you have them all.

Keep them in step with your `create_generated_clock` divide factors; nothing cross-checks them.

6.5.6. Also check

Hierarchy preservation. The reference flow applies `set_ungroup false` and `set_boundary_optimization false` to the CDC designs before `compile`. How much this matters depends on item 1: with leaf cell pins it is a safety margin, with wrapper ports it is load-bearing. Either way, your flow needs the equivalent, and it needs to name your cells rather than OCAH's.

Delay factors. $0.5 * T_{dst}$ and $1.0 * T_{dst}$ are policy, not physics. If your methodology wants more settling margin, change `cdc_sync_delay` and `cdc_data_delay` in `flows/synth/constraints/cdc_max_delay_procs.tcl`. Per-instance overrides are also available without editing procs: every proc takes an optional trailing delay in picoseconds, and re-stating a call in `<block>_cdc_max_delay.tcl` after the `source` wins.

Tool differences. Two Design Compiler specific choices to revisit under PrimeTime. `get_attribute <pin> clocks` is used because `get_clocks -of_objects` is PrimeTime-only, and generated clocks may expose a readable `period` in PrimeTime, making item 5 unnecessary.

6.6. Confirming it worked

Load the constraints and check the summary:

```
INFO: cdc_max_delay: SUMMARY <n> exceptions, 0 warnings
```

Zero warnings is necessary but not sufficient. A pin pattern that matches nothing warns, but one that matches the **wrong** thing does not. Two stronger checks:

- **Predict the count.** Multiply each proc's number of emitted exceptions by its number of calls and compare against the summary. A mismatch localises the failure, because the size of the shortfall identifies which proc did not fire.
- **Confirm the precedence.** Layer 2 is on pins and layer 1 on clocks, and the design assumes the more specific specification wins. Verify once in your tool:

```
report_timing -to [get_pins <a synchronizer>/i_d] -nworst 1
```

The `max_delay` line must show your $0.5 * T_{dst}$ value rather than the default. If it shows the default, layer 2 is inert and nothing else in this section applies.

`SUMMARY 0 exceptions` is treated as a setup failure and warns as such. It means the constraints did not load, not that the design has no crossings.

6.7. Scope and limitations

The 20 ns default is loose by design. Against a 1 ns clock it permits twenty clock periods of delay into a synchronizer input. It is a safety net that keeps `-allow_paths` usable, not a real constraint; any crossing that matters belongs in layer 2.

Enumeration works from a fixed list of module names, so a hand-written two-flop synchronizer that uses none of them will not be found. Structural detection of crossings is what a static CDC tool does, and this flow does not attempt it. These constraints time crossings already known to exist; they are not a substitute for CDC signoff.

Because the instance list is generated offline, removals are loud but additions are silent. A stale instance path warns, but a synchronizer added without regenerating produces no warning: the constraint never exists, the path falls back to the default, and it passes timing. Regenerate and diff the result against what is checked in whenever the RTL changes.